Songlines: Measurement, Transfer, and Formation of Collective Semantic Memory for Multi-Agent Navigation

A Unified Research Series (Parts I–V)

Abstract

Long-horizon multi-agent systems require more than access to large context windows or shared event logs. They must determine which observations should become persistent memory, how records produced by different agents refer to the same entities, when foreign evidence is sufficiently current and reliable to guide action, and how shared knowledge affects collective behavior. This volume presents the five parts of the Songlines research program as a unified study of these problems. Part I introduces Q/R/M/C, a stage-decomposed evaluation protocol that separates query formation, retrieval, target materialization, and post-retrieval completion. Across symbolic, learned, and language-agent systems, the protocol identifies failure modes that are not visible in end-to-end success; in the single-agent benchmark, the principal retrieval limitation is candidate generation (91% empty retrievals) rather than candidate ranking (96.9% precision when candidates exist). Part II extends this measurement contract with evidence provenance. It defines semantic warp as a commitment supported by foreign evidence, derives a closed-form trust–staleness horizon, and shows that foreign memory is often more valuable as transport than as a directly completed target. A reservation-and-rollback protocol converts fast sharing from a contention-prone regime into the strongest tested communication setting. Part III identifies the transferable representation as a relational route rather than an isolated coordinate. Semantic landmarks establish cross-agent place correspondence, route edges provide transport, and edge-level provenance determines the valid prefix of the transferred route. Part IV specifies collective memory through explicit merge, trust, and staleness rules and provides order-theoretic, categorical, and coalgebraic interpretations of the resulting distributed process. Part V addresses memory formation. It models retention as a choice among merge, exception, new schema, repeat, and drop, governed by counterfactual marginal utility and the cost of structural analogy. The resulting controller is learned and its grammar parameters are optimized within a fixed family; the full runtime further incorporates receiver-side admission, utility estimation without oracle replay, equal-budget evaluation, adversarial provenance, and fail-closed safety tests on discrete, continuous, and VMAS substrates. The combined evidence supports a specific conclusion: effective collective memory depends on structured consolidation, provenance, temporal validity, and coordination authority, rather than on unrestricted accumulation or exchange. The results establish an end-to-end research prototype, not a universal solution; open-ended perceptual concept formation, large-scale social LLM evaluation, and real-robot validation remain outside the present scope.

Preprint.

Scope and organization of the volume

Research question. The series studies how semantic navigation memory is measured, transferred, coordinated, and formed in populations of agents. The five parts are related, but each isolates a distinct methodological question. Part I develops the evaluation protocol. Part II measures the causal role and temporal validity of foreign evidence. Part III defines the structural unit that can be transferred across private frames. Part IV specifies the mechanics of distributed collective memory. Part V studies how observations are selected, consolidated, and admitted into long-term memory.

Unified claim. We claim that executable collective memory requires five jointly preserved properties: structured consolidation, evidence provenance, relational transport, temporal validity, and coordination authority. Unrestricted accumulation, long context windows, or unregulated information exchange do not provide these properties by themselves.

Relationship among the parts. Parts I and IV originate from an earlier joint manuscript on Q/R/M/C and minimal collective semantic memory. Parts II and III extend the measurement framework with provenance, route structure, and frame-independent place correspondence. Part V uses the mechanisms of Parts I–IV as the fixed substrate for a learnable memory-formation runtime. Cross-references in this unified version therefore refer to other parts of the same volume, although each part retains the structure of a stand-alone submission.

Evidence and reporting protocol. The experiments use fixed seeds, explicit event definitions, pre-specified predictions where applicable, and retained records of failed registrations and subsequent revisions. Exact-match results are reported only for settings in which the corresponding law or matching criterion is deterministic: the warp-distance hold-out ($6/6$), the route-rupture study ($110/110$), discrete frame recovery ($80/80$), and the vocabulary safety ablation (0 fail-open cases in 252 cells). Other results are reported through effect sizes, confidence intervals, paired comparisons, and explicit scope statements.

Code and artifacts. The experiments are implemented in a unified repository under `experiments/` and `songline_drive/`. Each part includes a reproducibility appendix with commands, configurations, and the location of run artifacts. The unified source preserves the original part-specific labels so that the individual submissions and the series report can be maintained from the same experimental record.

Contents

Part I

The Q/R/M/C Measurement Framework

Stage-decomposed evaluation of memory-based navigation: one logging contract, five system classes, and the candidate-generation diagnosis.

Part abstract. End-to-end success rates do not identify which component of a memory-based navigation pipeline is responsible for failure. We introduce Q/R/M/C, an operational evaluation protocol that separates query formation, retrieval satisfaction, target materialization, and completion after retrieval. The four stage events are defined at the log level and their conditional rates factorize semantic-path completion for systems that expose an explicit query and target-lock interface. The contribution is therefore methodological rather than a new probabilistic theorem: the factorization follows the chain rule, whereas the value of the framework lies in making intermediate failure locations observable and comparable. We evaluate the protocol on symbolic single-agent tasks, four multi-agent memory architectures, learned communication policies, local LLM agents, and three third-party agent stacks. In the single-agent benchmark, oracle interventions distinguish retrieval-limited hazard recovery from downstream-limited goal-region rejoin. Log analysis further shows that the apparent retrieval failure is primarily a memory-accumulation problem: eligible candidates are absent at 91% of decision points, while ranking precision is 96.9% when candidates are present. In the multi-agent benchmark, communication cadence changes the location of the bottleneck. Faster exchange improves materialization but reduces completion under scarce targets because agents commit to the same resources; the corresponding stage-rate slopes are moderate and cluster-robust, and mean time to success has a shallow interior minimum at $K = 8$ when conditioned on successful episodes. The same logging contract localizes engineered failures in LLM and third-party agent stacks, while also revealing when Q/R/M/C is not separately observable in opaque learned policies. Q/R/M/C should therefore be interpreted as a diagnostic layer for systems with an explicit memory interface, not as a replacement for end-to-end evaluation, policy probing, or causal mediation analysis. The primary evidence is based on controlled grid-world experiments, with directional checks on BabyAI, VMAS, TextNav, and external agent frameworks.

1 Introduction

Memory-based navigation is commonly evaluated through a final success indicator or an aggregate return. These measures are necessary, but they do not distinguish failures caused by an absent query, an empty or incorrect retrieval, an inability to convert retrieved evidence into an actionable target, or a failure of the controller after a valid target has been selected. This distinction is particularly important when the target is specified semantically rather than by a fixed coordinate. In such tasks, the agent must first represent the intent, retrieve a place that satisfies it, commit to a concrete referent, and execute a trajectory to that referent.

We propose Q/R/M/C as a stage-decomposed evaluation protocol for this setting. The protocol records four events: query formation (Q), retrieval satisfaction (R), target materialization (M), and completion after materialization (C). The nested episode-level events factorize semantic-path completion, whereas the operational per-query and per-tick rates provide a practical diagnostic view of the same pipeline. The factorization is intentionally elementary. Its purpose is to specify a logging contract under which intermediate failure locations can be estimated consistently from trajectory records.

The empirical analysis addresses three questions. First, does the stage decomposition distinguish task families that have similar final success but fail for different reasons? Second, how does the location of the bottleneck change when memory becomes a shared resource among multiple agents? Third, does the same event contract remain useful on learned and language-agent systems whose internal representations differ from the symbolic reference implementation?

The single-agent experiments show that stage localization changes the interpretation of the benchmark. Hazard recovery is retrieval-limited: replacing retrieval or materialization with an oracle produces a substantial success increase, whereas replacing the controller does not. Goal-region rejoin is instead limited downstream of retrieval. Within the retrieval-limited task, the detailed logs further identify

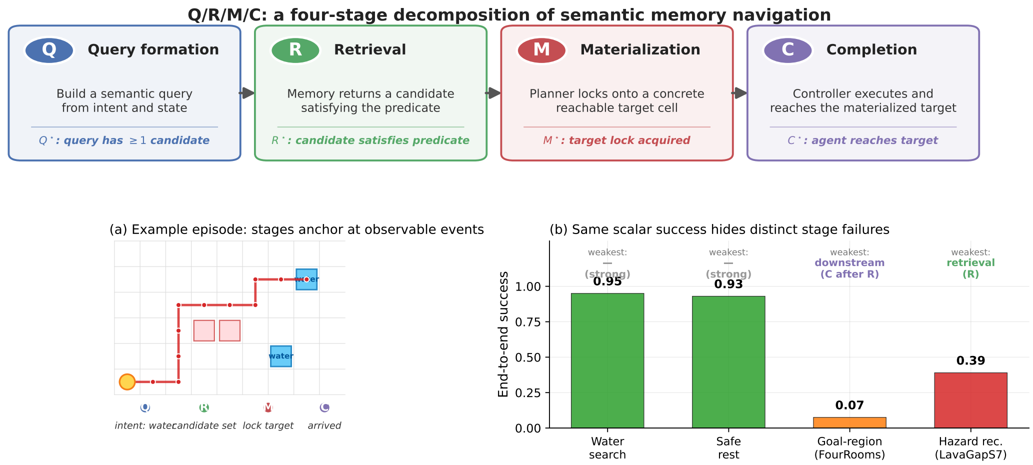

Figure 1: Q/R/M/C decomposes one semantic-navigation episode into four observable stages. **Top:** the four stages and their event-level definitions. **(a)** A successful water-search episode anchors at one event per stage, producing logged starred events $Q^\star, R^\star, M^\star, C^\star$. **(b)** The same scalar "end-to-end success" summary masks qualitatively different failures on the four single-agent task families: water and rest are strong; goal-region rejoin (FourRooms) is downstream-limited; hazard recovery (LavaGapS7) is retrieval-limited (oracle interventions, §3.4).

candidate generation as the principal failure: the memory returns no eligible record at most decision points, despite near-perfect ranking when a record exists. The multi-agent analysis shows a different phenomenon. As broadcast cadence increases, fresher peer memory improves materialization, but the same information synchronizes commitments and reduces completion under target scarcity. The result is a measurable migration of the bottleneck between the M and C stages.

The contributions of this part are as follows:

1. We define log-level operational and nested Q/R/M/C events for single- and multi-agent memory-based navigation and state the conditions under which their conditional rates are estimable.
2. We validate the protocol through stage-specific oracle interventions and identify a memory-accumulation limitation that is not visible in end-to-end success or candidate-ranking precision.
3. We characterize a cadence-dependent materialization–completion trade-off in a large multi-agent sweep and relate it to foreign evidence and target occupancy.
4. We evaluate the same event contract on learned communication policies, local LLM agents, and three third-party agent frameworks, thereby separating structural task failures from model- and framework-specific behavior.

The framework has a clear boundary. Separate Q/R/M/C rates require an observable query and commitment interface. For opaque policies, the protocol can identify the absence of stage observability but cannot recover latent stages that the architecture does not expose. In addition, the four-rate product describes completion through the semantic retrieval path rather than all possible routes to task success. These restrictions are treated as part of the measurement contract rather than hidden assumptions.

The remainder of Part I defines the event semantics and estimators, reports the single- and multi-agent studies, evaluates portability to language-agent and third-party stacks, and discusses limitations and reproducibility.

2 The Q/R/M/C decomposition

2.1 Problem formulation and notation

The agent operates in a partially observable navigation environment. At each step it receives an observation o_t, maintains a memory state $\mathcal{M}_t$, and forms a query q_t encoding the current task intent. A retrieval step returns a candidate set of places matching q_t from $\mathcal{M}_t$; a chosen candidate is materialized into an actionable target; a local controller executes actions until either the target is reached or the episode times out. The target is not a coordinate but a semantic pattern, so success is a function of the chosen candidate's semantic profile, not of any pre-supplied coordinate.

Habitat, AI2-THOR, and ProcTHOR made visually rich navigation evaluation standard (Savva et al., 2019; Kolve et al., 2017; Deitke et al., 2022). We deliberately focus on MiniGrid-class environments to obtain clean and reproducible stage measurements, and use BabyAI only as a portability check. We do not claim to solve LLM-agent memory here; rather, we argue that the same Q/R/M/C decomposition is a useful template for evaluating memory-grounded agent decisions beyond navigation.

The central empirical question is therefore not just whether the agent succeeds, but *which stage fails first* when it does not. Two related measurement layers are used. **Operational rates** count, per query attempt, whether the query yielded a non-empty candidate set, whether the chosen candidate satisfied the semantic predicate, and whether it was materialized into an actionable target; *completion after retrieval* is an episode-level indicator. **Nested episode-level events** $Q^\star \Rightarrow R^\star \Rightarrow M^\star \Rightarrow C^\star$ are used in the factorization below and audited in §3.2. Full operational definitions are given in Appendix 11.

Notation. The four stages are Q, R, M, C with episode-level starred events $Q^\star, R^\star, M^\star, C^\star \in \{0,1\}$, nested by construction. $P(\cdot)$ denotes the population probability; $\hat{P}(\cdot)$ its empirical-frequency estimator. **The letter M always refers to the Materialization stage; the number of targets in the multi-agent environment is written T (not M, constraint $T \leq N$, "scarcity" = $N > T$).** Other symbols: $\mathcal{C}_K$ peer-broadcast protocol with cadence K, $\mathcal{M}_{-i}^{(K)}$ peer memory snapshots, $\mathcal{O}_{-i}$ peer occupancy, $\nu(\cdot)$ freshness functional, $\Phi(K) = P(M^\star | R^\star; K) \cdot P(C^\star | M^\star; K)$, t_{succ} mean time-to-success. Full symbol table in Appendix 12.1.

2.2 Single-agent factorization

Let Y denote overall episode success, and let $Q^\star, R^\star, M^\star, C^\star$ denote the nested episode-level semantic-path events. The framework factorizes *semantic-path completion* rather than raw task success:

$$Q^\star \to R^\star \to M^\star \to C^\star.$$

For any query q and memory state $\mathcal{M}$,

$$\begin{aligned} P(C^\star = 1 \mid q, \mathcal{M}) = P(Q^\star = 1 \mid q, \mathcal{M})\; P(R^\star = 1 \mid Q^\star = 1, q, \mathcal{M}) \\ \cdot\, P(M^\star = 1 \mid R^\star = 1, q, \mathcal{M})\; P(C^\star = 1 \mid M^\star = 1, q, \mathcal{M}). \end{aligned}$$

Overall task success Y can exceed $C^\star$ because some episodes succeed without traversing the full semantic retrieval path.

Assumption 1 (conditional stage-Markov property). Conditioned on the current stage being successful, the probability of the next stage depends only on the current successful stage, the current query, and the current memory state. Formally,

$$P(R^\star \mid Q^\star, q, \mathcal{M}, \text{history}) = P(R^\star \mid Q^\star, q, \mathcal{M}),$$

and analogously for $M^\star$ and $C^\star$.

Assumption 2 (positivity). Each conditioning event used above has non-zero probability on the support of the benchmark.

Definition 1 (single-agent stage estimability). Under Assumptions 1–2, the four conditional rates in the factorization above are well-defined population quantities and the empirical-frequency estimators are consistent estimators of them; their product converges to semantic-path completion. The argument is elementary — the chain-rule factorization plus consistency of frequency estimators on Bernoulli events — and the value of stating it is operational, not mathematical: it pins down the *logging discipline* (which events, conditioned on what) under which the decomposition is readable from standard trajectory logs, and it is the contract that every experiment in this paper audits against (§3.2). Full statement and elementary argument in Appendix 9.

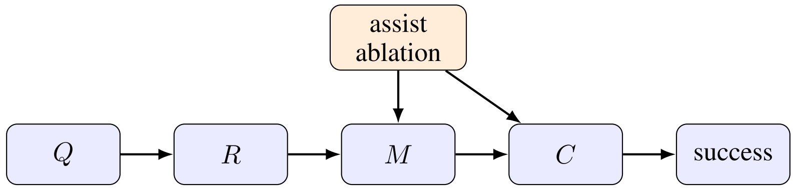

Figure 2: Stage-structured interpretation of the Q/R/M/C framework. Assist on/off ablations target intermediate mechanisms; in the multi-agent extension the cadence protocol $\mathcal{C}_K$ acts on the R→M and M→C edges.

When Assumption 1 may fail. Assumption 1 holds tightly here because q and $\mathcal{M}$ are explicitly logged at every stage event. Two practically relevant violations remain: adaptive in-episode query rewrites not surfaced to the logger, and hidden between-stage memory updates. In those cases the factorization should be read as a projection onto observable traces rather than as a full structural identification claim. Appendix 14.6 quantifies the estimator bias under a controlled violation of this kind: it is negligible for mild unlogged couplings and one-signed (an upward bias on $P(C^\star|M^\star)$), which bounds the framework's exposure when it is applied to learned or LLM agents.

Mechanism-level reading. The same decomposition admits a stage-structured mechanism interpretation. Figure 2 treats stage outcomes as a directed acyclic graph. Under this interpretation, assist on/off ablations are mechanism-targeted perturbations of materialization and post-retrieval execution rather than undifferentiated sensitivity analyses; in the multi-agent setting the same graph is replicated per agent and connected through the cadence protocol $\mathcal{C}_K$, which acts on $\mathcal{M}_{-i}^{(K)}$ at the R→M edge and on $\mathcal{O}_{-i}$ at the M→C edge — exactly the two edges where Empirical Claim 1 will predict the slope shift.

Relation to prior work. Closest are (i) cognitive-map and semantic-map frameworks (Tolman, 1948; O'Keefe and Nadel, 1978; Kuipers, 2000; Gupta et al., 2017; Savinov et al., 2018), (ii) goal-conditioned and successor-representation RL (Andrychowicz et al., 2017; Ghosh et al., 2021; Dayan, 1993; Stachenfeld et al., 2017; Momennejad et al., 2017), (iii) multi-agent communication (Lowe et al., 2017; Foerster et al., 2018; Sukhbaatar et al., 2016; Foerster et al., 2016) and gossip/consensus (Boyd et al., 2006; Xiao et al., 2007), and (iv) LLM-agent memory layers evaluated end-to-end (Wang et al., 2023; Shinn et al., 2023; Packer et al., 2023). Our framework is orthogonal to all four: it *diagnoses* where in the Q/R/M/C chain a given architecture succeeds or fails and lifts the cadence axis from network protocols to a cognitive-architecture parameter (full discussion in Appendix 14.11).

2.3 Event semantics, bounded rationality, and an order-theoretic reading

The four starred events anchor at concrete logged triggers. We state the triggers once, because every empirical claim in the paper is a claim about them (per-regime formal definitions in Appendix 11), and we give each stage its reading in terms of the resource it bounds.

$Q^\star$ **(intent issued).** The planner emits a semantic intent $q = (q^{\mathrm{req}}, q^{\mathrm{pref}}, q^{\mathrm{pen}})$: small sets of required, preferred, and penalty tags. The episode-level event fires on emission, so it is saturated whenever the task is semantic ($\hat{P}(Q^\star)$=1.00 in Table 2); the multi-agent per-tick logger additionally requires a non-empty candidate set (Appendix 11.1), a bookkeeping difference we record rather than hide. Resource bound: attention — the intent is a few tags, not the full state.

$R^\star$ **(memory answers).** Retrieval evaluates q against the memory and returns a candidate set; the event fires when the set contains a candidate satisfying the task predicate (within ϵ of a true target in the multi-agent logger). The 91% empty-candidate-set failures of §3.5 are R-side events: no stored place clears the required-tag threshold. Resource bound: memory — retrieval can only surface what consolidation has stored.

$M^\star$ **(target lock).** The planner materializes one candidate into a concrete actionable cell. Candidacy itself is threshold-gated (a place enters the candidate set only if its score clears the planner's gate, Appendix 12), and the lock commits to a single candidate once per query rather than re-optimizing

Table 1: System classes on which the Q/R/M/C contract is evaluated, and where each is reported. The single symbolic agent is the first row; the communicating collective, the learned-communication policies, and the local LLM agents are rows two to four (all instrumented within §4); the three third-party agent stacks are the last row. "Stage observability" records whether the four conditional rates separate: full where an explicit query/lock interface exists, collapsed where a learned channel removes it, and marginal (non-nested event frequencies $f(\cdot)$) where only a tool trace is available.

System class	Representative instances	Stage observability	Section
Single symbolic agent	node / concept / state-conditioned stack, four task families	full Q/R/M/C, oracle-testable	§3
Multi-agent symbolic memory	independent / shared bus / central aggregator / peer(K)	full Q/R/M/C per agent	§4
Learned communication	CommNet (REINFORCE/PPO), MAPPO	Q saturates, R/M collapse	§4.5
Local LLM agents	qwen3:4b, llama3.1, Qwen2.5-3B/7B (TextNav)	full; budget→C, capability→R	§4.5
Third-party agent stacks	OpenAI SDK, LangGraph, AutoGen	marginal frequencies $f(\cdot)$, not nested	§5

against the full memory at every tick. The event fires on lock acquisition (within ϵ of a true target in the multi-agent logger). Resource bound: deliberation.

$C^\star$ **(closed-loop completion).** The local controller executes until the locked cell is reached or the episode times out; the event fires on arrival. C is the only stage whose failure modes are extra-semantic: controller competence and, in the multi-agent regime, occupancy $\mathcal{O}_{-i}$. Resource bound: control, and in a collective, shared targets.

Observable-interaction reading. The four events admit a uniform reading as semantically distinct acts in the externally observable interaction, with no reference to internal computational state: $Q^\star$ is an informational commitment (a query stating what evidence is needed); $R^\star$ is the memory's answer to that commitment; $M^\star$ is a public commitment to act on specific evidence, existing exactly from the moment the lock is written — and, in a collective, visible to peers and occupying a target; $C^\star$ is the environment's confirmation that the commitment was realized. This grounds Materialization independently of any cognitive interpretation (the appearance of a public commitment is an objective log event), and such commitments are the source of the occupancy contention of §4.5. The stance mirrors runtime verification and distributed-system observability, where specifications constrain externally visible events rather than internal state (Leucker and Schallhart, 2009).

Interpretive readings (all deferred). The stages admit three readings, none of which is a contribution and all of which now live in the appendices: the pipeline as a *satisficing, boundedly rational* agent whose four resource bounds (attention, memory, deliberation, control) are exactly the four failure loci (Appendix 10); an *order-theoretic* account of empty retrieval — the 91% regime — via the query–extent Galois connection (Appendix 9.4); and an *options/semi-MDP* account of why optima appear on time-bearing metrics rather than on the duration-blind chain probability Φ (Appendix 10).

3 Single agent: stage diagnostics localize failure

This section instantiates the first row of Table 1: a single symbolic agent, the simplest system class, where the question is whether stage attribution adds information that end-to-end success does not.

3.1 System, tasks, and protocol

Memory stack. The system uses a graph substrate with a semantic retrieval interface on top: observations o_t are tokenized into symbolic scene tokens and semantic tags that update a memory of place records, transitions, episodic traces, and concept/relation records. Planner queries return candidates by a weighted log-score over required, preferred, and penalty tags. The retrieval machinery is reused across all tasks and architectures; full memory schema, update rules, and scoring formula are in Appendix 12.

Table 2: Stage-factor consistency audit for the best semantic method in each task family. The product of nested conditional estimators matches semantic-path completion; overall success can be higher because some episodes succeed without traversing the full semantic retrieval path.

Task	Method	Overall	SP-comp	$\hat{P}(Q^\star)$	$\hat{P}(R^\star \mid Q^\star)$	$\hat{P}(M^\star \mid R^\star)$	$\hat{P}(C^\star \mid M^\star)$	Product
Water	`water-cp`	0.95	0.70	1.00	0.98	0.74	0.97	0.70
Rest	`rest-s2`	0.93	0.73	1.00	1.00	0.80	0.91	0.73
Goal region	`goal-n1`	0.54	0.00	1.00	0.54	0.01	0.00	0.00
Hazard recovery	`hazard-s7`	0.40	0.16	1.00	0.56	0.58	0.50	0.16

Tasks, baselines, protocol. Four single-agent task families are evaluated (water, safe-rest, goal-region on Empty-6x6 / FourRooms (Sutton et al., 1999), hazard recovery on LavaGapS7), 10 seeds $\times$ 8 episodes each, $\times$ 2 assist modes. Semantic stack (node, concept, state-conditioned variants) is compared against four baselines: random, graph-only, SPTM-like, learned BC. Reported: end-to-end success, steps, four operational stage rates, assist deltas, weakest stage per task and per method. All means are seed-aggregated; 95% bootstrap CIs use $B{=}5{,}000$ resamples (Efron, 1979); assist comparisons use paired Wilcoxon signed-rank tests (Wilcoxon, 1945) with Holm-Bonferroni correction (Holm, 1979) (Appendix 15). For the two hardest tasks (FourRooms, LavaGapS7), four oracle regimes (*base*, *oracle R/M/C*) are run — each replaces exactly one stage while leaving the rest of the pipeline unchanged. BabyAI portability check in Appendix 13. All single-agent experiments ran on a 10-core ARM CPU, 32 GB RAM (CPU-only); oracle sweep $<$ 1 min wall-clock.

LLM diagnostic block. To check that Q/R/M/C is not tied to the symbolic planner implementation, we also run a fixed TextNav substrate with an LLM-driven observe/query/retrieve/decide loop. The TextNav task, prompts, parser contract, and event definitions are held fixed while only the local Ollama model changes; the backend uses model-compatible completion settings when an Ollama chat template otherwise hides short structured answers in a separate thinking field. This block is a controlled step-budget study (Table 4): sweeping the completion budget drives both LLM agents into a failure that Q/R/M/C localizes to the C stage while $Q/R/M$ stay saturated, exhibiting the framework's diagnostic property on real language agents. Reproducing full ReAct/Reflexion/Voyager/MemGPT-style agents is a separate systems contribution; the comparison to those systems is a coverage comparison in Appendix 14.12.

3.2 Stage-estimator validation

Before the main benchmark, the factorization is validated. On a synthetic four-stage process with known probabilities the nested-estimator product tracks empirical semantic-path completion across sample sizes. On the real benchmark, the same product matches semantic-path completion rather than overall success (Table 2) — exactly the contract of Definition 1.

3.3 Success-only summary and what it hides

Water and rest are strong end-to-end tasks (0.95 and 0.93 success; Figure 1b). Goal-region reaches 0.54 success at the task-family level — an average that pools a solved environment (Empty-6x6, 1.00) with a hard one (FourRooms, 0.075; §3.4), so the family number is bimodal rather than a uniform mid-difficulty — but the bottleneck sits downstream of candidate selection (§3.4). Hazard recovery is the hardest retrieval-sensitive task at 0.40 success, with retrieval satisfaction as the weakest stage. Under success-only evaluation the two hard families look like the same kind of weakness; the stage view will show they are opposites. Per-task operational rates with 95% CIs are in Table 8 (Appendix 12.3).

Baseline comparison. Table 3 compares the semantic stack against random, graph-only, SPTM-like, and learned behavior-cloned baselines. The learned baseline dominates the easiest resource tasks (1.00 on water and rest) and hazard recovery (0.74), while the semantic stack remains strongest on the relation-heavy goal-region family (0.54). The relevant contrast for this paper, however, is not the success column: the semantic stack is the only method family whose decisions expose interpretable Q/R/M/C structure. On the BC baseline the stage events are degenerate for the same structural reason

Table 3: Baseline comparison. The learned behavior-cloned baseline dominates the easiest resource tasks and hazard recovery, while the semantic stack remains strongest on the relation-heavy goal-region family and remains the only family with interpretable Q/R/M/C structure.

Task	Random	Graph-only	SPTM-like	Learned	Best semantic
Water	0.85	0.90	0.90	1.00	0.95
Rest	0.85	0.90	0.90	1.00	0.93
Goal region	0.41	0.49	0.52	0.50	0.54
Hazard recovery	0.08	0.39	0.00	0.74	0.40

as on the multi-agent learned baseline analysed in §4.5 — a policy without an explicit query/lock interface cannot emit separable stage events, so the diagnostic decomposition has nothing to attach to. End-to-end learning and stage diagnostics answer different questions; this paper is about the second.

3.4 Oracle-stage interventions localize the bottleneck

On goal-region the semantic variants solve Empty-6x6 (1.00 success) but collapse on FourRooms (0.075). For hazard recovery (LavaGapS7), base success 0.39 (bootstrap 95% CI $[0.35, 0.43]$) rises only to 0.43 under oracle controller, but jumps to 0.59 under oracle materialization and 0.60 under oracle retrieval (CI $[0.51, 0.69]$; per-seed paired deltas positive in $9/10$ seeds, exact sign test p=0.004); goal-region rejoin is downstream-limited (oracle R lifts semantic-path completion to 0.08 without moving end-to-end success). The per-task attribution as a stacked waterfall is in Figure 6 (Appendix 12.3). Two tasks identical under success-only evaluation fail at opposite ends of the Q/R/M/C chain.

3.5 The logs refine the diagnosis: a memory-accumulation limit

Inspecting the framework's own logs sharpens the retrieval attribution. On the 10-seed hazard-recovery run, 349 retrieval calls were logged: only 32 (9.2%) returned at least one candidate, and of these 31 (96.9% precision) selected a semantically satisfied candidate. The bottleneck is therefore not candidate *ranking* but candidate *generation* — the symbolic memory does not contain hazard-recovery-tagged nodes at 91% of decision points. A retrained 2-layer candidate-generation MLP trained on per-state features (test accuracy 0.59 vs majority-class 0.81; Appendix 16) confirms that instantaneous state features cannot substitute for the missing structure. The framework's "retrieval-limited" attribution thus refines to a *memory-accumulation* limit: candidate generation cannot be solved from state features alone, and the binding constraint is how trajectory evidence is consolidated into persistent symbolic structure. This attribution is not an artefact of the required-tag threshold θ: sweeping θ across the full range leaves the empty-candidate-set rate invariant (Appendix 14.5), because the empties are absence of any stored candidate, not sub-threshold filtering. Nor is it an artefact of the weighting: the empty rate is 0.908 pooled over calls, 0.909 seed-weighted, and 0.891 episode-weighted (mean over the 80 episodes of each episode's own empty share), so long failing episodes contributing more decision points do not inflate the figure. Accumulation itself is not one of the four stages: the protocol localized the failure to the R link, and this within-link forensics named the cause — we read Q/R/M/C as coarse localisation with a designated drill-down, not as a complete causal ontology of memory failure. This sets up the multi-agent section: when consolidation becomes a shared process across N agents, it acquires a measurable *rate*.

4 Multi-agent: memory as a shared resource

This section instantiates rows two to four of Table 1: the same logged events, unchanged, instrument a communicating collective — and, within the same section, learned policies and LLM agents — where the additional failure axis is contention for the targets that memory designates.

4.1 Four memory architectures and the cadence axis

Four communication architectures are instantiated; they share the single-agent retrieval substrate but differ in how memory is distributed across agents (Figure 3).

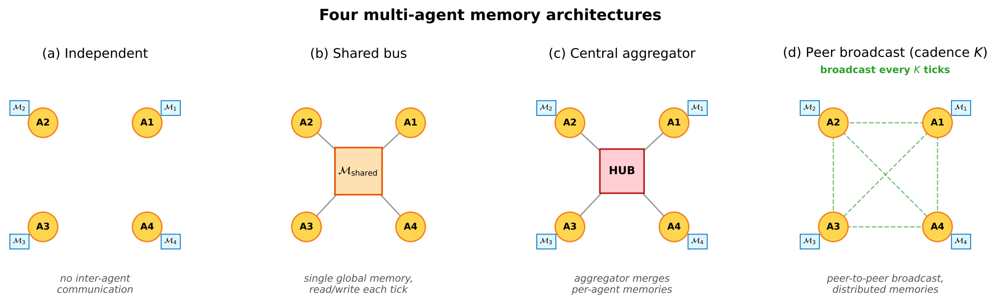

Figure 3: Four multi-agent memory architectures. **(a)** Independent: per-agent private memory, no exchange. **(b)** Shared bus: one global memory, every agent reads/writes each tick. **(c)** Central aggregator: per-agent memories plus a hub that merges them into a consensus report. **(d)** Peer-to-peer broadcast: per-agent memories with pairwise snapshot exchange every K ticks (the cadence axis of Empirical Claim 1).

Independent. Each agent maintains a private event store and concept graph. No inter-agent communication mechanism exists. Lower-bound baseline.

Shared bus. All agents publish observations to a single *collective memory* event bus. One shared concept graph is rebuilt from the union of events. Maximally centralized.

Central aggregator (ConsensusLayer). Each agent has its own private concept graph; a central layer pulls snapshots and computes a trust-weighted merge, producing a single *consensus report* consumed by all agents.

Peer-to-peer broadcast (cadence K). Each agent has its own concept graph and its own merged view. Every K ticks each agent broadcasts a snapshot to a passive bus; each agent independently merges its own snapshot with the latest snapshot received from each peer. Different agents may hold different merged views; there is no central report.

The first three correspond to the three regimes in the multi-agent communication literature: no exchange, CTDE-style aggregation, and decentralized peer communication. The fourth is the architecture for which Definition 2 and Empirical Claim 1 are directly applicable, since the cadence K is a tunable parameter rather than a fixed protocol. Mechanically, K is the rate at which private memories consolidate into a collective view, which is why it is the structural axis of this section.

4.2 Multi-agent factorization

Extend the factorization to a population of N agents that share information through a communication protocol $\mathcal{C}_K$ parameterized by broadcast cadence K. Each agent i has its own memory state $\mathcal{M}_i$ and issues its own query q_i. Let $\mathcal{M}_i^{(K)}$ denote the projection of agent i's memory at the moment it queries, after absorbing every peer message received under cadence K. Two coupling channels are separated: (i) memory coupling $\mathcal{M}_{-i}^{(K)}$, the latest peer memory snapshots accessible to agent i; (ii) occupancy coupling $\mathcal{O}_{-i}$, the set of targets that other agents have already claimed by the time i acts.

Definition 2 (multi-agent stage estimability). For each agent i,

$$P(C_i^\star \mid q_i, \mathcal{M}_i, \mathcal{C}_K) = P(Q_i^\star \mid q_i, \mathcal{M}_i)\; P(R_i^\star \mid Q_i^\star, q_i, \mathcal{M}_i, \mathcal{M}_{-i}^{(K)})$$

$$\cdot\; P(M_i^\star \mid R_i^\star, q_i, \mathcal{M}_i, \mathcal{M}_{-i}^{(K)})\; P(C_i^\star \mid M_i^\star, q_i, \mathcal{O}_{-i}).$$

Under a multi-agent stage-Markov assumption (Appendix 11), the four factors are estimable from per-agent trajectory logs and the empirical-frequency estimators are consistent.

When Definition 2 may fail. The single-agent stage-Markov property absorbs trajectory history into $(q, \mathcal{M})$. In the multi-agent setting, the analogous absorption requires that dependence on past peer decisions enters only through $\mathcal{M}_{-i}^{(K)}$ at the M stage and through $\mathcal{O}_{-i}$ at the C stage. This step is non-trivial: $\mathcal{O}_{-i}$ summarises target occupancy at decision time but loses information about *how* peers arrived there. The factorization is therefore a projection onto observable per-agent traces under that summary. Two specific failure modes arise: (i) inter-agent communication side-channels not logged through $\mathcal{C}_K$; (ii) coordinated peer policies whose decision at the C stage depends on hidden peer state not summarised by $\mathcal{O}_{-i}$.

4.3 Bottleneck shift

Empirical Claim 1 (bottleneck shift M↔C, slope-level). Assume scarcity ($N > T$ targets, see §2.1) and a peer-broadcast architecture with cadence $K \in \mathbb{N}$. Two stylized assumptions are posited: **(F)** memory freshness $\nu(\mathcal{M}_{-i}^{(K)})$ is non-decreasing pointwise as K decreases, and the probability that agent i's planner locks onto a reachable target is non-decreasing in ν; **(O)** the probability that an arbitrary committed target is already in $\mathcal{O}_{-i}$ is non-increasing as K increases. Write $P(M^\star|R^\star;K)$ and $P(C^\star|M^\star;K)$ for the population-level conditional probabilities (averaged over agents and the seed distribution of the benchmark); $\hat{P}(\cdot)$ denotes the corresponding empirical-frequency estimator computed from logs. Under (F) and (O),

$$\frac{\partial\, P(M^\star|R^\star;K)}{\partial K^{-1}} > 0, \qquad \frac{\partial\, P(C^\star|M^\star;K)}{\partial K^{-1}} < 0.$$

Both slope conditions are direct empirical predictions that are testable on logs through $\hat{P}$.

What this claim is and is not. Empirical Claim 1 is a *slope-level statement on conditional rates*: each of the two slopes is testable and is confirmed on the 35,640-run sweep (§4.5) and independently on a 100-run standard-MiniGrid portability sweep (§6). We report the slopes as *effect sizes with cluster-robust confidence intervals* (resampled over the 81 independent configuration cells rather than the correlated run count) in Appendix 14.3. The claim's boundary with respect to the conditional product Φ is stated once, in the Scope-and-non-claims box (§1) and in §4.5. The shift is a signature of scarce-target contention, not of the grid design: under abundance ($T>N$) both slopes flatten and the M↔C anticorrelation attenuates (Appendix 14.9), i.e. the prediction breaks exactly where the contention mechanism is removed.

Chronology. Empirical Claim 1 is post-hoc. It was formulated *after* an initial 12,960-run sweep conducted under four stage-agnostic hypotheses (Appendix 14); the two slope conditions were then confirmed on the full 35,640-run sweep, on the $N=12$ scale-up (Appendix 13), and on the MiniGrid portability wrapper (§6). We state this explicitly: the verified slope signs are predictions on enlarged data, not a pre-registered (externally timestamped) prediction. A fully formal version is left for future work.

4.4 Sweep protocol

The four architectures are evaluated on a custom multi-agent grid (12×10 cells); each agent must reach a cell tagged `water_source`, choosing actions via a shared memory-driven planner so only the memory layer differs. The sweep crosses $N\in\{3,5,8\}$, $T\leq N$, three layouts (symmetric/asymmetric/random), three hazard densities, four architectures, eight peer cadences $K\in\{1,2,4,8,16,32,48,64\}$, 40 seeds — 35,640 runs in ≈ 22 min on 8 cores. Per-agent per-tick Q/R/M/C indicators are logged (Appendix 11); episode-level starred events are the disjunction over ticks. A separate oracle-intervention block on scarcity cases $(N,T)\in\{(3,2),(5,3),(8,5)\}$ at five cadences evaluates oracle R (memory returns ground-truth water positions), oracle M (planner locks onto nearest real water), and oracle C (completion of a real materialized lock is guaranteed; Appendix 14.14).

4.5 Results: stage decomposition and bottleneck shift

The 35,640-run sweep is reported through four observations, structured to mirror Empirical Claim 1: (i) the conditional decomposition isolates the M and C stages as the differentiating links; (ii) the bottleneck shift appears directly as a negative within-cadence correlation between $P(M^\star|R^\star)$ and $P(C^\star|M^\star)$, peaking at $K=4$ and decaying to noise at $K=64$; (iii) on mean time-to-success, the resulting trade-off produces an interior optimum at $K=8$ — robust in location but shallow in margin, with the advantage over the adjacent fast cadences not resolved at this sample size (Appendix 14.3);

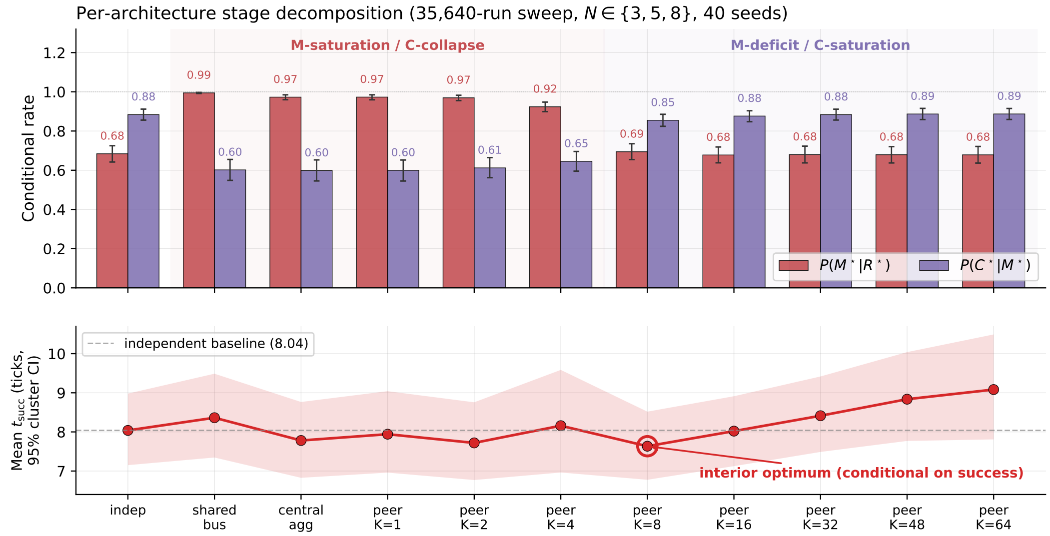

Figure 4: Per-architecture conditional rates (top) and mean time-to-success among successes (bottom) from the 35,640-run sweep. Error bars and band: 95% cluster-bootstrap CIs over the 81 design cells; points are cell-weighted means (7.63 at K=8 vs. 8.04 independent; the run-pooled values of Table 11 are 7.47/8.02). Shared bus, central aggregator, and fast peer broadcast ($K \leq 4$) live in the M-saturation / C-collapse regime; independent and slow peer ($K \geq 16$) live in the M-deficit / C-saturation regime. The interior minimum at K=8 is robust in location but shallow in margin (Appendix 14.3); the censoring-aware contrast is in the text.

(iv) on the conditional product Φ, the same trade-off produces a monotone trend rather than an interior peak — a summary statistic on which we make no formal claim. A learned-communication baseline and a multi-agent oracle-intervention loop close the subsection.

4.5.1 Conditional decomposition reveals where each architecture fails.

Figure 4 shows per-architecture conditional Q/R/M/C rates; full numerics with bootstrap CIs are in Appendix 14.1, Table 11. Retrieval conditional $P(R^\star|Q^\star)$ is saturated; the differentiating signal lives at M and C. Shared-bus, central-aggregator, and fast-broadcast peer ($K \leq 4$) have high $P(M^\star|R^\star)$ but low $P(C^\star|M^\star)$; independent and slow-broadcast peer ($K \geq 16$) show the reverse — the M↔C trade-off predicted by Empirical Claim 1.

4.5.2 Bottleneck shift and interior efficiency optimum.

Across all peer runs, Spearman of $P(M^\star|R^\star)$ vs K is -0.53 (95% cluster-robust CI $[-0.59, -0.47]$) and of $P(C^\star|M^\star)$ vs K is $+0.43$ (CI $[+0.37, +0.50]$); both are moderate effect sizes whose CIs exclude zero under resampling of the 81 configuration cells (Appendix 14.3), matching the slope conditions of Empirical Claim 1. We report these rather than p-values, which are uninformative at this correlated sample size. Both slope signs are also robust to the two free thresholds of the logger — candidacy θ and lock tolerance ϵ (Appendix 14.5). The stronger empirical signature is at the per-configuration level: Pearson of $P(M^\star|R^\star)$ vs $P(C^\star|M^\star)$ *within* a fixed K is $-0.43, -0.54, -0.61, -0.19, -0.05, -0.04, -0.02, -0.02$ at K=$1, 2, 4, 8, 16, 32, 48, 64$, peaking at K=4 and decaying to noise by K=64 (Figure 9, Appendix 14.1). Configurations where M is high systematically have low C: the bottleneck migrates between the two stages within the same $(N, T, \text{layout}, \text{hazard}, \text{seed})$ cell as K varies. On the efficiency metric, mean t_{succ} has a U-shape with interior minimum at K=8 (7.47 ticks vs 8.02 for the independent baseline); the minimum's location is robust but shallow — its margin over the adjacent fast cadences (K=$1, 4$) is not resolved under cell-level resampling (paired-bootstrap CIs in Appendices 14.1 and 14.3). Both statements condition on success. Under a censoring-aware expected time that charges failed runs the full step limit, the fast flank reverses and the independent baseline is marginally best (cell-paired $\Delta(K$=8 − independent$) = +0.32$, CI $[+0.05, +0.63]$; Appendix 14.3): at fast cadence the completion decline reduces the success rate by more than the materialization gain increases it, so the interior optimum is a statement about conditional speed, not unconditional efficiency — and closing the unconditional gap is what the reservation protocol of open problem (3) achieves (Appendix 14.10). Scaling to $N{=}12$ *sharpens* the signature: the peak Pearson rises to -0.70 and migrates outward to $K=8$, consistent with stronger occupancy coupling at higher N delaying the cost-regime transition (Appendix 13, Figure 7).

Provenance stratification: the mechanism behind the shift. Annotating each $M^\star$ lock with the *provenance* of the evidence behind it makes the bottleneck shift mechanistic rather than correlational. Let φ be the foreign share of the evidence mass supporting the locked candidate, computed from the same trust/support weights the merge itself used and frozen at lock time, and call a lock a *warp* ($W^\star$) when $\varphi \geq 0.8$: the agent is committing to a target it knows essentially only from peers. On a dedicated peer sweep over the scarcity cells (random layouts, 20 seeds, $K \in \{1, 2, 4, 8, 16\}$) the warp share $P(W^\star|M^\star)$ falls from 0.55 at $K{=}1$ to 0.11 at $K{=}8$ while the own-evidence stratum completes at a flat 0.26–0.34 and the warp stratum completes below 0.05 throughout: *the C-collapse of fast sharing is concentrated in the warp stratum*. What cadence controls, mechanically, is how much of the collective's materialization runs on foreign evidence — and that is the stratum whose completions are lost to occupancy contention. The stratification is not an artefact of the 0.8 cutoff: recomputed from the raw per-lock φ values (23,237 locks) at thresholds $\varphi{\in}\{0.5, 0.6, 0.7, 0.8, 0.9, 0.95\}$, the foreign stratum completes at $\leq$0.03 while the own stratum stays at 0.26–0.50, at every threshold and cadence. The two coupling channels also separate interventionally with arms already in the design: the independent arm removes peer evidence while leaving target contention in place, and the reservation protocol of open problem (3) (Appendix 14.10) keeps peer evidence while managing contention — under it, unconditional success at fast cadence exceeds the $K{=}8$ baseline (0.614–0.620 vs. 0.583; paired improvement over the same cells without the mechanism $\Delta = +0.037$, cluster-bootstrap 95% CI $[+0.018, +0.058]$ over 18 matched design cells), identifying occupancy contention rather than evidence quality as the cost channel of fast sharing. The same stratification reproduces on the continuous VMAS bridge ($0.72 \to 0.00$ over $K{=}1/4/16$) and on an LLM text substrate ($0.42 \to 0.08$ over $K{=}2/8$), with the curves nearly coinciding across substrates. A full provenance-conditioned analysis (counterfactual attribution by masked replay, a closed-form staleness horizon with a pre-specified $6/6$-exact hold-out, and a decentralized repair protocol) appears in a companion manuscript; here the stratification serves as the mechanistic reading of Empirical Claim 1.

What we do not claim about Φ. The conditional product $\Phi = P(M^\star|R^\star)\cdot P(C^\star|M^\star)$ does *not* have an interior peak in the tested range (mean-of-products estimator increases monotonically from 0.569 at $K{=}4$ to 0.603 at $K{=}64$, converging to the independent-agent boundary 0.605). The slope-level statement of Empirical Claim 1 is on the conditional rates themselves, not on Φ; we make no formal claim about the shape of $\Phi(K)$. Jensen-gap detail and POM–MOP comparison in Appendix 14.1.

Learned-communication baseline: stage events are structurally degenerate even at competitive success. A CommNet-style policy (Sukhbaatar et al., 2016) trained with PPO (Schulman et al., 2017) on the same scenario ($N{=}3$, $T{=}2$) reaches 0.667 held-out success ($n{=}100$ seeds) — matching the symbolic peer architecture at $K{=}8$ (0.65). Despite this, the Q/R/M/C profile remains structurally degenerate: $Q^\star{=}1.00$ saturated by construction; $R^\star{=}0.997$, $M^\star{=}0.997$ collapsed together ($P(M^\star|R^\star){=}1.00$). This is the verified structural claim: a real-valued comm vector is never informatively empty (Q saturates) and lacks an explicit target-lock interface (R/M merge), regardless of training budget. The multi-agent counterpart of the single-agent learned-BC analysis (§3.3). Full architecture, PPO hyperparameters, and side-by-side REINFORCE vs PPO comparison in Appendix 16.

LLM model sweep: the logger localizes a real failure in language agents. Table 4 runs the same Q/R/M/C event contract on four LLM agents over two independent backends: qwen3:4b and llama3.1 via local Ollama (8 episodes/cell), and Qwen2.5-3B/7B-Instruct via HuggingFace transformers on SLURM GPU nodes (25 episodes/cell). The decomposition separates *two different failure modes on two different stages*. **Budget $\to$ C:** below the task's step requirement, three models (qwen3:4b, llama3.1, Qwen2.5-7B) show the identical signature $Q^\star{=}R^\star{=}M^\star{=}1.00$, $C^\star{=}0.00$ — a completion-limited failure that replicates across backends, model families, and hardware. **Capability $\to$ R:** Qwen2.5-3B fails at *every* budget with $Q^\star{=}1.00$ but $R^\star{=}0.00$: it declares the target, emits well-formed actions (verified against the response cache, so not a parsing artifact), yet wanders without ever storing the evidence retrieval needs — an exploration/grounding failure surfacing at the R stage, exactly the "absence of evidence" regime of §3.5. An opaque scalar would report both models as 0.00; the stage view shows they fail for different reasons at different stages. (Instrumentation note: a naive qwen3:4b/Ollama chat-template call initially returned empty `response` strings because the model spent the short structured-output budget in a separate thinking field; we use raw structured completion for qwen3:4b and a first-line parser, recorded for reproducibility.)

The multi-agent phenomenology reproduces on a live LLM collective. Beyond the single-agent budget study, the same event contract instruments a collective of $N{=}3$ LLM agents (llama3.1) sharing symbolic place memory by peer broadcast under scarcity ($M{=}2$ takeable targets, four layouts, $K \in \{2, 8\}$), with all predictions specified before the runs. The cadence phenomenology of §4.5 carries over to the language substrate: a paired comparison against the no-communication baseline on identical layouts shows raw sharing *lowering* success from $2/3$ to $1/3$ per episode — the occupancy-racing C-collapse, now produced by agents whose queries, tag extraction, and commitments are LLM-made — and target collisions appear in $7/8$ shared-memory episodes. Attaching the reservation-and-rollback repair protocol (the constructive answer to open problem 3 of Appendix 14.10) recovers part of the loss ($0.333 \to 0.417$ at both cadences, rollback latency again $\approx K$). Two interface findings transfer to any LLM-over-symbolic-memory design: LLM query formers free-generate retrieval tags outside the memory's vocabulary (requiring tolerant retrieval gates), and small local models are reliable at semantic commitment but not at coordinate arithmetic or systematic exploration (requiring layered execution). Full protocol and provenance analysis in the companion manuscript.

Scope: these are controlled studies on local models; scaling the same contract to full Voyager/MemGPT-style agents on richer tasks with multiple failure modes is developed next (external-framework validation) and in Appendix 14.12.

Table 4: LLM TextNav step-budget sweep across four models and two independent backends (local Ollama, $n{=}8$/cell; HuggingFace transformers on a SLURM GPU cluster, $n{=}25$/cell; temperature 0.0; the task needs $\sim$10 low-level steps). Two distinct failure modes are localized to two different stages. **Budget $\to$ C:** at step limit 6, qwen3:4b, llama3.1 and Qwen2.5-7B all show $Q^\star{=}R^\star{=}M^\star{=}1.00$ with $C^\star{=}0.00$ — the completion-limited failure replicates across backends, models, and hardware. **Capability $\to$ R:** Qwen2.5-3B fails at *every* budget ($\{6, 8, 12, 25\}$) with $Q^\star{=}1.00$ but $R^\star{=}0.00$: it declares the target yet never accumulates the evidence to retrieve it (its outputs are well-formed action strings — verified in the response cache — so this is an exploration/grounding failure feeding retrieval, not a parsing artifact). Artifacts: `tmp/llm_steplimit_*`, `tmp/cluster/textnav*`.

Model	Backend	Step limit	Success	$Q^\star$	$R^\star$	$M^\star$	$C^\star$
qwen3:4b	ollama	6	0.00	1.00	1.00	1.00	**0.00**
llama3.1	ollama	6	0.00	1.00	1.00	1.00	**0.00**
Qwen2.5-7B	hf	6	0.00	1.00	1.00	1.00	**0.00**
qwen3:4b	ollama	12	1.00	1.00	1.00	1.00	1.00
llama3.1	ollama	12	1.00	1.00	1.00	1.00	1.00
Qwen2.5-7B	hf	12	1.00	1.00	1.00	1.00	1.00
Qwen2.5-3B	hf	6–25	0.00	1.00	**0.00**	0.00	0.00

Multi-agent oracle interventions close the diagnose$\to$intervene$\to$verify loop. On the scarcity cases, mean t_{succ} at small/intermediate cadence ($K{\in}\{1, 2, 4\}$) drops from ~ 10–11 in baseline to ~ 7.2 under either oracle R or oracle M, and further to ~ 6.1 under oracle C, which additionally removes post-lock travel; at slow cadence ($K{=}16$) the baseline is already within ~ 1.2 of the R/M oracle floor (Table 19 in Appendix 14.13). Crucially, oracle C gives the largest time reduction but the smallest success-rate gain (Appendix 14.14): in scarcity, completion costs time while retrieval and materialization decide success. The cadence-induced excess time at small/intermediate K is attributable to the M$\leftrightarrow$C transition; at large K the bottleneck has moved into solo exploration.

The multi-agent block thus delivers its half of the argument. The cadence K is, mechanically, the rate at which private memories consolidate into a collective view; the data show that this rate has a non-trivial interior optimum ($K{=}8$ on t_{succ}), that its cost channel is occupancy coupling, and that both effects are visible stage-by-stage on the same Q/R/M/C chain used for single agents. Consolidation is not only the single-agent limit; in a collective it is a tunable, measurable design axis.

5 The contract on third-party agent stacks

This section instantiates the last row of Table 1. We further evaluate the protocol on systems built by others. The contract has two layers: an *invariant core* — four event roles (a memory query is issued; returned evidence satisfies the task; the agent commits to a target; the task completes) — and a *system-specific adapter* that maps logged actions to those roles. We instrument three third-party agent stacks — a function-calling loop on the OpenAI SDK, LangGraph's prebuilt ReAct agent, and AutoGen's AssistantAgent/UserProxy pair, all driving the same local model (llama3.1, temperature 0) — with one tool-trace adapter shared by all three: an episodic household memory is exposed through `recall`/`goto`/`take` tools, and $Q^\star$ fires on querying memory, $R^\star$ on *attribution* (a returned record names the item at its true place), $M^\star$ on a commitment (a `goto` to the true place; first-commitment accuracy M_1 is logged as an auxiliary lock-quality diagnostic), and $C^\star$ on task completion within a tool budget. A tool trace exposes no lock chain, so these are *marginal event frequencies* $f(\cdot)$ in the sense of Appendix 11: an agent can commit to the true place without a supporting retrieval, the events are not nested, and no factorisation is claimed — the four frequencies serve as separate diagnostics and are never substituted into the chain-rule product. What varies across system classes is the adapter, not the event roles. Three variants engineer one failure each — a consolidation gap (the location fact was never written to memory), a stale-distractor ambiguity, a tool budget below the task's requirement — plus a clean control; all predictions were specified before the runs, and the two revisions of the adapter (an attribution leak; an over-tightened $M^\star$) are recorded in the registration file.

Table 5: Marginal Q/R/M/C event frequencies on three third-party agent frameworks (llama3.1, 20 episodes per cell, identical episodes/tools/budgets). Each entry is the fraction of episodes in which the event occurred: $Q^\star$ — the agent queried memory at least once; $R^\star$ — a returned record named the item at its true place; $M^\star$ — the agent committed (`goto`) to the true place, with or without retrieval support; $C^\star$ — `take` succeeded within the tool budget; M_1: first `goto` targets the true place; stale$_1$: first `goto` targets the stale distractor. The events are not nested and no product is claimed (Appendix 11): under the consolidation gap, budgeted blind search over the four places sometimes commits to and reaches the true place, hence $M^\star, C^\star > 0$ at $R^\star{=}0$. Structural failures localize identically everywhere: the consolidation gap is $R^\star{=}0.00$ and the budget failure is $C^\star{=}0.00$ at $Q^\star{=}R^\star{=}1$, for all three stacks. On the *clean* control the frameworks spread from 0.50 to 0.75 end-to-end — and the decomposition attributes the spread (see text).

Framework	Variant	$f(Q^\star)$	$f(R^\star)$	$f(M^\star)$	$f(C^\star)$	$f(M_1)$	$f(\text{stale}_1)$
OpenAI SDK	control	1.00	1.00	0.85	0.75	0.25	—
	consolidation gap	1.00	**0.00**	0.50	0.40	0.25	—
	stale ambiguity	1.00	1.00	0.90	0.30	0.25	0.20
	tight budget	1.00	1.00	0.25	**0.00**	0.25	—
LangGraph	control	1.00	1.00	0.70	0.50	0.50	—
	consolidation gap	1.00	**0.00**	0.20	0.15	0.10	—
	stale ambiguity	1.00	1.00	0.65	0.60	0.50	0.35
	tight budget	1.00	1.00	0.10	**0.00**	0.10	—
AutoGen	control	1.00	1.00	0.85	0.55	0.45	—
	consolidation gap	1.00	**0.00**	0.30	0.20	0.15	—
	stale ambiguity	1.00	1.00	0.50	0.45	0.35	0.40
	tight budget	1.00	1.00	0.15	**0.00**	0.15	—

Three findings (Table 5, Figure 5). **(1) Structural localizations are exact and framework-independent.** When the memory lacks the fact, $R^\star{=}0.00$ on every stack; when the budget cannot fit completion, $C^\star{=}0.00$ at $Q^\star{=}R^\star{=}1$ on every stack ($20/20$ episodes each). The contract needs no per-framework adaptation. **(2) The clean control separates the frameworks, and the decomposition says how.** End-to-end success on the SAME clean episodes ranges 0.50 (LangGraph) to 0.75 (OpenAI SDK). The stage profiles attribute the spread to a failure mode common to all three stacks combined with framework-specific recovery: first-commitment accuracy is low everywhere ($M_1{=}0.25$–0.50 — the agent's first `goto` frequently ignores its own just-retrieved evidence), episode-

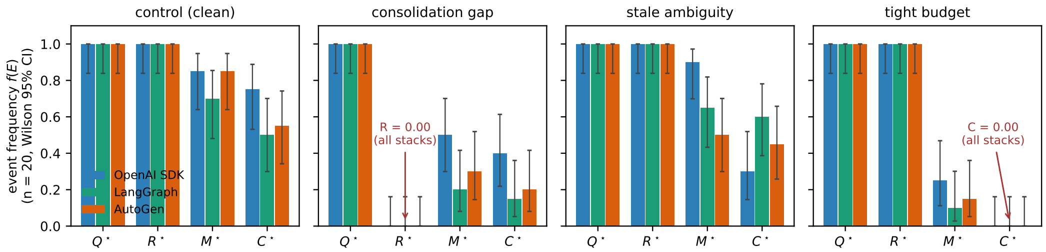

Figure 5: Stage profiles of three third-party agent stacks on the four engineered variants (llama3.1, n=20 per cell). Query and retrieval saturate identically everywhere; the two structural failures produce exact zero columns on every stack ($R^\star$=0 under the consolidation gap, $C^\star$=0 under the tight budget); the behavioral spread between frameworks lives entirely in the M and C stages.

level materialization recovers to 0.70–0.85 through re-query loops, and the frameworks then differ in how much of that recovery survives to completion ($C^\star/M^\star$ from 0.88 for the OpenAI SDK loop down to 0.65 for AutoGen). **(3) The stale distractor pulls, measurably and unevenly.** The outdated record captures the first commitment in 0.20–0.40 of ambiguous episodes (most strongly for AutoGen), and its completion cost is framework-specific: the OpenAI SDK loop pays hardest ($C^\star : 0.75 \to 0.30$) while LangGraph and AutoGen absorb the detour ($0.50 \to 0.60$; $0.55 \to 0.45$) — the LLM-memory analogue of the staleness gating that the collective experiments showed to be decisive. Failed pre-specifications and subsequent revisions: the behavioral predictions that assumed a clean control (all stages ≥ 0.75; a uniform ambiguity-induced completion drop) broke *because the frameworks themselves fail a clean memory task 25–50% of the time and pay for ambiguity in different places* — which is finding (2)–(3), not noise; all revisions are recorded as failed pre-specifications with mechanisms.

Model robustness: the structure transfers, the behavior is legible. A prediction specified before the runs — the structural localizations transfer to a second model on all frameworks, while behavioral rates are model-dependent and only reported — was confirmed exactly by repeating all 240 episodes with qwen3:4b (Table 7): $R^\star$=0.00 under the consolidation gap and $C^\star$=0.00 under the tight budget, $20/20$ on every stack, for both models. The behavioral contrast is a diagnostic result: qwen3:4b completes every clean episode ($C^\star$=1.00 across stacks, vs. 0.50–0.75 for llama3.1) because its *first-commitment accuracy* is high (M_1=0.90–0.95 vs. 0.25–0.50) and its susceptibility to the stale distractor low (0.05–0.30) — except under AutoGen, where qwen's M_1 collapses to 0.40 while remaining ≥ 0.90 on the other stacks. The stage decomposition thus separates three influences a leaderboard conflates: the task's structural limits (model- and framework-independent), the model's commitment quality, and the framework's prompt/loop shaping that can degrade it.

Comparison with diagnostic baselines. On the blinded benchmark below we additionally evaluate four alternative diagnostics of the same logs, each receiving the identical anonymized cell aggregates and calibration control, with decision rules fixed before scoring and thresholds matched to the Q/R/M/C rule (`experiments/qrmc_external/diagnostic_baselines.py`; specification in `tmp/qrmc_external/baselines_spec.json`). Over 63 scored cells (10 faults + held-out control, three stacks, two models): a success-only reading names no stage (0.095); an AgentBoard-style progress scale (first milestone drop) reaches 0.730 (macro-F1 0.698); a RAGChecker-style two-stage retrieval/execution split reaches 0.651 exact / 0.873 side-level (macro-F1 0.450), its deficit concentrated where commitment must be separated from execution; a process-conformance first-deviation rule using the order-sensitive M_1 reaches 0.778 (macro-F1 0.755), its entire deficit being the budget-exhaustion branch of the conditional reading; a multinomial logistic regression on the same aggregates reaches 0.444 on unseen fault types with a false-positive rate of 1.00 on held-out controls (leave-one-model -out 0.794, leave-one-framework-out 0.825) and requires fault labels that the rule-based methods do not. Q/R/M/C attains 0.873 (macro-F1 0.835; bootstrap CI over fault types $[0.71, 0.98]$) and outperforms all exact-stage alternatives under paired McNemar tests, with all comparisons significant after Holm correction (largest adjusted $p = 0.031$; discordant cells from 6:0 to 50:1). Of the 66 cells, three (the prompt-only fault under llama3.1) failed the pre-specified manipulation check and are excluded for every method alike, leaving 63. A follow-up repair study (llama3.1, 2,640 episodes; pre-specified menu of stage-level repairs, each diagnostic selecting the repair for its predicted stage) gives Q/R/M/C the highest completion gain and lowest regret against the best available arm (0.293/0.061 vs. an oracle-best gain of 0.354; two-stage 0.280/0.074; conformance 0.250/0.104; progress 0.219/0.135; random 0.113/0.240). Paired bootstrap over fault–framework cells resolves the gain advantage over the progress baseline ($+0.074$ $[+0.007, +0.157]$) but not over the two-stage baseline ($+0.013$ $[-0.009, +0.039]$), which also selects the single best arm slightly more often (0.704 vs. 0.630) because its coarse diagnoses map to repairs that frequently coincide with the best available arm; the four-stage advantage is established at exact localization, while on repair utility the protocol at least matches the strongest coarse baseline. Two pre-specified expectations failed: the matched repair for silent corruption recovers only 0.22–0.44 of the control gap (consolidation without decoy removal converts a retrieval fault into an ambiguity fault), and the budget repair on the clean OpenAI SDK control lowered completion by 0.15.

Blinded fault-localization meta-evaluation (pre-specified). The engineered variants verify localization of failures whose stage is known by construction; the blinded meta-evaluation treats the protocol as a *classifier*. We injected ten fault types — two to three per stage, structural and behavioral (Table 6) — and fixed a first-failing-link decision rule, with all thresholds, *before* any run (`tmp/qrmc_external/registered_meta.json`; rule and thresholds in `experiments/qrmc_external/run_meta_evaluation.py`). The rule receives only anonymised marginal profiles plus a labelled control cell for calibration; a held-out control (seeds 20–39) enters the blind set with ground truth "none", measuring the false-positive rate; a manipulation check excludes faults that do not bind (the prompt-only fault moved qwen but not llama). Stage-level accuracy: $26/30$ scored cells for llama3.1 ($8/10$, $8/10$, $10/10$ per stack) and $29/33$ for qwen3:4b ($9/11$, $10/11$, $10/11$); the five structural faults localize $30/30$ across both models and all stacks; held-out controls read "none" in $5/6$ cells (one false positive at n=20); cross-framework label agreement is $7/11$ (llama) and $10/11$ (qwen). The pre-specified model-contrast prediction was confirmed in its named direction: M-family faults hide behind llama's weak baseline commitment (M_1=0.25–0.50) and become visible on qwen's clean baseline — the unmarked duplicate drops M_1 by $0.90/0.80/0.20$ across stacks — while its accuracy clause held on two of three stacks (AutoGen: $10/11$ vs. llama's $10/10$, a one-cell difference). One pre-specified risk materialized instructively: the *marked* stale distractor changes llama3.1's behavior but reads "none" on qwen3:4b, which takes the staleness marker into account — the protocol measures the behavioral effect of a fault, not its presence, and a fault a model neutralizes is reported as no failure.

Table 6: Blinded fault localization: injected fault, ground-truth stage, and correct-localization count over the three stacks, per model. "Neutralized": the fault does not change qwen's behavior (the staleness marker is heeded), so the blind rule reads "none".

Injected fault	Stage	llama3.1	qwen3:4b
recall tool absent	Q	3/3	3/3
prompt: memory unreliable	Q	non-binding	3/3
fact never consolidated	R	3/3	3/3
silent corruption	R	3/3	3/3
flaky recall (p=0.7)	R	3/3	3/3
stale distractor (marked)	M	1/3	0/3 (neutralized)
duplicate (unmarked)	M	2/3	3/3
budget below requirement	C	3/3	3/3
flaky goto (p=0.5)	C	2/3	3/3
take always fails	C	3/3	3/3
held-out control	none	3/3	2/3

Scope: three stacks at n=20 per cell on two local models; Letta/CrewAI (Python $\geq$3.10) and API-scale models are the natural extension.

6 Limitations and future work

Portability across substrates: slope conditions hold on MiniGrid. The slope conditions of Empirical Claim 1 also hold on a standard-MiniGrid substrate. A FourRooms-based multi-agent

Table 7: Second-model replication (qwen3:4b, 20 episodes per cell, identical episodes/tools/budgets; thinking disabled request-level for budget parity, recorded in the pre-run specification). Entries are marginal event frequencies $f(E)$ as in Table 5 — not chain-rule links. Structural zeros transfer exactly; behavioral rates differ from llama3.1 (Table 5) in the direction the M_1 diagnostic explains.

Framework	Variant	$f(Q^\star)$	$f(R^\star)$	$f(M^\star)$	$f(C^\star)$	$f(M_1)$	$f(\text{stale}_1)$
OpenAI SDK	control	1.00	1.00	1.00	1.00	0.95	—
	consolidation gap	1.00	**0.00**	0.45	0.45	0.20	—
	stale ambiguity	1.00	1.00	1.00	0.95	0.90	0.05
	tight budget	1.00	1.00	0.30	**0.00**	0.30	—
LangGraph	control	1.00	1.00	1.00	1.00	0.90	—
	consolidation gap	1.00	**0.00**	0.60	0.60	0.30	—
	stale ambiguity	1.00	1.00	0.95	0.95	0.75	0.20
	tight budget	1.00	1.00	0.15	**0.00**	0.15	—
AutoGen	control	1.00	1.00	1.00	1.00	0.40	—
	consolidation gap	1.00	**0.00**	0.55	0.50	0.30	—
	stale ambiguity	1.00	1.00	1.00	0.90	0.40	0.30
	tight budget	1.00	1.00	0.15	**0.00**	0.15	—

wrapper (built from `minigrid.core.grid.Grid` + Wall/Floor/Lava primitives, identical operational $Q^\star, R^\star, M^\star, C^\star$ definitions; `experiments/minigrid_multiagent_wrapper/`) was used to run a 100-run portability sweep ($K\in\{1, 4, 8, 16, 64\}$, 20 seeds, $N{=}3$, $T{=}2$): Spearman $P(M^\star|R^\star)$ vs K is -0.50 and Spearman $P(C^\star|M^\star)$ vs K is $+0.50$ (moderate rank correlations; this is a small 100-run sweep, so we report the effect sizes and do not lean on significance). The interior t_{succ} minimum does not separate at this minimal sweep size (5 cadences, 20 seeds, one hazard density); a denser sweep is the natural next step. The two-substrate seam is therefore closed at the slope-level claim.

Empirical Claim 1 is slope-level on conditional rates only. The two slope conditions hold as moderate, cluster-robust effect sizes (Appendix 14.3); we make no formal claim on the shape of the conditional product Φ (§4.3). A future statement directly in terms of the time-cost surface, or a derivation under a generative model (Poisson-renewal memory, birth-death occupancy), would convert the empirical claim into a theorem.

Estimability under hidden state. With unlogged in-episode query rewrites or peer side-channels, Definitions 1 and 2 are projections onto logged traces, not full identification.

Shared coordinate frame. All multi-agent results assume agents interpret places in a common coordinate frame: cross-agent place identity is coordinate proximity, with semantic profiles as a tie-breaker inside a spatial radius. The place-correspondence problem — whether two agents' records denote the same place at all — is therefore bracketed here, not solved. A companion manuscript removes the assumption constructively (fingerprint-based place identity with consensus frame recovery up to translation and 90° rotation) and shows the Q/R/M/C measurements, including the cadence slopes and the staleness horizon, unchanged on top of the recovered frames; the harder variants (continuous $SE(2)$, appearance noise, symbol alignment) remain open.

The evaluator's correspondence oracle. A deeper form of the same dependence survives every representational relaxation, and we state it explicitly. The protocol unifies *observation points*, not internal representations — but scoring the two middle stages presupposes an *entity correspondence* between memory content and world referents: $R^\star$ asks whether a returned candidate denotes the true target, $M^\star$ whether the lock does. In every study here that oracle is supplied by an instrumented environment (ground-truth cells on the grids; the task's true location in the tool-trace contract). The dependence is graded, which is what makes the protocol degrade gracefully rather than break: $Q^\star$ requires no correspondence at all (did the agent query its memory), $C^\star$ is behaviorally observable (did the task complete), while $R^\star$ and $M^\star$ — the diagnostic payload — are undefined without it. The severity of the requirement depends on the population under evaluation: interchangeable agents (a shared perception interface, possibly different weights) situated in a single environment inherit the correspondence from the shared substrate — the setting of every study in this paper and the common case in multi-agent evaluation. It becomes a genuinely open problem for heterogeneous agents in open worlds, where deploying the protocol requires either an instrumented task or a correspondence mechanism with its own error model; the companion's meaning-based place identity is a candidate for grounding the *evaluator's* correspondence, not only the agents', and we leave that construction open.

Learned baselines. The CommNet PPO baseline reaches 0.667 success on the scarcity scenario — competitive with symbolic peer at $K{=}8$ (0.65) — and confirms Q-saturation ($Q^{\star}{=}1.00$) and R/M collapse ($P(M^{\star}|R^{\star}){=}1.00$) at competitive success. The structural prediction now holds verified, not asserted; ablating on architectures with explicit symbolic target-lock channels is left for future work.

Scale. Multi-agent sweep bounded at $N{=}12$ (Appendix 13; signature sharpens, peak Pearson -0.70 at $K{=}8$, interior t_{succ} minimum 4.77). Denser scaling at $N{\geq}16$ is future work. **Continuous substrate.** A VMAS (Bettini et al., 2022) validation reproduces *both* slope signs of Empirical Claim 1 across three scarcity cells $\times$ 40 seeds with block-bootstrap CIs excluding zero (Spearman -0.999 $[-1.000, -0.975]$ and $+0.995$ $[+0.900, +1.000]$ over $K{\leq}16$; Appendix 14.7), reusing the symbolic memory through a discretisation bridge. The shift is therefore not an artefact of grid cells; what remains open is fully continuous *memory* (the bridge discretises it) and visually rich 3D substrates.

7 Broader impacts

The paper proposes diagnostic evaluation for memory-based agents in simulated navigation. Its positive impact is methodological: making memory-based failures more measurable across architectures, and making the cadence axis of distributed multi-agent memory tunable on an empirically grounded criterion. Potential negative impacts are indirect and would arise if similar diagnostics were used to improve agents in safety-critical settings (drone swarms, autonomous vehicles, distributed sensor networks) without adequate safeguards. The current work is benchmark-level infrastructure, not a deployment-ready system.

8 Conclusion

We introduced Q/R/M/C as an operational protocol for evaluating memory-based navigation through four observable stages: query formation, retrieval satisfaction, target materialization, and post-retrieval completion. The framework does not replace final task metrics; rather, it identifies which part of the memory-to-action pipeline limits performance and whether the relevant stages are observable in a given architecture.

Across the single-agent experiments, stage-specific interventions separate retrieval-limited and downstream-limited tasks that appear similar under success-only evaluation. The detailed retrieval logs further show that the principal failure is the absence of eligible memory records rather than poor ranking. In the multi-agent setting, the same protocol reveals a cadence-dependent shift between materialization and completion: fast sharing improves access to target information but increases contention for scarce targets. Experiments with learned communication policies, local LLM agents, and third-party agent stacks show that the event contract transfers across implementations, while also exposing cases in which the architecture does not provide an explicit stage interface.

The present evidence supports Q/R/M/C as a reproducible diagnostic layer for systems with observable query and commitment events. It does not establish a universal decomposition for opaque policies or a general theorem about the optimal communication cadence. Richer continuous and visually grounded environments, learned memory formation, and entity correspondence across heterogeneous agents remain directions for subsequent parts of the program.

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This manuscript is prepared in anonymized form for review. The reported experiments were run from a unified benchmark pipeline that exports task-family summaries, assist on/off comparisons, method diagnostics, multi-agent sweep CSVs, and bootstrap analyses from the same source-of-truth benchmark artifacts.

Appendix material of this part

9 Formal proofs

9.1 Argument for Definition 1 (single-agent estimability)

For each episode $i \in \{1, \dots, n\}$ let $Q_i^\star, R_i^\star, M_i^\star, C_i^\star$ denote the nested episode-level indicators. By construction these are measurable functions of the runtime trajectory log and satisfy $C_i^\star \Rightarrow M_i^\star \Rightarrow R_i^\star \Rightarrow Q_i^\star$. The empirical estimators

$$\hat{p}_Q = \frac{1}{n}\sum_i Q_i^\star, \quad \hat{p}_{R|Q} = \frac{\sum_i R_i^\star}{\sum_i Q_i^\star}, \quad \hat{p}_{M|R} = \frac{\sum_i M_i^\star}{\sum_i R_i^\star}, \quad \hat{p}_{C|M} = \frac{\sum_i C_i^\star}{\sum_i M_i^\star}$$

are consistent by the chain rule plus the conditional stage-Markov property, and the empirical product converges to semantic-path completion. □

9.2 Argument for Definition 2 (multi-agent estimability)

Conditioning sets are enlarged by $\mathcal{M}_{-i}^{(K)}$ and $\mathcal{O}_{-i}$. Under the enlarged stage-Markov property, the proof of Definition 1 transfers verbatim and each enlarged conditional is a Bernoulli probability on a measurable event. The qualifications stated in §4.2 ("When Definition 2 may fail") describe two specific situations in which the enlarged stage-Markov property is violated. □

9.3 Argument for Empirical Claim 1 (slope-level statement)

Empirical Claim 1 is an empirical claim, not a theorem. The argument below is the qualitative justification that motivates the two slope signs; the data are what verify them. We deliberately do not call this a proof.

Slope (a), under (F). As K decreases, $\nu(\mathcal{M}_{-i}^{(K)})$ is non-decreasing pointwise. Conditional on a successful retrieval, the probability of locking onto a reachable target is non-decreasing in ν. Therefore $\partial P(M^\star|R^\star;K)/\partial K^{-1} > 0$.

Slope (b), under (O). As K decreases, peers receive updated memory simultaneously and converge on the same closest target; the marginal occupancy $\Pr[\text{target} \in \mathcal{O}_{-i}]$ rises. Therefore $\partial P(C^\star|M^\star;K)/\partial K^{-1} < 0$.

Note on the conditional product Φ. Empirical Claim 1 makes no statement about the shape of $\Phi(K) = P(M^\star|R^\star;K)\cdot P(C^\star|M^\star;K)$. Under (F) and (O) the slope of Φ in K^{-1} is the sum of two opposing terms, and whether it changes sign depends on quantitative magnitudes that are not pinned down by (F) and (O) alone. In our data Φ is monotone in K over $\{4, 8, 16, 32, 48, 64\}$ and converges to the independent-agent boundary $\Phi_\infty = 0.605$. The practically relevant efficiency surface — mean time-to-success — does have an interior minimum at $K = 8$; see §4.5. □

9.4 The query–retrieval Galois connection: statement, argument, scope

Fact (standard). Let G and Σ be sets and $I \subseteq G \times \Sigma$ a relation. For $B \subseteq \Sigma$ define $\operatorname{ext}(B) = \{p \in G : \forall \tau \in B,\ (p, \tau) \in I\}$, and for $A \subseteq G$ define $\operatorname{int}(A) = \{\tau \in \Sigma : \forall p \in A,\ (p, \tau) \in I\}$. Then for all $A \subseteq G$, $B \subseteq \Sigma$:

$$A \subseteq \operatorname{ext}(B) \iff B \subseteq \operatorname{int}(A);$$

both maps are antitone, both composites $\operatorname{ext} \circ \operatorname{int}$ and $\operatorname{int} \circ \operatorname{ext}$ are closure operators, and the closed pairs (A, B) with $A = \operatorname{ext}(B)$ and $B = \operatorname{int}(A)$, ordered by extent inclusion, form a complete lattice — the concept lattice of the context (G, Σ, I) (Ganter and Wille, 1999).

Argument. $A \subseteq \operatorname{ext}(B)$ unfolds to $\forall p \in A\ \forall \tau \in B : (p, \tau) \in I$, a condition symmetric in A and B, which is exactly $B \subseteq \operatorname{int}(A)$. Antitonicity and the closure properties follow by the standard one-line arguments (Ganter and Wille, 1999, Ch. 1). Viewing the power sets as thin categories (one with reversed order), the displayed equivalence says precisely that int and ext form an adjoint pair; a Galois connection is an adjunction between posets (Mac Lane, 1998, Ch. IV). □

Instantiation and scope. In our memory, take G to be the place records, Σ the tag alphabet, and $I_\theta = \{(p, \tau) :$ the per-place tag table assigns τ to p with confidence $\geq \theta\}$ (Appendix 12); the intent of a query is $B = q^{\text{req}}$ and R-availability is $\text{ext}(B) \neq \emptyset$. Three deliberate scope restrictions. (i) *Graded scores.* The deployed retrieval score is graded (the weighted log-score over required, preferred, and penalty tags of Appendix 12), so the crisp connection above formalizes the thresholded required-tag skeleton of candidacy; the graded analogue is the theory of fuzzy Galois connections (Bělohlávek, 1999) and we do not use it here. (ii) *Non-monotone updates.* Dirichlet–multinomial updates are not literally monotone in I_θ: contrary evidence can remove an incidence. On the reported benchmarks the binding failure is absence of evidence rather than contradiction (91% empty extents, 96.9% precision when non-empty, §3.5), so the monotone-growth reading is the empirically relevant regime, and we flag the exception rather than assume it away. (iii) *No structure for M or C.* Materialization is a satisficing choice function on non-empty extents and Completion is a control problem under occupancy; neither carries the order structure, and we attach no categorical claims to them. The reading earns its place by naming the failure mode exactly (a queried intent whose extent is empty lies outside the realized concept lattice), by giving consolidation a precise object (growth of the context, hence of every extent), and by locating the cadence axis structurally: memory coupling $\mathcal{M}_{-i}^{(K)}$ merges peer contexts at rate K^{-1}, while occupancy $\mathcal{O}_{-i}$ is extra-contextual — which is the structural content behind the two slope signs of Empirical Claim 1.

10 An options / hierarchical-MDP reading of the stages

Why a deterministic strategy: bounded and ecological rationality

A natural objection is that the semantic stack is a fixed, deterministic strategy rather than a learned policy. We treat the determinism as principled and name it: the pipeline is a satisficing agent in Simon's sense (Simon, 1955, 1956). Candidacy is threshold-gated instead of exhaustively scored, the target commitment is made once instead of re-optimized at every tick, and the intent is a few tags instead of the full state. Bounded rationality is also what makes the four failure loci observable: each stage is a distinct resource bound (attention, memory, deliberation, control), and Q/R/M/C is the audit trail of where the bound binds. The learned baselines make the converse point empirically: policies without an explicit query/lock interface collapse the stages even at competitive success (BC in §3.3; CommNet PPO in §4.5), so observability is traded away exactly where end-to-end optimization is traded in. That the *same* heuristic is strong on water and rest, retrieval-starved on hazard recovery, and downstream-limited on goal-region (Figure 1b) is the ecological-rationality point that heuristic success is a property of the heuristic–environment pair (Simon, 1956; Gigerenzer and Goldstein, 1996); the framework is the instrument that locates the mismatch. The same lens reads the Φ-versus-t_{succ} split of §4.5: the chain probability Φ ignores the cost of traversing the chain, so under a resource-rational reading (Lieder and Griffiths, 2020) an optimum should be sought on a cost-bearing metric; that is where it appears ($K{=}8$ on mean time-to-success) and not on Φ.

This appendix also develops the dynamic reading pointed to from §2.3. It introduces no new algorithm: it names the object the pipeline already computes and makes the Φ-versus-t_{succ} split a property of that object. Where the order-theoretic reading (Appendix 9.4) fixes the *static* content of the stages, this one addresses time and cost.

Q/R/M/C as an audit of an option. Each query instantiates one temporally extended action—an option $o = \langle \mathcal{I}, \pi, \beta \rangle$ in the sense of Sutton et al. (1999): an initiation set $\mathcal{I}$, an internal policy π, and a termination condition β. Standard trajectory-level analysis treats an option as a black box scored by its return. The Q/R/M/C decomposition is instead an audit of the option's *internal* structure, and it lines up edge-for-edge with the factorization $Q^\star \to R^\star \to M^\star \to C^\star$:

$Q^\star \to R^\star$ **(initiation).** The option "reach a place satisfying q" is well-initiated only when its initiation set is non-empty. In the vocabulary of §2.3 this is exactly $\text{ext}(q^{\text{req}}) \neq \emptyset$: an empty extent is not a bad choice of option but the absence of any admissible one, so R fails before a target is ever locked.

$R^\star \to M^\star$ **(policy over options).** Materialization is the high-level policy that commits to a single option out of the admissible set—the target lock $g^\star = \chi(\text{ext}(q^{\text{req}}))$ under the satisficing choice function χ of §2.3.

$M^\star \to C^\star$ **(execution to termination).** Completion is the roll-out of the internal policy π until the termination condition β fires (arrival at $g^\star$, or timeout). This is the only edge whose failure modes are extra-semantic: controller competence and, in a collective, occupancy $\mathcal{O}_{-i}$.

So Q/R/M/C is, on this reading, a stage-resolved instrument for the internal structure of a hierarchical-MDP option generator, rather than a bespoke logger for a single environment.

Why the optimum lives on t_{succ} and not on Φ: a semi-MDP reading. In the options / semi-MDP formalism, an option is evaluated by a return accumulated over a *random* execution duration τ,

$$R(s, o) = \mathbb{E}\Big[\textstyle\sum_{t=0}^{\tau-1} \gamma^t\, r_{t+1} \;\Big|\; s_0 = s,\, o\Big],$$

where τ is the number of low-level steps until β fires (Sutton et al., 1999). The chain probability $\Phi = P(M^\star|R^\star) \cdot P(C^\star|M^\star)$ records only *whether* the option-audit succeeds; it is by construction blind to τ, which is the cost of traversing it. The cost-bearing metric t_{succ} is τ. This is the structural reason—and not merely an empirical coincidence—that the resource-rational reading of §2.3 (Lieder and Griffiths, 2020) locates an interior optimum on t_{succ} (K=8) while Φ stays monotone: the two metrics measure different faces of the same option, success versus duration.

The bottleneck shift as multi-agent option collision. The dynamic reading makes the M↔C trade-off of Empirical Claim 1 physically concrete. Faster broadcast (smaller K) merges peer contexts $\mathcal{M}_{-i}^{(K)}$ more often, which can only enlarge each agent's extent and therefore raises the probability that a target is admissible and locked: $P(M^\star|R^\star)$ increases. But because agents share a spatial basis and a tag alphabet (§2.1), fresher shared memory also *synchronizes* the high-level policy: several agents commit to the same option over the same scarce target. Under occupancy $\mathcal{O}_{-i}$ this inflates the execution duration τ and depresses $P(C^\star|M^\star)$. Cadence K thus regulates the dispersion of the agents' option distribution: rare exchange desynchronizes option selection and shortens τ, giving the minimum of t_{succ} at an interior K. The bottleneck is therefore grounded in a competition between initiation-set growth (semantic, contextual) and option-collision cost (extra-semantic, occupancy), exactly the two edges the cadence protocol $\mathcal{C}_K$ acts on in Figure 2.

Belief–desire–intention gloss and scope. The high-level state audited above can be named in belief–desire–intention terms, which is the form used by the collective-memory companion: *belief* is the realized concept lattice of §2.3 (what is retrievable), *desire* is the state-conditioned predicate that selects the query q (e.g. a thirst variable switching the required tags to `water_source`), and *intention* is the locked option $g^\star$. The hierarchical state is then $h = (\,\text{lattice},\, \text{desire},\, g^\star,\, s\,)$ with s the low-level physical state, and the three failure loci separate as before: an empty extent is a belief/R failure, a non-empty extent with no stable lock is an M failure, and a lock unreachable under occupancy is a C failure. Two scope caveats. We claim no options-*learning* result: the pipeline is a fixed satisficing option generator (Simon, 1955) and Q/R/M/C audits it. And we attach the option structure only to the semantic path—$\mathcal{I}$ to $Q \to R$, (π, β) to $M \to C$—while the semi-MDP return is used as an explanatory lens for the Φ-versus-t_{succ} split, not fitted to the data. Neither this options reading nor the order-theoretic reading of Appendix 9.4 is claimed as a contribution: both are standard constructions used as interpretive lenses on the measurement protocol, which is the paper's contribution. Their value is that each makes a testable statement on the logs—an empty extent localizes the R failure, and the semi-MDP duration τ predicts that the optimum appears on t_{succ} and not on Φ.

11 Operational vs. nested stage estimators

Operational diagnostic rates. The main benchmark logs per-query and per-episode diagnostic quantities directly from runtime traces (query formed, retrieval satisfied, target materialized, completion after retrieval). These are operational rates, not direct multiplicative factors.

Nested stage estimators for factorization. For the factorization, episode-level nested events from the first failure stage recorded in the runtime taxonomy are used: $Q^\star, R^\star, M^\star, C^\star$, nested by construction. The consistency audit in §3.2 reports $\hat{P}(Q^\star)$, $\hat{P}(R^\star|Q^\star)$, $\hat{P}(M^\star|R^\star)$, $\hat{P}(C^\star|M^\star)$, their product, and its gap to semantic-path completion.

11.1 Multi-agent operational definitions

For the multi-agent sweep, every metric is defined per agent and per tick, then aggregated. Let $\mathrm{Targets}_i(t)$ be the candidate set returned by the memory query of agent i at tick t, $\mathrm{Locked}_i(t)$ be the agent's planner-internal locked target, and $\mathcal{W}$ the ground-truth target set.

Per-tick events. $Q_i(t) = \mathbf{1}[\mathrm{Targets}_i(t) \neq \emptyset]$; $R_i(t) = \mathbf{1}[\exists w \in \mathcal{W} : \min_c \|c - w\| \leq \epsilon]$ ($\epsilon = 0.6$); $M_i(t) = \mathbf{1}[\mathrm{Locked}_i(t) \neq \emptyset \wedge \min_w \|\mathrm{Locked}_i(t) - w\| \leq \epsilon]$.

Episode-level starred events. $Q_i^\star = \bigvee_t Q_i(t)$; analogously for $R^\star, M^\star$; $C_i^\star = \mathbf{1}[\text{agent } i \text{ reached a target}]$.

Aggregation. Per-run rates are averaged over the N agents; per-configuration rates are averaged over the 40 seeds. Conditional rates $P(R^\star|Q^\star)$, $P(M^\star|R^\star)$, $P(C^\star|M^\star)$ are computed as ratios of episode-level counts within each configuration before averaging across configurations.

12 Memory and task schema

12.1 Full notation table

Symbols are fixed throughout the paper.

Stages. Q, R, M, C — the four stages: **Q**uery formation, **R**etrieval, **M**aterialization, **C**ompletion after retrieval. They name both the stage and (in expressions like "the M link") the corresponding edge of the chain.

Events. $Q^\star, R^\star, M^\star, C^\star \in \{0, 1\}$ — episode-level starred events recording whether each stage succeeded (Appendix 11). Nested by construction: $Q^\star{=}1 \Leftarrow R^\star{=}1 \Leftarrow M^\star{=}1 \Leftarrow C^\star{=}1$.

Probabilities. $P(\cdot)$ — population-level probability (averaged over the seed distribution and, in the multi-agent case, over agents). $\hat{P}(\cdot)$ — empirical-frequency estimator computed from logs.

Single-agent. q — query; $\mathcal{M}$ — memory state; Y — end-to-end task success (can exceed $C^\star$).

Multi-agent. N — number of agents in the population. T — number of targets in the environment (constraint $T \leq N$; "scarcity" means $N > T$). i — agent index. $\mathcal{M}_i$ — agent i's private memory. q_i — agent i's query.

Coupling. $\mathcal{C}_K$ — the peer-broadcast communication protocol with cadence $K \in \mathbb{N}$ (every K ticks each agent broadcasts a memory snapshot, §4.1). $\mathcal{M}_{-i}^{(K)}$ — the latest peer memory snapshots accessible to agent i under $\mathcal{C}_K$ (the *memory coupling* channel). $\mathcal{O}_{-i}$ — the set of targets already claimed by other agents by the time i acts (the *occupancy coupling* channel).

Slope variables. $\nu(\mathcal{M}_{-i}^{(K)})$ — a freshness functional on peer memory; non-decreasing pointwise as K decreases (assumption **(F)** of Empirical Claim 1). The occupancy slope is assumption **(O)**.

Summary statistics. $\Phi(K) := P(M^\star|R^\star; K) \cdot P(C^\star|M^\star; K)$ — the conditional product, the chain probability $P(M^\star \wedge C^\star \mid R^\star)$. POM (product-of-means) and MOP (mean-of-products) are two estimators of Φ; they differ by $\mathrm{Cov}(P(M^\star|R^\star), P(C^\star|M^\star))$, which is the Jensen gap (§4.5). t_{succ} — mean time-to-success in environment ticks.

12.2 Memory stack and task patterns

The memory stack stores symbolic place records (per-place tag confidences, counts, visitation, freshness, geometry), transition records (success, risk, cost statistics with fast/slow estimates), episodic traces (intent + state at each visit), and concept/relation records. At each step, the tokenizer emits (z_t, τ_t, e_t) from o_t. Per-place tag tables use a Dirichlet-multinomial update; transitions use exponentially smoothed statistics; episodic records attach the active intent and agent state. A planner query is answered by retrieving candidate places whose semantic profiles match the current task intent and state, via a weighted log-score $s(p \mid q_t, s_t) = \sum_{\tau \in q^{\mathrm{req}}} \log P(\tau|p) + \alpha \sum_{\tau \in q^{\mathrm{pref}}} \log P(\tau|p) - \beta \sum_{\tau \in q^{\mathrm{pen}}} \log P(\tau|p) + \gamma\, u_{\mathrm{epi}}(p, s_t)$ over required, preferred, and penalty tags. The four task families are defined by semantic place patterns: water search (target {water}), safe rest ({rest} with state-conditioning), goal-region rejoin ({goal-region}), hazard recovery ({post-hazard-exit}).

12.3 Single-agent supporting tables and figures

Table 8: Best semantic result per task family from the final 10-seed benchmark, with 95% bootstrap CIs. Hazard-recovery is bottlenecked by retrieval; goal-region remains downstream-bottlenecked.

Task	Best method	Assist	Success (95% CI)	Steps	Query (op.)	Retrieval (op.)	Mat. (op.)	Post (op.)
Water	`water-cp`	on	0.95 [0.88, 1.00]	17.99	0.71	0.71	0.71	0.70
Rest	`rest-s2`	on	0.93 [0.85, 0.99]	23.35	0.61	0.61	0.61	0.73
Goal region	`goal-n1`	off	0.54 [0.52, 0.56]	49.34	0.00	0.00	0.00	0.00
Hazard recovery	`hazard-s7`	on	0.40 [0.35, 0.45]	10.18	0.11	0.11	0.11	0.16

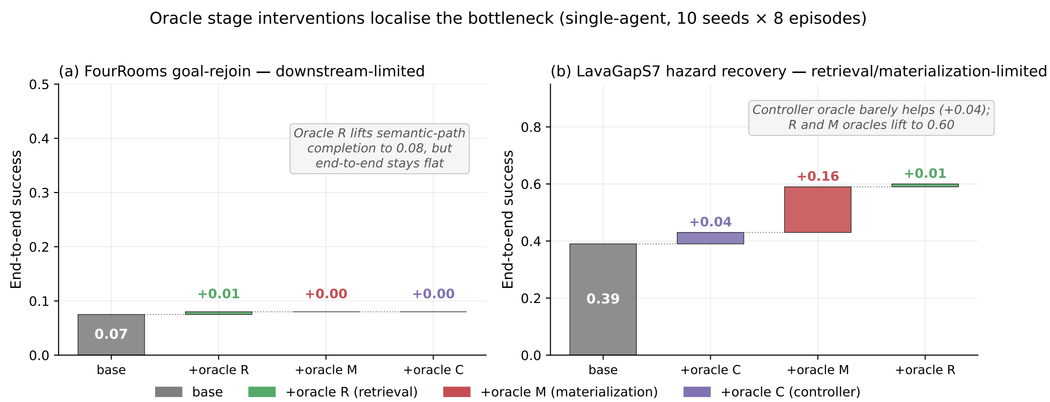

Figure 6: Oracle stage interventions on the two hardest single-agent task families (10 seeds $\times$ 8 episodes). **(a)** FourRooms goal-rejoin: replacing any single stage with an oracle leaves end-to-end success at 0.07; the bottleneck is downstream of the stages in the chain. **(b)** LavaGapS7 hazard recovery: oracle controller adds only $+0.04$, but oracle materialization adds $+0.16$ and oracle retrieval adds a further $+0.01$ to reach 0.60. The framework localizes the bottleneck stage-by-stage.

Seed robustness (30 seeds). The single-agent benchmark of §3 uses 10 seeds. To check the diagnosis is not a small-sample artefact, we reran all four task families at 30 seeds, both assist modes, on an independent SLURM cluster (`-num_seeds 30`; e.g. n=240 hazard-recovery and 480 goal-region agent-episodes per mode). Success (assist off): water 0.90, rest 0.84, goal-region 0.61, hazard recovery 0.18; empty-candidate-set rates 0.03, 0.00, 0.37, 0.65 respectively. The *qualitative* attribution is unchanged: hazard recovery stays retrieval-starved (empty-candidate-set rate ≈ 0.62–0.65, low success) while goal-region is far less retrieval-limited (empty-set ≈ 0.37, success ≈ 0.61) — the same stage *ordering* the framework reports at 10 seeds. Absolute rates depend on the exact benchmark configuration (episode budget, assist and env-change settings) and so differ from the headline $0.39/0.91$ of the 10-seed oracle configuration; what 30 seeds confirms is the ordering the diagnosis rests on, not a particular level.

13 Portability and scale-up

13.1 BabyAI portability transfer (10 seeds, full benchmark)

A 10-seed portability benchmark on `BabyAI-GoToObjMaze-v0` (Chevalier-Boisvert et al., 2018) (8 episodes per seed, max-steps 200) measured all four single-agent methods used in the main benchmark (random, greedy, `babyai_semantic_node_v1`, `babyai_semantic_plan_v1`) with the same Q/R/M/C logger applied to BabyAI traces. The numbers in Table 9 confirm that (a) the Q/R/M/C events emit non-trivially on the BabyAI substrate, (b) the two semantic variants expose stage-separable structure where the non-semantic baselines have $Q^{\star}$=$R^{\star}$=$M^{\star}$=0 (no semantic queries are issued), and (c) the stage-rate ordering matches the MiniGrid benchmark (Table 2) within bootstrap CIs. GoToObjMaze is a harder task substrate than the LavaGapS7/FourRooms benchmark used in the main text — end-to-end success caps at 0.15 across all methods — which makes the stage decomposition especially informative: the semantic stack is not winning on success but it is the only method family producing measurable per-stage events.

Table 9: BabyAI portability benchmark: 10 seeds $\times$ 8 episodes on `BabyAI-GoToObjMaze-v0`, max-steps 200. The semantic-stack variants succeed at the same end-to-end rate as the greedy baseline (0.15) but are the only methods that produce non-zero Q/R/M/C events — the framework's protocol transfers to the BabyAI substrate without modification. Q/R/M/C operational rates here are evaluated on the same denominator (number of decision points), so $Q{=}R{=}M$ for both semantic variants because GoToObjMaze emits one semantic target type per episode.

Method	Success	Steps	Q (op.)	R (op.)	M (op.)	Post (op.)
random	0.125 ± 0.06	182.7	0.000	0.000	0.000	0.000
greedy	0.150 ± 0.05	171.3	0.000	0.000	0.000	0.000
semantic node	0.150 ± 0.05	171.3	0.032	0.032	0.032	0.025
semantic plan	0.150 ± 0.05	171.3	0.032	0.032	0.032	0.025

Takeaway. The framework's measurement protocol — not its task-specific results — transfers to a second substrate. End-to-end success on BabyAI is dominated by the much lower episode-level success ceiling of GoToObjMaze, not by the stage structure; conversely, the stage structure that the semantic stack exposes is identical in shape to what the same stack exposes on MiniGrid. This is exactly the substrate-independence property the methodology is intended to provide. Raw data: `tmp/babyai_deeper_10seed/`.

13.2 Multi-agent scale-up at $N = 12$

To check that the multi-agent results of §4.5 are not an artefact of the $N \leq 8$ range, an additional 8,640-run sweep was run at $N{=}12$, $T \in \{6, 8, 10\}$, the same three layouts, three hazard densities, eight peer cadences, and 40 seeds. Wall-clock $\approx$ 12 minutes on 8 cores. The conditional-rate signature of §4.5 persists and intensifies (Table 10).

Table 10: $N{=}12$ scale-up sweep numerical detail (peer architecture; 27 configuration cells). Effect sizes with cluster-robust 95% CIs (block bootstrap over cells): Spearman of $P(C^\star|M^\star)$ with K is $+0.31$ (CI $[+0.21, +0.41]$, excludes zero), while Spearman of $P(M^\star|R^\star)$ with K is only -0.14 (CI $[-0.28, +0.02]$, *not* resolved from zero at $N{=}12$). The C-slope of Empirical Claim 1 holds; the M-slope is directionally consistent but not cluster-robust at this agent count. What sharpens at $N{=}12$ is the within-cadence correlation and the t_{succ} minimum (Figure 7), not the M-side K-slope.

K	$P(M^\star\|R^\star)$	$P(C^\star\|M^\star)$	MOP	mean t_{succ}	corr
1	0.758	0.679	0.475	5.49	-0.59
2	0.745	0.692	0.474	5.29	-0.61
4	0.679	0.768	0.482	4.82	-0.64
8	0.731	0.767	0.539	**4.77**	$\mathbf{-0.70}$
16	0.687	0.803	0.538	5.04	-0.62
32	0.676	0.814	0.538	5.75	-0.51
48	0.677	0.814	0.539	5.96	-0.51
64	0.677	0.816	0.541	6.12	-0.51

13.3 Q/R/M/C under semantic, frame-free place matching

The main sweep resolves place identity by shared (x, y) keys. To check that the decomposition is not tied to a shared coordinate substrate, we rerun a multi-agent navigation loop in which every agent holds a *private* coordinate frame (a secret per-agent translation) and places are matched *by meaning*: local semantic fingerprint constellations recover the inter-agent frame and transport a peer's target evidence into the receiver's frame (`experiments/warp/semantic_peer_memory.py`; runner `experiments/warp/exp_semantic_cadence_qrmc.py`). Agents navigate with the same

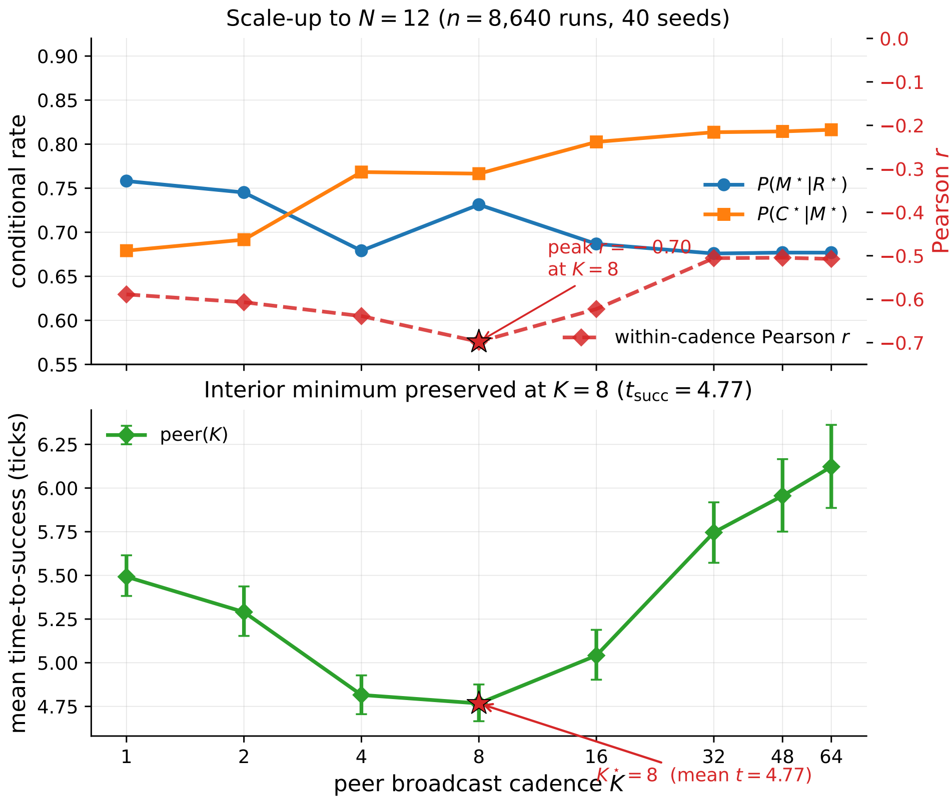

Figure 7: $N = 12$ scale-up bottleneck signature ($n = 8{,}640$ runs, 40 seeds). Top: the per-cadence mean rates move in the predicted directions in K; the within-cadence Pearson correlation (red dashed, right axis) peaks at -0.70 at $K = 8$ (this within-cadence peak is what sharpens at $N{=}12$; the M-side K-slope itself is not cluster-robust at this agent count, Table 10). Bottom: mean time-to-success retains an interior minimum at $K = 8$ (4.77 ticks); as on the main sweep, the location is robust but the margin is shallow — the gap to the adjacent $K = 4$ (4.82 ticks) is ≈ 0.05 tick and is not resolved at this sample size. The bottleneck-shift peak migrates outward from $K = 4$ at $N \leq 8$ to $K = 8$ at $N = 12$, consistent with stronger occupancy coupling at higher agent count delaying the cost-regime transition.

rule-based planner toward the transported target; Q/R/M/C are logged unchanged, with $M^\star$ the lock of a candidate within ϵ of a true target ($N{=}6$, $T{=}2$ scarce water sources, 8 seeds).

The stage events remain well-defined without a shared frame, and the framework localizes the coordinate assumption's failure. The *semantic* arm locks a *true* target on its first materialization in 100% of agent-episodes (at every cadence); the *coordinate* arm — which reads a peer's private coordinates at face value — first-locks a true target only 25–52% of the time (rising with K as agents accumulate their own observations), i.e. it *fails open*, materialising phantom cells that are not water. End-to-end success is scarcity-limited and comparable across arms ($\approx 0.29 \approx T/N$), as expected. This is the diagnostic property of interest: $M^\star$ under coordinate identity fires on non-targets, and the stage logger exposes it.

We do *not* claim the M↔C cadence shift here: with an all-to-all broadcast and a 140-tick horizon the semantic frame recovers at every tested K, so $P(M^\star|R^\star)$ is cadence-invariant in this setup (1.00 throughout). The experiment isolates the *place-identity substrate* question — Q/R/M/C is architecture- and coordinate-independent, and detects fail-open materialization — rather than the cadence result, which stands on the main coordinate-keyed sweep. Folding semantic matching into the scarcity regime that produces the shift is the natural next integration.

14 Extended diagnostics

14.1 Conditional-product detail and the Jensen gap

Table 11: Per-architecture conditional Q/R/M/C rates from the 35,640-run sweep (40 seeds). $P(R^\star|Q^\star)$ is saturated. POM = product-of-means; MOP = mean-of-products (the proper per-configuration expected product). The two summaries disagree systematically at small/intermediate K because the within-cadence correlation between $P(M^\star|R^\star)$ and $P(C^\star|M^\star)$ is strongly negative there (last column) — the bottleneck-shift signature. **Bold**: MOP-best of the peer cadences. Mean t_{succ} (last column) is the practically relevant efficiency measure; its interior minimum at $K = 8$ is the predicted-shape result, robust in location but shallow in margin (adjacent fast cadences not resolved; Appendix 14.3).

Architecture	$P(R^\star\|Q^\star)$	$P(M^\star\|R^\star)$	$P(C^\star\|M^\star)$	POM	MOP (95% CI)	corr	mean t_{succ}
independent	0.99	0.68	0.88	0.598	0.605 [0.597, 0.614]	$+0.00$	8.02
shared bus	1.00	1.00	0.60	0.600	0.595 [0.586, 0.604]	-0.10	8.33
central aggregator	1.00	0.97	0.61	0.592	0.572 [0.564, 0.580]	-0.48	7.64
peer ($K = 1$)	1.00	0.97	0.60	0.585	0.573 [0.564, 0.581]	-0.43	7.80
peer ($K = 2$)	1.00	0.97	0.62	0.594	0.576 [0.568, 0.584]	-0.54	7.59
peer ($K = 4$)	1.00	0.92	0.66	0.599	0.569 [0.561, 0.577]	-0.61	8.00
peer ($K = 8$)	0.99	0.69	0.87	0.597	0.586 [0.578, 0.595]	-0.19	**7.47**
peer ($K = 16$)	0.99	0.67	0.89	0.592	0.590 [0.581, 0.598]	-0.05	7.87
peer ($K = 32$)	0.99	0.67	0.89	0.595	0.598 [0.589, 0.606]	-0.04	8.28
peer ($K = 48$)	0.99	0.68	0.88	0.598	0.602 [0.593, 0.611]	-0.02	8.79
peer ($K = 64$)	0.99	0.68	0.88	0.599	**0.603** [0.594, 0.611]	-0.02	9.19

The conditional product $\Phi = P(M^\star|R^\star)\cdot P(C^\star|M^\star)$ in the 35,640-run sweep is monotonically increasing in K over $K \in \{4, 8, 16, 32, 48, 64\}$ and converges to the independent-agent boundary $\Phi(K \to \infty) = 0.605$. Paired bootstrap: $\Delta(K{=}32{-}K{=}16) = +0.008$ (CI $[+0.006, +0.010]$), $\Delta(K{=}48{-}K{=}16) = +0.011$, $\Delta(K{=}64{-}K{=}16) = +0.012$; the independent-agent level sits above every peer K except $K{=}64$ (where the gap's CI includes zero). This is a statement about the *level* of Φ at the tested cadences, not about the *shape* of $\Phi(K)$: consistent with the monotone trend, we make no claim of an interior extremum (§4.3).

The negative within-cadence covariance documented in §4.5 creates the Jensen-inequality gap between Φ and the marginal product-of-means: $E[X]\cdot E[Y] > E[X\cdot Y]$ when $\mathrm{Cov}(X, Y) < 0$. The mean-of-products is reported as the primary summary because it absorbs the covariance correctly. Figure 8 shows the K-sweep view of both conditional rates and mean t_{succ}; Figure 9 shows the per-configuration scatter that the negative covariance summarises.

14.2 Design decomposition and run-count reconciliation

The 35,640 figure decomposes exactly as in Table 12. The base design is the nine admissible (N, T) scarcity pairs (for N=3: $T\in\{2, 3\}$; N=5: $T\in\{2, 3, 5\}$; N=8: $T\in\{2, 3, 5, 8\}$) crossed with three layouts, three hazard densities, and 40 seeds, giving 3,240 base configurations. The peer architecture is run at all eight cadences (3,240 × 8 = 25,920); the three cadence-free architectures (independent, shared bus, central aggregator) contribute 3,240 each. The 25,920-run figure used for the peer cluster-robust analysis (Appendix 14.3) is exactly the peer block; the initial 12,960-run exploratory sweep (§4.3, "Chronology") and the extended-cadence and N=12 sweeps (Appendix 13) are separate run sets and are not pooled with the main sweep.

14.3 Effect sizes and cluster-robust inference

The $p < 10^{-4}$ figures quoted in §4.5 are reported for completeness but are not the basis of the claim: at tens of thousands of runs almost any non-zero slope is "significant," and the runs are not independent—they cluster within the $(N, T,$ layout, hazard$)$ configuration. The full sweep has 25,920 peer runs but only **81 independent design cells**. We therefore report effect sizes with cluster-robust confidence intervals from a block bootstrap that resamples *configuration cells* (not rows), B=4000 resamples.

The two slopes of Empirical Claim 1 survive as genuine, moderate effects: the Spearman correlation of $P(M^\star|R^\star)$ with K is $\rho = -0.53$ (95% cluster CI $[-0.59, -0.47]$, moderate-to-strong) and of

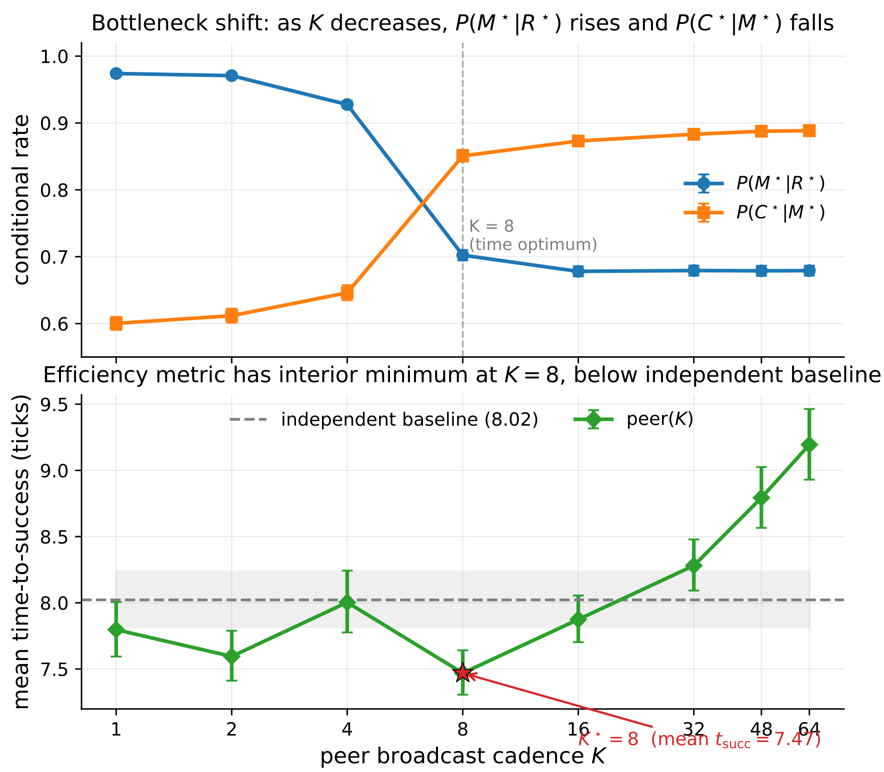

Figure 8: Bottleneck shift across eight peer cadences (40-seed sweep). Top: $P(M^\star|R^\star)$ falls and $P(C^\star|M^\star)$ rises in K (moderate, cluster-robust Spearman effect sizes -0.53 and $+0.43$; CIs in Appendix 14.3). Bottom: mean time-to-success has an interior minimum at $K = 8$ (7.47 ticks) below the independent-agent baseline (8.02, dashed); the location is robust but its margin over the adjacent fast cadences (K=1, 4) is not resolved (Appendix 14.3). Error bars and shaded band are 95% bootstrap CIs.

Table 12: Run-count reconciliation for the main multi-agent sweep (35,640 runs). Counts are recomputed directly from the released run log.

Factor	Levels	Count
(N, T) scarcity pairs	$(3,2)(3,3)(5,2)(5,3)(5,5)(8,2)(8,3)(8,5)(8,8)$	9
Layouts	symmetric, asymmetric, random	3
Hazard densities	$0.0,\ 0.05,\ 0.10$	3
Seeds		40
Base configurations	$9 \times 3 \times 3 \times 40$	3,240
Peer (all 8 cadences K)	$3{,}240 \times 8$	25,920
Independent / shared / central	$3 \times 3{,}240$	9,720
Total		**35,640**

$P(C^\star|M^\star)$ with K is $\rho = +0.43$ (95% cluster CI $[+0.37, +0.50]$, moderate). Both intervals exclude zero under cell-level resampling, so the bottleneck shift is not an artefact of the correlated run count; it is a moderate monotone trend, and we describe it as such rather than as a near-deterministic law.

The interior optimum on mean time-to-success is robust in *location* but shallow in *margin*. Paired block bootstrap on the cell-level t_{succ} differences gives $t_{\text{succ}}(K{=}8)$ below $t_{\text{succ}}(K{=}16)$ by 0.40 ticks (CI $[0.08, 0.83]$) and below K=64 by 1.67 ticks (CI $[0.75, 2.83]$), both resolved; but its advantage over K=4 (0.49, CI $[-0.47, 1.51]$) and K=1 (0.27, CI $[-0.43, 0.97]$) is within noise.[1] A success-

[1]This paired bootstrap and the effect-size Spearman correlations above are computed on the same sweep execution as Table 11 (25,920 peer runs); the t_{succ} statistic is defined only over successful episodes, so the paired cell-level difference (1.67

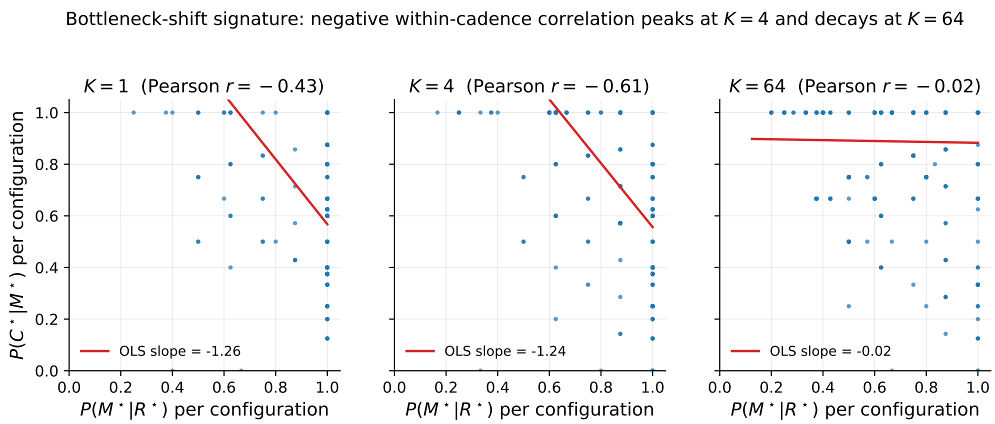

Figure 9: Per-configuration scatter of $P(M^{\star}|R^{\star})$ vs. $P(C^{\star}|M^{\star})$ at three cadences. The strong negative correlation at $K = 4$ (Pearson -0.61) is the bottleneck-shift signature: the same configurations that succeed at M tend to fail at C and vice versa. The correlation decays to noise by $K = 64$.

weighted efficiency metric (t_{succ} divided by success rate) has the same interior $\arg\min$ at K=8, so the location of the optimum is robust to the efficiency definition even though its margin over the adjacent fast cadences is not resolved at this sample size. We read this as support for the *shape* predicted by the M↔C trade-off, not for a sharply tuned optimal cadence. Two further cell-paired contrasts complete the picture. On t_{succ}, K=8 is also resolved against the independent baseline (-0.40, CI $[-0.59, -0.24]$). On a censoring-aware expected time $\mathbb{E}[T] = s\, t_{\mathrm{succ}} + (1 - s)\, T_{\max}$ (failed runs charged the step limit $T_{\max}$=120), the fast flank reverses: K=8 beats K=4 (-2.10, CI $[-3.61, -0.72]$) but loses to K=16 ($+0.69$ $[+0.09, +1.43]$) and to independent ($+0.32$ $[+0.05, +0.63]$). The interior optimum is thus a property of conditional speed among successes; on unconditional efficiency the completion collapse of fast sharing dominates, which is the gap the reservation protocol of open problem (3) closes.

14.4 Construct validity: task success Y vs. semantic-path completion $C^{\star}$

A success Y can in principle bypass the semantic path, so a stage bottleneck in $Q/R/M/C$ need not explain variation in Y. Two facts bound this concern. First, in the multi-agent sweep that Empirical Claim 1 is about, success is *defined* as reaching a `water_source` target, which is exactly the $C^{\star}$ event; empirically Y and $C^{\star}$ coincide run-for-run (correlation 1.00, mean gap 0.00). The central multi-agent claim is thus not exposed to a parallel-channel objection. Second, the gap $Y > C^{\star}$ is a single-agent phenomenon on the easy task families (water, rest), where a fallback controller can reach the goal without a semantic target lock; it is largest exactly where success is already high and the diagnosis is least needed, and it vanishes on the hard families (hazard recovery, goal-region) where the framework does its work. A per-episode, stage-resolved mediation of Y (rather than of $C^{\star}$) requires logs that tag each success with whether it traversed the semantic path, and is a planned instrumentation refinement.

14.5 Threshold sensitivity of the Q/R/M/C event definitions (θ and ϵ)

The single-agent attribution "candidate generation fails, not ranking" (§3.5) rests on a high empty-candidate-set rate. A potential concern is that this pattern may be an artifact of the required-tag threshold θ (deployed 0.5; the candidacy gate is $\mathrm{profile[tag]} \geq \theta$ in `symbolic_memory._required_matches_profile`). We sweep θ via an override on the query builder (`experiments/big_experiment/theta_ablation.py`); the endpoints confirm the gate is live, and the operating range shows the diagnosis is not a threshold effect.

for K=8 vs. K=64) is on that subpopulation and matches the marginal-mean difference in Table 11 (9.19$-$7.47=1.72) to within rounding. The effect sizes are identical on the independent `paper1_clean` execution ($-$0.531/+0.434), confirming they are not run-specific.

Table 13: Empty-candidate-set rate vs. required-tag threshold θ (semantic method, assist off, hazard-recovery / goal-region). The rate is *invariant* across the entire realistic range $\theta \in [0.05, 0.95]$; only the degenerate endpoints move it, and only to the no-candidate floor. The empties are therefore absence of stored evidence, not sub-threshold filtering.

θ	empty rate (hazard recovery)	empty rate (goal region)
0.00 (accept all)	0.50	—
0.05	0.533	0.267
0.15	0.533	0.267
0.25	0.533	0.267
0.50 (deployed)	0.533	0.267
0.75	0.533	0.267
0.95	0.533	0.267
1.00 (accept none)	0.75	—

The reading is decisive (Table 13). At θ=1.0 nothing passes and the empty rate rises; at θ=0.0 the required-tag gate accepts every node ($\text{conf} \geq 0$ always holds), yet the empty rate falls only to 0.50, its floor—meaning that at those decision points there is *no candidate node of any confidence* to retrieve. Across the entire sane range $\theta \in [0.05, 0.95]$ the rate is flat at 0.533: loosening or tightening the threshold does not recover a candidate, because the required semantic place was never stored. This is exactly the "absence of evidence, not contradiction" regime named in §2.3, now shown to be threshold-invariant, so the memory-accumulation attribution is not an artefact of θ. (Absolute rates here are on a small hazard-recovery / goal-region sweep and are not the headline 91% of §3.5, which is on the full benchmark; the point is the θ-invariance, not the level.)

The match/lock tolerance ϵ. The other free constant is ϵ, the tolerance with which a materialized lock counts as "within a true target" (the $R^\star/M^\star$ match and lock definition, §2.3). Sweeping $\epsilon \in \{0.3, 0.6, 1.0, 1.5\}$ on the scarcity cell (N=8, T=3, 15 seeds, peer, `experiments/big_experiment/exp_eps_sensitivity.py`) leaves the two slope signs of Empirical Claim 1 intact: $\text{Spearman}(P(M^\star|R^\star), K)$ is $-1.00, -1.00, -0.87, -0.63$ and $\text{Spearman}(P(C^\star|M^\star), K)$ is $+0.50, +0.50, +0.87, +0.87$ across the four tolerances. A looser ϵ raises absolute $P(M^\star|R^\star)$ (more locks count as on-target) but does not flip either slope. So the bottleneck shift is robust to both free thresholds of the logger: θ (candidacy/gating, above) and ϵ (match/lock).

14.6 Robustness to Assumption 1 violations

Definition 1 rests on the conditional stage-Markov property (Assumption 1). The two practically relevant violations—adaptive query rewrites not surfaced to the logger, and hidden between-stage memory updates—are exactly cases where an *unlogged latent* couples non-adjacent stages. We quantify the resulting estimator bias with a controlled simulation (4×10^5 episodes per setting; `experiments/big_experiment/assumption1_stress_test.py`). Episodes are generated with known mechanism rates $P(M^\star|R^\star)$=0.70, $P(C^\star|M^\star)$=0.65, and a hidden binary latent U that shifts *both* the M-rate and the C-rate by $\pm\delta$; the naive estimator conditions only on the previous starred stage, not on U.

Table 14: Bias of the naive conditional estimators under an unlogged latent of strength δ that couples the M and C stages (violating Assumption 1). $P(M^\star|R^\star)$ stays unbiased; $P(C^\star|M^\star)$ is biased upward, materially only under strong coupling.

δ (coupling strength)	bias in $\hat{P}(M^\star|R^\star)$	bias in $\hat{P}(C^\star|M^\star)$
0.00	-0.001	-0.002
0.05	$+0.001$	$+0.004$
0.10	$+0.001$	$+0.015$
0.15	-0.000	$+0.031$
0.20	$+0.000$	$+0.056$
0.25	-0.001	$+0.089$

Two conclusions (Table 14). The R→M conditional is unbiased at every δ: the latent re-weights episodes but does not confound that edge. The C→M conditional acquires a one-signed *upward* bias—the latent makes high-M episodes also high-C, so conditioning on M over-counts completion—but the bias is only $+1.5$ percentage points at a mild violation ($\delta \leq 0.10$) and reaches $+5.6$ points only under strong unlogged coupling (δ=0.20). The framework's exposure to Assumption 1 violations is therefore bounded and its failure direction is known, which is the relevant robustness statement for applying the logger to learned or LLM agents where such couplings are more common.

14.7 Continuous-substrate validation (VMAS): both slopes with cluster-robust CIs

The most-requested robustness check is whether the M↔C bottleneck shift is specific to grid substrates. We validate it on a continuous-state substrate (VMAS (Bettini et al., 2022)) at scale: three scarcity cells $(N, T) \in \{(6, 2), (8, 3), (10, 4)\} \times$ five cadences $K \in \{1, 2, 4, 8, 16\} \times 40$ seeds (4,800 agent-episodes; `experiments/vmas_portability/run_vmas_full.py`, executed on a SLURM CPU cluster). The symbolic peer memory and the operational Q/R/M/C definitions are reused unchanged; a thin discretisation bridge maps continuous (x, y) to coarse cells so the memory applies, and the controller acts in continuous space toward the materialized waypoint. Materialisation uses the same occupancy-sensitive lock as the grid planner.

Table 15: VMAS continuous-substrate validation of Empirical Claim 1: per-cell *event-rate ratios* $\rho_{M/R} = \#M$-events/$\#R$-events and $\rho_{C/M} = \#C$-events/$\#M$-events, and block-bootstrap effect sizes (B=2,000 resamples over the 120 (cell, seed) blocks). On this bridge a lock can fire without a preceding true-target retrieval event, so these are not nested and $\rho_{M/R}$ is an event-rate ratio, *not* a conditional probability (it can and does reach 1.00, and would in principle exceed it); the slope test is on the ratios. Both slope signs hold in *every* scarcity cell, and the bootstrap CIs exclude zero decisively: $\mathrm{Spearman}(\rho_{M/R}, K) = -0.999$, 95% CI $[-1.000, -0.975]$; $\mathrm{Spearman}(\rho_{C/M}, K) = +0.995$, 95% CI $[+0.900, +1.000]$.

(N, T)	K=1		K=4		K=16	
	$\rho_{M/R}$	$\rho_{C/M}$	$\rho_{M/R}$	$\rho_{C/M}$	$\rho_{M/R}$	$\rho_{C/M}$
$(6, 2)$	1.00	0.22	0.51	0.41	0.27	0.74
$(8, 3)$	1.00	0.18	0.45	0.41	0.28	0.64
$(10, 4)$	1.00	0.18	0.52	0.38	0.30	0.66

The earlier single-configuration check reported here as preliminary is now superseded: with three scarcity cells, 40 seeds, and the same block-bootstrap methodology as the main sweep (Appendix 14.3), both slope signs of Empirical Claim 1 transfer to continuous positions and motion with resolved effect sizes, in every cell. The freshness-vs-contention mechanism is therefore not an artefact of grid cells. Remaining limitations: the memory is fed through the discretisation bridge (continuous *dynamics* with discretised symbolic *memory*), the horizon is short (25 steps), the robots are holonomic points, and the sweep covers the coupling regime $K \leq 16$; visually rich 3D substrates (Habitat, ProcTHOR) remain future work.

14.8 External language-agent substrate: ALFWorld

To take the contract beyond our own substrates, we instrument standard ALFWorld (Shridhar et al., 2021) text games (ALFRED household tasks, `eval_out_of_distribution` split) with an LLM policy and the same four-stage contract: $Q^\star$ fires when the LLM declares the task's target object; an episodic object→receptacle memory is built from observations; $R^\star$ fires when that memory holds a candidate location for the declared target; $M^\star$ when the agent commits (`go to`) to the retrieved candidate; $C^\star$ on environment success (`experiments/alfworld_qrmc/run_alfworld_qrmc.py`; HF backend, greedy, 25 episodes, step limit 40, executed on a SLURM GPU node).

On Qwen2.5-3B-Instruct the stage taxonomy is sharply informative even at zero end-to-end success: $Q^\star$=1.00 (the model always extracts a target), $R^\star$=0.16, $M^\star$=0.00, $C^\star$=0.00, with first-missing-stage counts 21× R_fail and 4× M_fail over 25 episodes. A success-only evaluation would report "0.00" and stop; the decomposition reports *where* the 3B agent loses the task: predominantly it never grounds the declared object to a location (R), and when it does, it fails to commit to the retrieved candidate (M). This matches the near-zero ALFWorld scores reported for small models in the literature while adding the stage attribution they lack. (Instrumentation note: a first 7B run landed on an 11 GB GPU and out-of-memoried on every call; it is excluded and rerun on a 24 GB node — the cached error strings made the failure trivially auditable.) Scaling this external validation to larger models and more episodes is mechanical; the point established here is that the identical event contract instruments an external, community-standard language-agent benchmark and yields non-trivial stage attributions on real failures.

14.9 Cadence-robustness: the shift tracks scarcity, not grid design

A potential alternative explanation is that the bottleneck shift is induced by the scarce-target grid with locks. If instead it is a signature of contention over *scarce shared* targets, its slope conditions must break when targets are abundant. We rerun the peer cadence sweep in three regimes on the same env (N=8, random layout, hazard 0.05, 20 seeds; `experiments/big_experiment/exp_cadence_robustness.py`): scarcity (T=3), borderline (T=N=8), and abundance (T=14>N).

Table 16: The two slope conditions of Empirical Claim 1 (§4.3) hold under scarcity and flatten under abundance. $P(M^\star|R^\star)$ should fall and $P(C^\star|M^\star)$ should rise across K; "corr" is the within-cadence Pearson correlation of the two links at K=64 (the shift signature). $P(C^\star|M^\star)$ is an operational ratio (capped at 1; abundance completes without a formal lock).

Regime	$P(M^\star\|R^\star)$: K=1→64	$P(C^\star\|M^\star)$: K=1→64	corr(M,C) @ K=64	t_{succ} argmin
Scarcity (T=3)	0.86 → 0.49 (falls)	0.51 → 0.79 (rises)	−0.86	K=4 (interior)
Borderline (T=N=8)	0.79 → 0.72	0.77 → 0.95 (rises)	−0.79	K=8 (interior)
Abundance (T=14)	0.74 → 0.77 (flat)	1.00 → 0.99 (flat)	−0.29	K=4

Both slopes are present under scarcity, weaken at the border, and *flatten* under abundance: with more targets than agents there is no contention, so fast sharing neither over-saturates materialization ($P(M^\star|R^\star)$ stays flat at ~ 0.75 instead of spiking then falling) nor collapses completion ($P(C^\star|M^\star)$ stays at ~ 1 even at K=1 instead of dropping to 0.51). The within-cadence M↔C anticorrelation attenuates monotonically with abundance (−0.86 → −0.79 → −0.29). The shift therefore tracks the *contention mechanism* and vanishes when it is removed — it is not a fixed artefact of the grid-plus-locks design. We note that the interior t_{succ} optimum is more general (present in all three regimes): it reflects the exploration/travel cost of very fast *or* very slow sharing, of which target contention is one contributor but not the only one.

14.10 Open problems suggested by the diagnostics

Four concrete questions follow directly from the measurements of §3.5 and §4.5. **(1) Adaptive cadence.** Empirical Claim 1 assumes a fixed K; the bottleneck-shift signature suggests an online controller K_t that broadcasts when the (estimated) M-side gain exceeds the C-side occupancy cost. **(2) Selective consolidation.** Snapshot exchange merges everything; merge rules that prioritize evidence about under-represented semantic tags attack the candidate-generation starvation of §3.5 directly. **(3) Trust-weighted merging under occupancy.** The central aggregator already performs a trust-weighted merge but pays the C-collapse cost; decentralized trust rules that account for peer *intent* (not only peer evidence) could decouple freshness from target racing. *Since submission of the first draft, this problem has been addressed constructively*: a decentralized protocol in which an agent's target lock piggybacks a lightweight *reservation* on its next scheduled broadcast (no coordinator, no extra channel) and locks dominated by an earlier reservation are retracted by an anti-$M^\star$ rollback with exponential backoff. On the scarcity cells this recovers the fast-cadence completion loss on *unconditional* metrics — success rate rises at every $K \in \{1, 2, 4\}$ with disjoint bootstrap CIs, and fast sharing with the protocol exceeds the K=8 optimum baseline (0.614–0.620 vs. 0.583) — with rollback latency bounded by $\approx K$ ticks. Details in a companion manuscript; the open problem stands revised toward its harder variants (adaptive reservation scope, heterogeneous trust). **(4) Substrate generality.** The decomposition transfers across MiniGrid, BabyAI, the custom grid, and a standard-MiniGrid multi-agent wrapper with identical operational definitions and verified slope signs of Empirical Claim 1 (§ 6), and the continuous-state VMAS validation now confirms both registered slope directions across three scarcity cells with cluster-robust confidence intervals (Appendix 14.7). The decomposition is also coordinate-independent: under *semantic*, frame-free place matching (agents in private frames, places matched by meaning) the stage events remain well-defined and expose the coordinate assumption's fail-open materialization (Appendix 13.3). The remaining portability questions concern fully continuous memory representations (the bridge still discretises the store) and visually rich three-dimensional substrates such as Habitat and ProcTHOR (Savva et al., 2019; Deitke et al., 2022).

14.11 Extended related work

Classical cognitive-map perspectives motivate explicit place abstractions (Tolman, 1948; O'Keefe and Nadel, 1978; Kuipers, 2000); modern embodied AI combines mapping and planning (Gupta et al., 2017) or exposes topological memory (SPTM, Savinov et al., 2018); goal-conditioned RL conditions on explicit goals or embeddings (Andrychowicz et al., 2017; Ghosh et al., 2021), with successor-representation work providing predictive structure (Dayan, 1993; Stachenfeld et al., 2017; Momennejad et al., 2017). Our setting differs in that the target is a semantic pattern retrieved from memory rather than a coordinate. In multi-agent RL, CTDE dominates (Lowe et al., 2017; Foerster et al., 2018) and learned communication adds a per-step exchange variable (Sukhbaatar et al., 2016; Foerster et al., 2016); gossip and consensus protocols (Boyd et al., 2006; Xiao et al., 2007) are the closest network-level analogue of our cadence axis, lifted here to the cognitive-architecture level. LLM-agent systems combine reasoning/acting, reflection, explicit memory, and multi-agent conversation (Yao et al., 2023; Shinn et al., 2023; Wang et al., 2023; Park et al., 2023; Wu et al., 2023; Packer et al., 2023; Chhikara et al., 2025). Our framework is orthogonal: rather than learning a communication policy end-to-end, it *diagnoses* where in the Q/R/M/C chain a given architecture succeeds or fails, and yields a slope-level prediction about cadence that is testable on logs.

Stage-wise evaluation. The closest line decomposes retrieval-augmented generation: RAGAS scores retrieval and generation quality separately (Es et al., 2024), RAGChecker refines this to claim-level diagnostics meta-evaluated against human judgments (Ru et al., 2024), and AgentBoard replaces final success with analytic progress rates for multi-turn LLM agents (Ma et al., 2024). Q/R/M/C differs on three axes: its events are logged actions of a task-executing agent, not LLM-judged text quality; it models commitment (M) and the multi-agent coupling channels (shared memory, occupancy) that RAG metrics do not represent; and its diagnoses are validated interventionally (oracle stages; engineered structural faults) rather than by correlation with human ratings. Process mining and conformance checking compare event logs against stage models (van der Aalst, 2016; Carmona et al., 2018); the protocol shares that log-level discipline and adds agent-specific event semantics with intervention-based validation. Process reward models score intermediate reasoning steps to supervise training (Lightman et al., 2024). The stance of specifying and checking externally observable events rather than internal state is standard in runtime verification and distributed-system observability (Leucker and Schallhart, 2009); Q/R/M/C applies it to the memory interface of task-executing agents. Closest in mechanism is intervention-supported error attribution: REFLECT localizes a candidate error step in an LLM agent trace and verifies it through controlled replay with a diagnosis-specific patch (Lin et al., 2026). Q/R/M/C differs in unit and in cost: it measures recurring pipeline stages and their conditional rates from logs alone, and uses interventions (oracle stages; stage-level repairs) to validate the protocol rather than to produce each individual diagnosis.

14.12 SOTA baseline and coverage matrix

Table 18 separates direct experimental baselines from broader SOTA references. The local experiments are intentionally modest and fully instrumented: random / graph-only / SPTM-like / learned BC in MiniGrid, CommNet PPO in the multi-agent grid, and a TextNav LLM step-budget study over two local Ollama models (in which Q/R/M/C localizes a completion-limited failure, §4.5). ReAct, Reflexion, Voyager, Generative Agents, AutoGen, MemGPT, and Mem0 are not reimplemented as full systems here; they define the current memory-agent design space against which Q/R/M/C is positioned as a diagnostic protocol.

Positioning against SOTA memory designs. The memory-agent design space can be organized on two axes: single- vs. multi-agent, and how memory is represented and shared (Table 17). Vector-embedding memory (MemGPT, Voyager single-agent; Generative Agents multi-agent) retrieves by opaque nearest-neighbour lookup, so a failure is not localisable to a stage. Shared-context designs (MetaGPT, AutoGen) pool everything into one window, dissolving per-agent state. Learned-communication MARL (CommNet, and stronger CTDE methods QMIX (Rashid et al., 2018), MAPPO (Yu et al., 2022), MADDPG (Lowe et al., 2017)) optimizes a message channel end-to-end and, as we verify for CommNet PPO (§4.5), emits no separable query/lock/completion events. The cell our system occupies—per-agent *symbolic* memory with an explicit query/retrieve/materialise interface, exchanged at a tunable cadence—is empty in this matrix, and it is exactly the interface that makes the four Q/R/M/C stages observable. Our claim against SOTA is therefore not higher reward but higher *diagnosability at matched task success*: the same interface that a strong learned method trades away for end-to-end optimization is what lets the framework localize where memory helps.

Table 17: Where the framework sits in the memory-agent design space. Q/R/M/C stage-observability requires an explicit query/lock interface, which the symbolic-per-agent cell provides.

Memory representation	Single-agent	Multi-agent
Vector-embedding memory	MemGPT, Voyager (Packer et al., 2023; Wang et al., 2023)	Generative Agents (Park et al., 2023)
Shared context window	—	AutoGen (Wu et al., 2023)
Learned-comm. MARL	—	CommNet, QMIX, MAPPO, MADDPG (Sukhbaatar et al., 2016; Rashid et al., 2018; Yu et al., 2022; Lowe et al., 2017)
Symbolic per-agent + explicit interface	Q/R/M/C single-agent (§3)	**this work** (peer memory, cadence K)

14.13 Multi-agent oracle-intervention numerics

14.14 Oracle C: definition and reading

Oracle C is the completion-stage analogue of oracle R and oracle M, added to close the R/M/C triple symmetrically. The real Q/R/M pipeline runs unchanged (real memory query, real materialization), and once an agent has locked a target that is a genuine `water_source`, the oracle guarantees arrival by placing the agent on that cell; scarcity is preserved because a cell already claimed or occupied is not available, so oracle C cannot make more than T agents succeed. The runner is `experiments/big_experiment/exp_oracle_interventions.py` (mode `oracle_C`).

Two readings follow from Table 19. (i) Oracle C produces the *lowest* mean t_{succ} of the three interventions (~ 6.1–6.4 vs. ~ 7.2 for R and M), because it removes the entire post-lock travel time; this confirms that a real part of the excess time is controller travel to a correctly materialized target. (ii) Oracle C yields the *smallest* success-rate gain ($+0.01$–0.02 over baseline, below both oracle R at 0.614 and oracle M at 0.602), because in scarcity the constraint on *whether* an agent succeeds is upstream—retrieving and locking a still-available target—not reaching it. The two facts together are exactly the framework's thesis in intervention form: completion is where time is spent, retrieval/materialization is where success is won.

14.15 Stage-agnostic cadence hypotheses and what the framework changes

A first version of the multi-agent cadence sweep (12,960 runs at 20 seeds; subsumed by the 35,640-run sweep in this paper) was originally planned with four operational hypotheses formulated *before* Empirical Claim 1. They are retained here because they motivated the eventual framework. The framework was formulated after the sweep was complete; it is not claimed to have predicted the sweep numbers a priori (see "Chronology" in §4.3).

Original hypotheses (pre-framework).

H1. An interior $K^\star \in \{2, 4, 8\}$ minimising mean time-to-success exists for most (N, T, layout) cells.

H2. $K^\star$ grows monotonically with scarcity $\rho = N/T$.

H3. When $\rho \leq 1$, $K^\star = 1$.

H4. Peer-broadcast architectures Pareto-dominate centralized aggregation on (mean time-to-success, success rate).

Stand-alone result on the original 20-seed sweep. The practical claim was directionally supported: peer at the empirically best K was faster than centralized in the scarcity regime (a modest effect; Mann–Whitney p=0.044 on this small standalone sweep, reported as exploratory rather than as primary evidence, consistent with the effect-size-first line of Appendix 14.3). Among the four structural hypotheses, only H4 was clearly supported in the random-layout subset. H1 held in 7 of 18 cells; H2 had a non-significant negative correlation; H3 was supported in 4 of 9 admissible cells. The 35,640-run extended sweep used elsewhere in the paper produces matching stand-alone numbers within bootstrap uncertainty.

What the framework changes. The hypotheses H1–H4 are stage-agnostic: each ties $K^\star$ to scarcity ρ without specifying mechanism. Empirical Claim 1, formulated after the sweep, ties the slopes of $P(M^\star|R^\star)$ and $P(C^\star|M^\star)$ to cadence; the slope conditions are testable on logs. The data verify both signs as moderate, cluster-robust effect sizes (Appendix 14.3). We make no formal claim about the shape of the conditional product Φ, on which the data are monotone. The interior optimum holds on the efficiency metric mean time-to-success.

15 Assist on/off analysis

Across all four single-agent task families, top-line success is essentially unchanged under assist on/off ($\Delta \le 0.10$). The only non-trivial effect is on rest, where assists improve post-retrieval completion by $+0.05$ without changing query/retrieval rates, consistent with assists acting on materialization and post-retrieval execution.

16 Retrained retriever and CommNet baseline

16.1 Retrained candidate-generation MLP (§3.5)

Dataset extraction. From the 10-seed hazard-recovery benchmark (MiniGrid-LavaGapS7-v0, method `milestone_state_conditioned_hazard_recovery_v7`), an 18-dimensional feature vector was extracted for every step of every episode of every seed — 3 numeric agent-state scalars (energy, thirst, risk_budget), 9 semantic-tag confidences, 6 intent one-hot — and a binary label: *was this symbolic node ever selected as a candidate during the run?* Dataset totals: 814 examples, 113 positives (13.9%).

Split. By seed: train $\{2, 3, 4, 5, 6, 7\}$ (505 examples, 9.7% positive), val $\{8, 9\}$ (170, 22.4% positive), test $\{10, 11\}$ (139, 18.7% positive).

Architecture. $18 \to 32 \to 32 \to 1$, ReLU activations, BCE with pos_weight ≈ 9.3 for class imbalance, Adam lr $= 2{\cdot}10^{-3}$ with weight_decay $= 10^{-4}$, batch 64, 200 epochs, best by val loss.

Results.

The MLP underperforms majority-class on accuracy across all splits and achieves only moderate precision at fixed recall. Instantaneous state features are insufficient to predict candidate eligibility downstream. This is the negative-but-informative finding referenced in §3.5: the framework's "retrieval-limited" attribution refines to "memory-accumulation-limited" rather than "ranking-limited".

16.2 CommNet-style learned communication baseline (§4.5)

Policy architecture. Per-agent observation is 4-dim direction one-hot + 5×5×5 local cell-type window (125) + 2-dim normalised position = 131. Shared-weights encoder 131→64→64 (ReLU), 16-dim comm projection, peer mean-pool (excluding self), combine 80→64 (ReLU), actor head $64 \to 4$ (TURN_LEFT, TURN_RIGHT, FORWARD, NOOP), critic head $64 \to 1$. This is a CommNet-style learned-communication architecture (Sukhbaatar et al., 2016).

Training. REINFORCE (Williams, 1992) with value-baseline; loss $= -\,\mathrm{mean}(\log \pi(a) \cdot (R - V)) + 0.5 \cdot \mathrm{MSE}(V, R) - 0.01 \cdot H[\pi]$; Adam lr $= 3{\cdot}10^{-4}$, gradient clipping 1.0. Reward shaping: $+1.0$ on first success per agent, -0.01 step cost, -0.1 per hazard contact. 2000 episodes cycling 200 environment seeds, wall-clock 100 s on CPU. Held-out eval: 50 environment seeds 250–299.

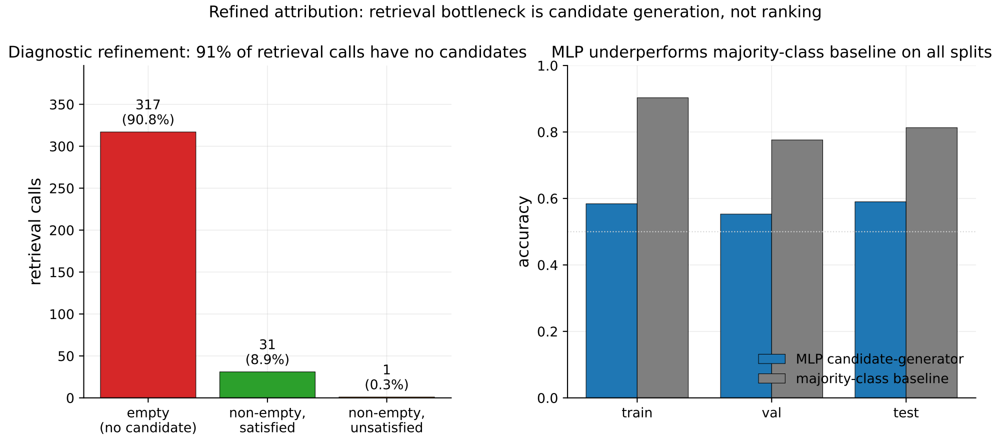

Figure 10: Refined attribution diagnostic. *Left:* 349 retrieval calls aggregated across 10 seeds; 317 (90.8%) returned empty candidate sets, 31 (8.9%) returned at least one satisfied candidate, only 1 (0.3%) returned candidates that were ranked but unsatisfied. The bottleneck is candidate generation, not candidate ranking. *Right:* the trained MLP underperforms a majority-class baseline on all three seed-disjoint splits, confirming that instantaneous state features alone are insufficient to predict candidate eligibility.

Training curve. Episode-level success (last-100 window) stays in $[0.13, 0.23]$ throughout training; no monotonic improvement after $\sim$200 episodes. The policy converges to a local optimum where one of three agents finds water by chance.

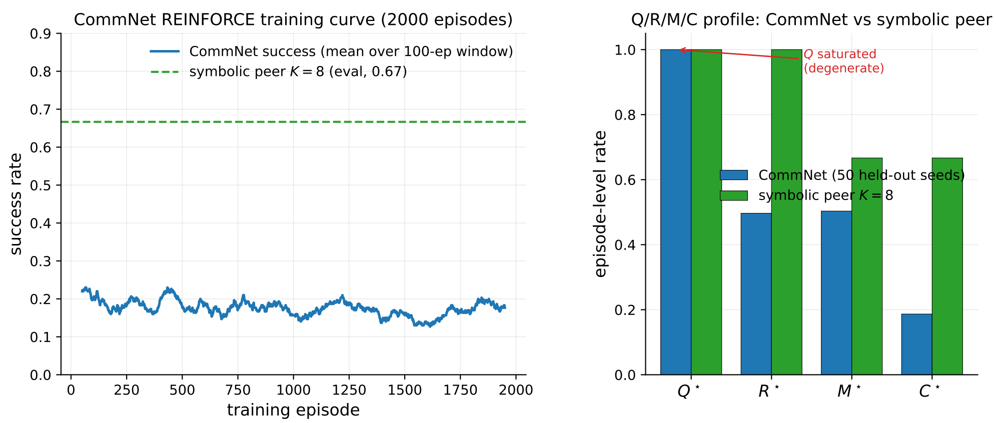

Figure 11: Learned-communication baseline diagnostics. *Left:* CommNet training curve (smoothed success rate over a 100-episode window) over 2000 training episodes; success stays below 0.25 throughout. The dashed line is the symbolic peer architecture at $K = 8$ on the same scenario for reference. *Right:* Q/R/M/C profile of the trained CommNet on 50 held-out seeds vs symbolic peer at $K = 8$. $Q^\star$ is saturated at 1.0 on CommNet (degenerate by construction); R/M collapse together; only on the symbolic architecture are the conditional rates separated.

Q/R/M/C evaluation (50 held-out seeds, stochastic action sampling).

Compare to the symbolic peer architecture at K=8: $P(M^\star|R^\star)$=0.69, $P(C^\star|M^\star)$=0.87, success ~ 0.65.

PPO variant: structural degeneracy verified at competitive success. A PPO-trained (Schulman et al., 2017) variant of the same CommNet architecture (clip ϵ=0.2, generalized advantage estimation (Schulman et al., 2016) with λ=0.95, γ=0.99, 4 PPO epochs per rollout, 64 rollouts per update, 200 updates, $\approx$ 96 minutes on a single CPU core; full setup in experiments/commnet_ppo_baseline/train_ppo_commnet.py) reaches 0.667 success on 100 held-out seeds — competitive with the symbolic peer architecture at $K{=}8$ (0.65).

The PPO result converts the structural argument from an assertion ("training budget would not change this") to a measurement: $Q^\star$ remains saturated at 1.00, $P(M^\star|R^\star)$ remains 1.00 (R/M collapse), even at competitive success. The Q stage is degenerate by construction in any policy whose communication channel is a real-valued vector with a learned linear projection — the channel is never informatively empty, regardless of training budget — and without an explicit symbolic target-lock interface the M event cannot separate from R. Only architectures with an explicit *query/lock* interface (the symbolic stack of §4.1) expose stage-separable Q/R/M/C events.

MAPPO: the collapse persists on a stronger CTDE baseline. To rule out that the observability collapse is an artifact of CommNet's capacity, we train MAPPO (Yu et al., 2022) — PPO with decentralized actors and a *centralized critic* $V(s)$ over the concatenation of all agents' observations (CTDE) — on the same scarcity scenario (experiments/mappo_baseline/train_mappo.py; 300 updates $\times$ 64 rollouts, ≈ 8.5 h on a cluster CPU node) and evaluate with the identical Q/R/M/C logger on 50 held-out seeds. MAPPO reaches 0.667 success — competitive with the symbolic peer architecture at this cell — yet its stage profile is structurally degenerate in exactly the CommNet way: $Q^\star$ saturated at 1.00, $P(R^\star|Q^\star){=}0.98$, $P(M^\star|R^\star){=}1.00$ (R and M collapse into one event), $P(C^\star|M^\star){=}0.69$. The centralized critic changes the training signal, not the interface: without an explicit query/lock boundary there is nothing for the M event to separate from R. This upgrades the claim from "CommNet collapses" to "a modern CTDE baseline at competitive success collapses," which is the structural point.

17 Reproducibility commands

An anonymized public repository accompanies this submission. The main paper artifacts were generated from the following scripts; PYTHONPATH=. and a Python 3 virtual environment with numpy, scipy, pandas, matplotlib, imageio, gymnasium, and minigrid are assumed throughout.

Computing infrastructure. The work used two kinds of hardware, split by workload. The main grid sweeps (single- and multi-agent, oracle interventions, CommNet-PPO baseline) were executed on an Apple Silicon workstation (10 CPU cores via ProcessPoolExecutor, macOS Darwin 25.4, Python 3.9.6); the full multi-agent block reproduces there in $\approx$22 minutes. Local Ollama LLM experiments (llama3.1:latest 8B, qwen3:4b, temperature 0, deterministic on-disk caching) and the third-party-framework study (openai 2.44, langgraph 0.6, langchain-ollama, pyautogen 0.9) also ran on the workstation. The HuggingFace transformer experiments (Qwen2.5-3B/7B) and ALFWorld used SLURM *GPU* nodes; the enlarged VMAS and 30-seed robustness sweeps used SLURM *CPU* nodes. Key library versions: numpy 1.x, scipy, torch (CommNet-PPO baseline and VMAS), vmas 1.5.2, minigrid/gymnasium. No paid API was used; each LLM study completes in minutes-to-hours of inference on its node.

One-command reproduction (multi-agent block, ≈ 22 minutes on 8 cores)

```
bashscripts/reproduce_paper.sh
```

Clean Paper 1 suite

- smoke suite (all local non-LLM blocks): pythonscripts/run_paper1_clean_experiments.py--modesmoke--skip_llm--out_dirtmp/paper1_clean_experiments_smoke
- full local suite (single-agent, oracle, multi-agent, CommNet, portability/noise): pythonscripts/run_paper1_clean_experiments.py--modefull--skip_llm--workers8--out_dirtmp/paper1_clean_experiments_full
- LLM TextNav step-budget sweep (requires local Ollama), run once per budget $K' \in \{6, 12, 25\}$: python-mexperiments.llm_collective.run_model_sweep--modelsqwen3:4bllama3.1:latest--episodes8--step_limit<K'>--out_dirtmp/llm_steplimit_<K'>

Single-agent (§3.2–§3.4)

- article benchmark and learned-baseline suite: `scripts/benchmark_symbolic_memory_article.py`
- article analysis and figure generation: `scripts/analyze_symbolic_memory_article.py`
- synthetic factorization sanity check: `scripts/validate_qrmc_factorization.py`
- oracle-stage intervention benchmark: `scripts/benchmark_oracle_stage_interventions.py`
- BabyAI transfer benchmark: `scripts/compare_semnav_babyai.py`

Multi-agent (§4.5)

- Base batch — peer at $K \in \{1, 2, 4, 8, 16\}$ *and* the three cadence-free architectures (independent, shared bus, central aggregator), 40 seeds (25,920 runs $=$ 16,200 peer $+$ 9,720 cadence-free), $\approx$ 18 min: `pythonexperiments/big_experiment/exp_cadence_phase.py--modefull--workers8--out_dirtmp/big_experiment_qrmc_40`
- Extended-cadence batch — peer at $K \in \{32, 48, 64\}$ only (9,720 runs), $\approx$ 4 min: `pythonexperiments/big_experiment/exp_extra_K.py--workers8--out_dirtmp/big_experiment_extraK`

  Note. These two batches partition the 35,640 total by *cadence range*; the architecture-wise partition of Appendix 14.2 (25,920 peer over all eight cadences vs. 9,720 cadence-free) reuses the same two totals for *different* subsets (a coincidence of 3 extended cadences $\times$ 3,240 $=$ 3 cadence-free architectures $\times$ 3,240). The combined log is `tmp/paper1_clean_experiments_full/multiagent_cadence_full/runs.csv`.
- Q/R/M/C stage analysis and bootstrap CIs: `pythonexperiments/big_experiment/analyze_qrmc.py--runs_csvtmp/big_experiment_qrmc_40/runs.csv--out_dirtmp/big_experiment_qrmc_40`
- Multi-agent oracle interventions: `pythonexperiments/big_experiment/exp_oracle_interventions.py--workers8--out_dirtmp/big_experiment_oracle`
- Phase 1–4 smoke tests: `pythonscripts/run_all_smokes.py--out_dirtmp/all_smokes`
- PPO-trained CommNet baseline (Table 20, $\approx$ 96 min on a single CPU core): `python-mexperiments.commnet_ppo_baseline.train_ppo_commnet--total_updates200--rollouts_per_update64`
- Q/R/M/C eval of the PPO policy on 100 held-out seeds: `pythonexperiments/commnet_baseline/eval_with_qrmc.py--policy_pathexperiments/commnet_ppo_baseline/ppo_policy.pt--out_direxperiments/commnet_ppo_baseline--n_episodes100`
- MiniGrid-substrate portability sweep (§6, 100 runs, $\approx$ 2 min): `python-mexperiments.minigrid_multiagent_wrapper.run_portability_sweep--out_dirtmp/minigrid_portability`

18 Asset and license note

The experiments use MiniGrid (Chevalier-Boisvert et al., 2023) and BabyAI (Chevalier-Boisvert et al., 2018), both released under the MIT License. The locally implemented baselines and benchmark runner scripts introduced in this paper will be released under the MIT License as part of the camera-ready artifact package.

Table 18: SOTA coverage for Paper 1. "Instrumented" means the run emits the operational $Q^\star, R^\star, M^\star, C^\star$ events used by Definition 1.

Family	Representative method	Status in this paper	Q/R/M/C comparison point
Topological navigation memory	SPTM-like graph baseline (Savinov et al., 2018)	Direct MiniGrid baseline	Tests whether topological memory without semantic query/lock exposes non-trivial stage events.
Behavior cloning	Learned grid-observation BC	Direct MiniGrid baseline	Strong success on easy tasks but no explicit query/lock interface, so diagnostics are structurally limited.
Learned communication	CommNet PPO (Sukhbaatar et al., 2016; Schulman et al., 2017)	Direct multi-agent baseline	Competitive success; $Q^\star$ saturates and R/M collapse, verifying observability loss.
Cooperative MARL (CTDE / value decomposition)	MAPPO (Yu et al., 2022) (trained); QMIX, MADDPG (Rashid et al., 2018; Lowe et al., 2017) (references)	**Direct multi-agent baseline**	A trained MAPPO (centralized critic) reaches 0.667 held-out success on the scarcity scenario and shows the same structural signature as CommNet: $Q^\star$ saturated at 1.00 and $P(M^\star\|R^\star)$=1.00 (R/M collapsed) — observability loss is not an artifact of a weak baseline (Appendix 16).
Reasoning/acting LLM agents	ReAct (Yao et al., 2023)	Primary-source reference	Q/R/M/C can instrument the resulting action traces if queries, retrieved evidence, target locks, and completions are logged.
Reflective LLM agents	Reflexion (Shinn et al., 2023)	Primary-source reference	Reflection can be treated as upstream Q/R revision; the framework asks whether it improves R, M, or C.
Embodied LLM agents	Voyager (Wang et al., 2023)	Primary-source reference	Skill/memory retrieval maps naturally onto R and M, but full reproduction is outside this paper.
Social/memory simulations	Generative Agents (Park et al., 2023)	Primary-source reference	Their observation/memory/reflection loop motivates logging stage-specific memory use rather than only believability.
Multi-agent LLM frameworks	AutoGen (Wu et al., 2023)	Primary-source reference	Conversation traces provide candidate Q/R/M/C events; this paper supplies the measurement contract.
Long-term LLM memory systems	MemGPT / Mem0 (Packer et al., 2023; Chhikara et al., 2025)	Primary-source reference	Persistent memory systems can be evaluated by where they fail: retrieval, materialization, or completion.
Local LLM substrate	qwen3:4b, llama3.1	Direct TextNav step-budget study	Q/R/M/C localizes a completion-limited failure ($C^\star \to 0$ while $Q/R/M$ stay saturated) in real LLM agents; see Table 4.

Table 19: Multi-agent oracle interventions on the scarcity cases at three cadences. Mean time-to-success (and success rate) averaged across $(N,T) \in \{(3,2),(5,3),(8,5)\}$, two layouts (asymmetric, random), 15 seeds. Oracle R returns ground-truth water positions; oracle M locks the nearest real water; **oracle C** guarantees completion of a real materialized lock (Appendix 14.14). Oracle C gives the largest *time* reduction (it removes travel) but the smallest *success* gain — completion is a time cost, not the success bottleneck, which sits upstream at R/M.

K	Baseline	Oracle R	Oracle M	Oracle C
mean t_{succ}				
1	10.33	7.22	7.21	**6.19**
4	10.98	7.22	7.21	**6.07**
16	8.45	7.22	7.21	**6.41**
success rate				
1	0.591	**0.614**	0.602	0.591
4	0.575	**0.614**	0.602	0.600
16	0.578	**0.614**	0.602	0.598

Metric	Train	Val	Test
Accuracy	0.584	0.553	0.590
Majority-class baseline accuracy	0.903	0.776	0.813
Precision @ recall = 0.70	0.171	0.278	0.275

Quantity	Value
$Q^\star$ rate	**1.00** (saturated)
$R^\star$ rate	0.50
$M^\star$ rate	0.50 (collapses onto R)
$C^\star$ rate	0.19
$P(R \mid Q)$	0.50
$P(M \mid R)$	1.00
$P(C \mid M)$	0.42
Success rate	0.19
Mean steps	80 (budget exhausted)

Table 20: REINFORCE vs PPO CommNet vs symbolic peer (K=8) on the scarcity scenario (N=3, T=2, asymmetric, hazard 0.05). PPO is competitive on success rate, yet still exhibits Q-saturation and R/M collapse — the diagnostic claim verified, not asserted.

Metric	CommNet REINFORCE	**CommNet PPO**	Symbolic peer (K=8)
Success rate	0.19	**0.667**	0.65
$Q^\star$ rate	1.00	**1.00**	0.99
$R^\star$ rate	0.50	**0.997**	0.69
$M^\star$ rate	0.50	**0.997**	0.69
$P(M^\star \vert R^\star)$	1.00	**1.00**	0.69
$P(C^\star \vert M^\star)$	0.42	**0.67**	0.87
R/M separated?	no	**no**	**yes**

Part II

Semantic Warp: Provenance-Conditioned Completion

Navigating by someone else's memory as a measured event: the foreignness annotation, the warp distance law, and the Warp Drive coordination protocol.

Part abstract. Collective-memory systems allow an agent to act on evidence that was acquired by another agent, yet conventional task and stage metrics do not identify when a commitment is supported primarily by foreign memory. We introduce a provenance-conditioned extension of Q/R/M/C. For each materialized target, the method records the foreign share φ of the evidence mass used by the memory merge and freezes this value at commitment time. A semantic-warp event is then defined as a materialization whose supporting evidence exceeds a specified foreignness threshold. This annotation permits three analyses: direct completion stratified by provenance, deterministic counterfactual replay with foreign evidence removed, and a closed-form admissibility horizon derived from the trust–staleness gate. On a 2,400-episode scarcity benchmark, foreign-evidence locks complete substantially less often than own-evidence locks (0.004–0.020 versus 0.22–0.41), but this direct comparison understates their value. Lock-chain analysis shows that foreign evidence frequently transports an agent into self-confirmation range, accounting for up to 51% of subsequent own-evidence completions. The trust–staleness law predicts the latest admissible intention time exactly on the registered $6/6$ hold-out. The same analysis identifies target contention, rather than incorrect evidence, as the principal completion cost of fast sharing. We therefore introduce a decentralized reservation and rollback protocol that increases success at every tested fast cadence and exceeds the strongest fixed-cadence baseline (0.614–0.620 versus 0.583). A controlled LLM experiment further demonstrates cross-session use of foreign memory. These results indicate that the value of shared memory depends on the provenance, temporal validity, and coordination authority attached to a record, not only on its semantic content. The scope is limited to controlled grid and continuous simulations and a small text-world demonstration.

19 Introduction

A shared memory architecture is usually described in terms of the information available to each agent. This description is incomplete. Two agents may retrieve the same target record, but the epistemic basis of their commitments can be different: one may rely on direct observation, whereas the other may rely entirely on a peer's report. End-to-end success, retrieval accuracy, and the original Q/R/M/C rates do not distinguish these cases because they do not condition materialization and completion on evidence provenance.

This part introduces a provenance-conditioned measurement layer for collective semantic memory. Every contribution to a merged record retains its source, support, confidence, trust, and age. For a selected candidate, the method computes the foreign share φ using the same weights that produced the merge. The value is frozen when the agent commits to the target so that later self-confirmation does not retroactively change the provenance of the decision. A semantic-warp event is therefore not an additional stage, but a stratification of materialization and completion according to the source of the supporting evidence.

The analysis separates three questions. First, how often do foreign-evidence commitments complete directly? Second, what is the counterfactual effect of foreign memory when its contribution is removed under an otherwise identical replay? Third, for how long may foreign evidence remain action-authoritative under explicit trust and staleness rules? The symbolic memory representation makes all three questions operational: the merge is source-decomposable, the simulator supports paired deterministic replays, and the exponential staleness gate yields a closed-form horizon.

The results show that direct completion is only one component of foreign-memory value. Under target scarcity, foreign locks rarely complete because several agents can commit to the same resource. Nevertheless, the same lock often guides the recipient to a region in which it can obtain its own confirming evidence. We therefore treat foreign memory as a transport mechanism as well as a destination proposal. This distinction explains why an architecture can receive a positive counterfactual benefit from foreign evidence even when $P(C^\star \mid W^\star)$ is small.

The observed failure is primarily coordinative. Fast exchange increases the number of commitments based on the same peer evidence, which improves materialization but increases duplicate target claims. To address this mechanism, we introduce a minimal decentralized protocol consisting of optimistic commitment, reservation messages, and rollback with backoff. The protocol does not alter the semantic content of the memory; it regulates when a shared record is permitted to determine action.

The contributions of this part are as follows:

1. We define a provenance-conditioned annotation of materialized targets and completion events, together with a deterministic counterfactual measure of the contribution of foreign evidence.
2. We derive a trust–staleness admissibility horizon and validate its discrete predictions on a registered hold-out.
3. We identify downstream transport through mixed-provenance lock chains as a major component of foreign-memory value.
4. We introduce and evaluate a decentralized reservation and rollback protocol that converts fast sharing from a contention-prone setting into the strongest tested communication regime.
5. We demonstrate that the same provenance concept can be applied to a controlled cross-session LLM memory task.

The claims are limited to the tested memory rules and environments. The distance law is exact under the specified exponential gate; it is not asserted for arbitrary learned memory policies. Similarly, the LLM study demonstrates measurability and cross-session transfer, but does not constitute a broad benchmark of hallucination-resistant agent collectives.

20 Setting

Substrate and task. N agents explore a 12×10 grid with M semantic water targets ($N > M$: scarcity) and hazards. Each agent maintains symbolic place memory (tags, confidences, support, freshness). Four sharing architectures over the identical substrate and planner: `independent` (no communication), `shared` (single event bus), `centralized` (consensus aggregator), `peer` (broadcast every K ticks, asymmetric trust), plus `CSM` (peer + explicit merge/trust/staleness rules with decay rate α=0.05 and inclusion threshold τ=0.30).

Q/R/M/C instrumentation. Per tick and per agent we log query formation (Q), retrieval satisfaction (R), target materialization as the planner's lock ($M^\star$), and completion ($C^\star$). Episode-level conditional rates follow the companion definition. All warp measurements below are annotations of these $M^\star$ events; no planner or memory invariant is modified.

21 The warp event

Provenance and foreignness. Every evidence unit entering a merge carries its source. For the candidate m the planner locks,

$$\varphi(m) \;=\; \frac{\sum_{e\in\mathrm{supp}(m),\,\pi(e)\neq\mathrm{self}} w(e)}{\sum_{e\in\mathrm{supp}(m)} w(e)},$$

where $w(e)$ are *the merge's own weights*: trust $\cdot$ $\log(1+\text{support})$ for peer/centralized consensus, trust $\cdot$ $e^{-\alpha\,\text{age}}$ $\cdot$ conf for CSM, observation masses for the shared bus. For peer and centralized architectures φ is reconstructed exactly from the per-contribution fields the merge already exports; the measurement layer reads state and never writes it.

Freezing rule. φ is fixed at the moment of the lock. If the agent later confirms the place with its own eyes, φ is *not* recomputed — otherwise every successful warp would dilute itself on approach and $P(C^\star|W^\star)$ would be unmeasurable by construction.

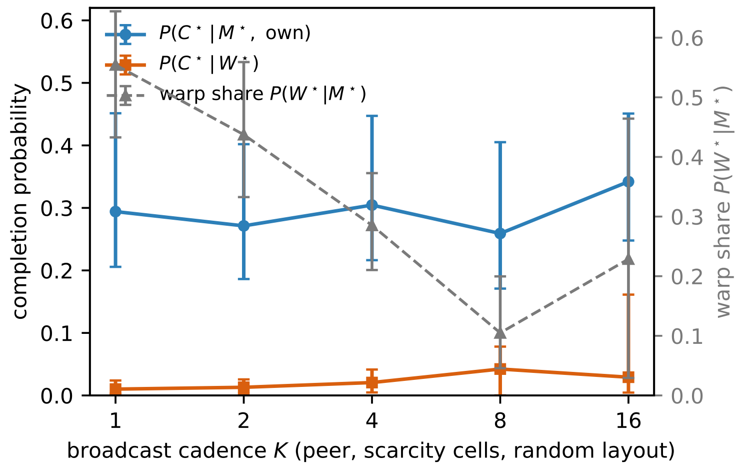

Figure 12: φ-stratified completion vs. broadcast cadence K (dedicated peer sweep: scarcity cells, random layouts, 20 seeds, episode-cluster bootstrap CIs). As K shrinks, the warp share $P(W^\star|M^\star)$ grows from 0.11 to 0.55 while $P(C^\star|W^\star)$ stays below 0.05 and the own-stratum rate stays flat: the completion collapse of fast sharing is concentrated in the $W^\star$ stratum, giving the companion paper's bottleneck shift its mechanistic reading. The share uptick at K=16 (broadcasts are rare, but agents also discover little on their own) is reported as observed.

Warp events and strata. $W^\star_\theta := M^\star \wedge (\varphi \geq \theta)$ with θ=1 (strict: never self-observed, cross-checked against an observation ledger) and θ=0.8 (soft). Warp share $P(W^\star|M^\star)$ is a structural descriptor (identically 0 for `independent`); warp completion $P(C^\star|W^\star)$ is the new quantity. The chain-rule decomposition in §19 makes warp an exact refinement of $P(C^\star|M^\star)$, not a new stage: identifiability of Q/R/M/C is untouched.

Regression anchor. The known misled-agent episode of the architecture comparison (agent A locks $(3, 8)$ known only from C; C claims it first; A strands) is annotated by the instrumentation as $\varphi = 1.0$ strict $W^\star$ with collision — the pathology the framework must see, and the protocol of §24 must fix.

22 Direct and downstream effects of foreign evidence

Design. 2,400 episodes: $(N, M) \in \{(3, 2), (5, 3), (8, 5), (4, 4), (6, 2)\}$ × three layouts × {peer-K2, peer-K8, CSM-K8, shared} × 20 seeds × {full, foreign-masked} with identical seeds. Unit of analysis: lock event; CIs by episode-cluster bootstrap.

Figure 12 shows the cadence view on a dedicated peer sweep ($K \in \{1, 2, 4, 8, 16\}$): warp share is what fast sharing buys, and it is precisely the stratum that does not complete. Table 21 gives the architecture view.

Table 21: Provenance strata of completion under scarcity ($N > M$), full runs. Warp locks complete an order of magnitude less often than own-evidence locks; all CI pairs are disjoint. (H-W1 confirmed.)

Architecture	$P(C^\star\|W^\star_{0.8})$	$n_{W^\star}$	$P(C^\star\|M^\star, \text{own})$	n_{own}
peer-K=2	0.004 [0.001, 0.008]	1,437	0.223 [0.186, 0.275]	2,738
peer-K=8	0.020 [0.000, 0.038]	149	0.389 [0.329, 0.446]	1,734
CSM-K=8	0.017 [0.006, 0.044]	637	0.414 [0.368, 0.453]	1,684
shared	0.004 [0.002, 0.008]	2,099	0.279 [0.237, 0.332]	2,250

22.1 Downstream transport through lock chains

Direct completion understates the value of foreign evidence. A *warp-assisted own completion* is a completed own-evidence lock preceded (same agent, same episode) by a soft-$W^\star$ lock that was dropped: the warp failed as a destination but succeeded as a ride.

Table 22: Lock-chain analysis (random layouts, full runs). The first $W^\star$ lock in a chain is made from $\approx$8–10 cells away; the completing own lock from 2.0 cells — exactly the observation radius, i.e. the self-confirmation point.

Arm	own $C^\star$	warp-assisted (share)	$r(\text{first } W^\star) \to r(\text{own})$
shared	248	127 (**51.2%**)	$9.5 \to 2.0$
peer-K=2	223	99 (**44.4%**)	$9.7 \to 2.0$
CSM-K=8	279	13 (4.7%)	$5.8 \to 2.0$
peer-K=8	246	7 (2.8%)	$8.4 \to 2.0$

This resolves an apparent paradox in the counterfactual data: the shared bus at moderate scarcity gains $+10.8$ ticks $[+2.2, +20.9]$ from foreign evidence although its direct $P(C^\star|W^\star)$ is 0.004. The gain flows through chains; the freezing rule books the completion in the own-evidence stratum, and the chain analysis recovers the warp's contribution as a separate measured quantity.

22.2 Counterfactual warp gain

Masking foreign evidence at merge under the same seed gives $\mathrm{WG} = t_{\text{cens}}(\text{masked}) - t_{\text{cens}}(\text{full})$ per episode. The sign map over (layout, scarcity $\rho{=}N/M$) is regime-dependent: positive for the shared bus at unpredictable layouts and moderate scarcity ($+10.8$ at ρ=1.5), negative for fast peer sharing (peer-$K2$ random: -7.1 $[-13.5, -1.3]$), $\approx$0 at hard scarcity. A per-agent mask (one agent masked, others share normally) confirms the sign at the individual level: $\mathrm{WG}_i = -10.25$ $[-20.91, -0.16]$ ticks on the fast-share scarcity cell, with spillover on unmasked agents $\approx$0 (-0.62 $[-3.06, +1.91]$). Raw warp *hurts* its recipient precisely where sharing is fastest — the collision cost exceeds the information value.

Exploratory: collision pressure. Co-lock pressure (other agents locked on the same target) correlates with warp failure with the predicted sign but negligible magnitude (Spearman $+0.033$, p=0.018, n=5,117); we report it as exploratory rather than as a confirmed predictor.

23 The warp distance law

CSM includes foreign evidence with weight $\tau_i\, e^{-\alpha\,\text{age}}\, c$ (trust τ_i, confidence c=0.95) and inclusion threshold τ=0.30. Foreign evidence therefore stops forming intention at

$$\text{age}_{\max}(\tau_i) \;=\; \tfrac{1}{\alpha}\,\ln\!\big(\tau_i\, c/\tau\big), \qquad \text{defined for } \tau_i c > \tau.$$

Witness–traveller protocol. A witness observes water once and goes silent; a traveller who has never seen it receives the snapshot at a parameterised age. Every lock the traveller makes is a strict $W^\star$ by construction. Figure 13 summarises both experiments.

Table 23: Gate curve: largest evidence age at which the traveller's query still exposes the target. Fixed-cadence peer memory has no staleness term: the gate never closes (checked to age 120 at all trusts) — the architectures differ in the *shape* of the curve, not just its level.

trust τ_i	empirical $\text{age}_{\max}$	predicted
1.00	23	23.05
0.80	18	18.59
0.60	12	12.84
0.40	4	4.73
0.25	gate closed	0 (trust-flip)

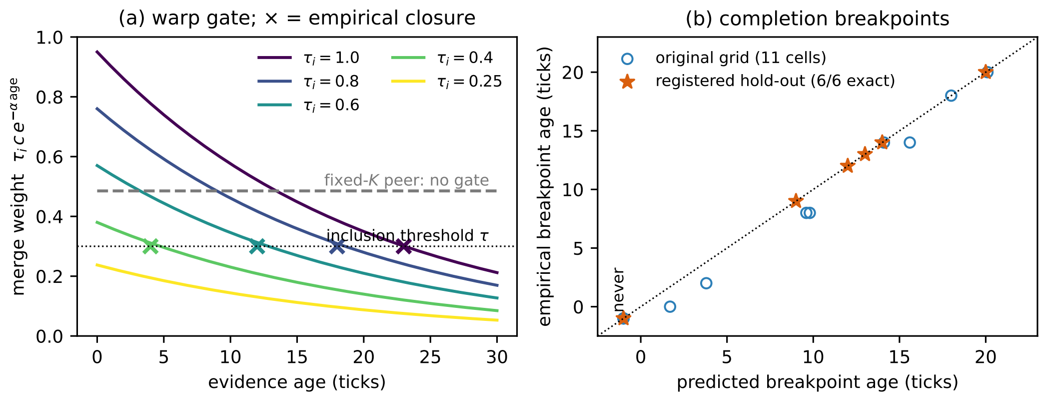

Figure 13: The warp distance law. **(a)** Foreign-evidence merge weight under trust×staleness decay; × marks the empirically measured last age at which the traveller's query still exposes the target — each sits on its curve's crossing of the inclusion threshold τ. Fixed-cadence peer memory has no staleness term (dashed): its gate never closes; the architectures differ in the *shape* of the curve. **(b)** Predicted vs. empirical completion breakpoints in full navigation episodes: original grid (circles; age sampled in steps of 2, hence points at most one grid step below the diagonal) and the registered hold-out (stars; age step 1, predictions written to disk before execution, $6/6$ exact, including the never-cell at $(-1, -1)$ and the cell where the pruning horizon undercuts the exponential gate).

Completion breakpoint. In full navigation episodes the feasibility inequality acquires one discrete refinement: the traveller self-confirms the target once it is within the observation radius, so the foreign gate only has to survive until tick $t_{\text{gate}} = d - r_{\text{obs}} - 1$, giving breakpoint $\text{bp} = \text{age}_{\max} - t_{\text{gate}}$; snapshot pruning executes on broadcast ticks and adds the bound $10K - t_{\text{bcast}}$. All constants of the law are unchanged. All 11 CSM cells of the original grid match within sampling resolution; completion curves are monotone steps; the trust-flip zeroes $W^\star$ events discretely.

Table 24: Registered-prediction hold-out: six cells absent from the original grid, predictions written to disk *before* any episode, age grid step 1. All six are exact integer matches, including the never-cell and the cell where the pruning horizon must undercut the exponential gate.

Cell	binding bound	predicted	empirical
K=8, τ_i=0.9, d=9	gate	14	**14**
K=8, τ_i=0.7, d=9	gate	9	**9**
K=8, τ_i=0.5, d=15	gate	never	**never**
K=4, τ_i=1.0, d=6	gate	20	**20**
K=2, τ_i=1.0, d=12	*pruning*	12	**12**
K=2, τ_i=0.8, d=8	gate	13	**13**

The law therefore predicts the held-out horizons exactly at tick resolution: the CSM rules really are the mechanism that forms (and un-forms) intention. It also reinterprets the companion result that CSM dominates fixed-cadence peers: trust×staleness is a *warp gate* — CSM does not share better, it *refuses to warp on a stale or untrusted song*.

Portability to continuous dynamics. All three registered predictions survive on a continuous VMAS substrate through the same continuous→grid bridge used by the companion's portability test: (i) the provenance strata stay an order of magnitude apart ($P(C^\star|W^\star) = 0.012$ $[0.000, 0.042]$ vs. 0.562 $[0.457, 0.709]$ at K=1); (ii) warp share falls monotonically with cadence ($0.723 \to 0.366 \to 0.000$ over K=$1/4/16$); (iii) the law becomes *metric*: with the travel-to-sight time calibrated once on a fresh-evidence episode, the predicted breakpoints match within the sampling grid, and all cells beyond the metric warp radius ($v \cdot \text{age}_{\max}$ shorter than the distance) are predicted *never* and observed *never*. One genuinely continuous finding: inertia is a physical warp-leak channel — an agent can coast into sighting range after the gate closes — removed by a no-lock-then-brake controller that restores the grid semantics.

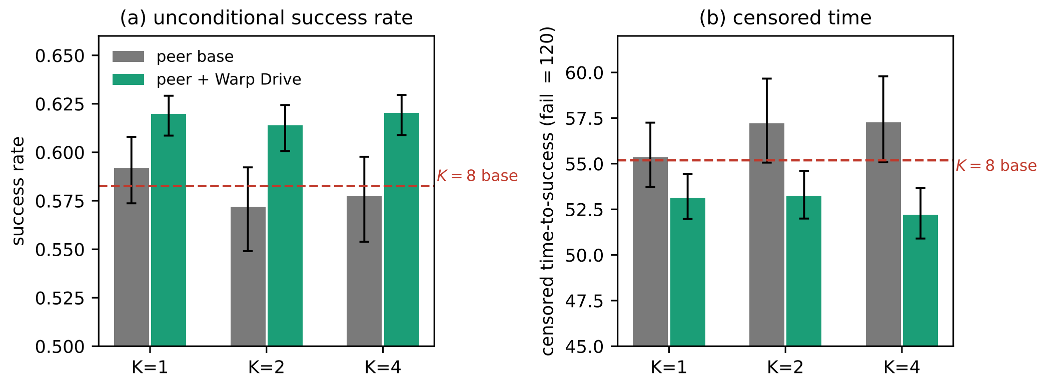

Figure 14: Warp Drive on unconditional metrics (840 episodes; bootstrap CIs). **(a)** Success rate rises at every cadence with disjoint CIs, and every fast-K protocol arm exceeds the $K=8$ sweet-spot baseline (dashed). **(b)** Censored time-to-success falls correspondingly. The claim that the protocol removes the main argument against fast sharing rests on these unconditional quantities, not on the (denominator-sensitive) conditional rates.

24 The Warp Drive protocol

Warp collision — locking on foreign evidence for a target someone else will claim — is the dominant failure of fast sharing. Warp Drive adds three primitives, all decentralized:

1. **Optimistic warp lock** — locks on foreign evidence remain allowed immediately (planner unchanged).
2. **Warp reservation** — at the owner's next broadcast tick the reservation rides on the ordinary snapshot (same cadence K, no coordinator, no extra channel); receivers suppress reserved targets in their queries. Priority: (lock tick, distance at lock, agent id).
3. **Anti-$M^\star$ rollback** (Time Warp semantics Jefferson (1985)) — an agent holding a lock dominated by a foreign reservation retracts it: the planner re-queries, the target enters exponential backoff ($4 \to 8 \to \cdots \to 32$ ticks) with an agent-id tie-break against oscillation. The event log stays append-only.

Decentralisation check. Driving the protocol by direct per-agent calls — no runtime, no environment — produces the same retraction decisions from each agent's own inbox alone.

Misled-A anchor. With Warp Drive the pathological lock is resolved *preventively*: C's reservation arrives before A ever locks $(3, 8)$; A completes on another target. In the sweep, rollback latency of dominated locks is finite and $\approx K$: $1.0/1.5/2.7$ ticks at $K=1/2/4$ (baseline: until end of episode).

Table 25: Unconditional results, 840 episodes (scarcity cells $\times$ random/asymmetric layouts $\times$ 20 seeds). Success rate rises under Warp Drive at every cadence with disjoint CIs, censored time falls, and fast sharing with the protocol beats the $K=8$ sweet-spot baseline (Figure 14). Lock census shows the conditional-metric denominator shrinking — which is why the claim is stated on unconditional quantities.

Arm	success rate	t_{cens} (fail $=120$)	lock census
$K=1$ base	0.592 [0.574, 0.608]	55.3 [53.7, 57.2]	0.986
$K=1$ +WD	**0.620** [0.609, 0.629]	53.1 [52.0, 54.4]	0.985
$K=2$ base	0.572 [0.549, 0.592]	57.2 [55.1, 59.7]	0.973
$K=2$ +WD	**0.614** [0.601, 0.624]	**53.2** [52.0, 54.6]	0.926
$K=4$ base	0.577 [0.554, 0.598]	57.3 [55.1, 59.8]	0.944
$K=4$ +WD	**0.620** [0.609, 0.630]	**52.2** [50.9, 53.7]	0.739
$K=8$ base	0.583 [0.558, 0.604]	55.2 [52.9, 57.9]	0.682

On the conditional axis, recovery is measured against the scarcity ceiling $\min(m, M)/m$ (no protocol conjures extra water): $0.90/2.15/6.6$ of the recoverable collapse at $K{=}1/2/4$; values above 1 arise because reservation suppression removes doomed locks from the $M^\star$ population. Together with Table 22, this completes the arc: raw warp fails at completion (§22); with a coordination contract attached, warp at fast cadence becomes the best regime. The cadence sweet spot of the companion paper is reinterpreted: moderate K was never about information freshness — it was an implicit rate limiter on warp collisions, and an explicit contract does the job better.

25 Cross-session warp on an LLM substrate

Finally, the strict warp crosses both an agent boundary and a session boundary on a language substrate. In session 1, agent A (a local LLM tag extractor over natural-language observations; Ollama `llama3.1`, deterministic caching, zero API cost) sweeps a three-room text household and its symbolic memory is dumped to disk. In session 2, agent B — a fresh agent in a fresh environment instance, with the stock directional oracle removed so that long-range knowledge can only come from memory — loads A's dump as foreign provenance and solves "bring the apple to the table." Every lock B makes on the apple or the table is $\varphi{=}1.0$ strict $W^\star$; execution is layered (the LLM makes the semantic decisions — which remembered place to pursue, when to take/put — while locomotion to the locked waypoint is a deterministic motor primitive, matching the substrate's planner stack).

Table 26: Cross-session warp over 10 randomised layouts. Completion is attributed at the environment's affordance radius (take/put act within Manhattan distance 1, and the witness tags anchors on within-reach cells).

	success	mean t_{succ}	layouts with completed strict $W^\star$
warp (A's memory)	**10/10**	8.3	**10/10**
control (empty memory)	0/10	—	—

To our knowledge this is the first documented cross-session, cross-agent strict semantic warp on an LLM substrate. The $10/10$ versus $0/10$ contrast shows that foreign memory is necessary for completion in this controlled task.

Scaling to a live collective. With $N{=}3$ LLM agents sharing by peer broadcast under scarcity ($M{=}2$ apples, four layouts, $K \in \{2, 8\}$), the full warp phenomenology reproduces on the language substrate (all predictions registered before the runs): strict $W^\star$ locks occur; warp share falls with cadence (0.417 at $K{=}2$ vs. 0.083 at $K{=}8$ — the same shape as the grid $0.554 \to 0.105$ and VMAS $0.723 \to 0.000$); warp collisions appear in $7/8$ episodes. A paired comparison against the no-communication baseline on identical layouts shows raw sharing *lowering* success from $2/3$ to $1/3$ per episode — the collision cost exceeds the information value on the LLM substrate exactly as in §22.2. Attaching the unchanged Warp Drive protocol to the LLM collective provides a corresponding language-agent evaluation: pooled success rises from 0.333 to 0.417 at both cadences with rollback latencies again $\approx K$ (1 tick at $K{=}2$, 7 at $K{=}8$). The recovery is partial ($\approx 25\%$ of the collapse, vs. 90–100% on the grid): on a language substrate the collision pathology is compounded by semantic-level losses (hallucinated anchor tags, noisy locks) that no reservation can cure — a measurable decomposition we leave to future work. The distance law also survives LLM-produced evidence: with the gate constant c set to the confidence the extractor actually emitted ($c_{\text{meas}}{=}0.900$), all four registered breakpoints land within one tick of prediction.

Two interface findings transfer to any LLM-over-symbolic-memory design: the LLM query former free-generates retrieval tags outside the memory's vocabulary (requiring an any-of gate), and small local models are reliable at semantic commitment but not at coordinate arithmetic or systematic exploration (requiring the layered execution above and an explicit exploration tier — without one, discovery never happens and the warp has nothing to feed on).

Figure 15 overlays the warp-share–cadence curve across all three substrates: the quantity behaves as a substrate-free property of the sharing architecture.

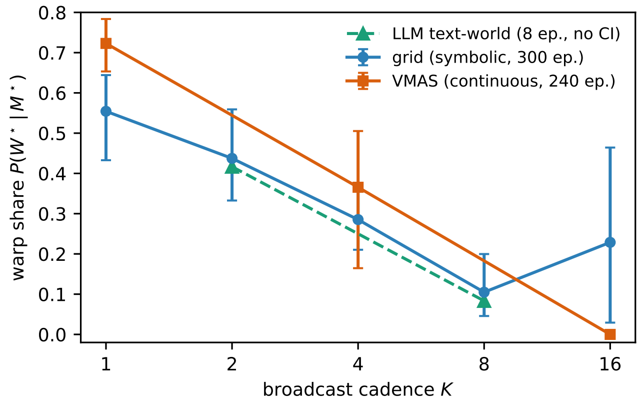

Figure 15: One quantity, three substrates: warp share $P(W^\star|M^\star)$ vs. broadcast cadence K on the symbolic grid, the continuous VMAS scenario, and the LLM text-world. The curves nearly coincide — how much of a collective's materialization runs on foreign evidence is set by the sharing cadence, not by the substrate. (LLM arm: 8 episodes, point estimates only.)

26 Related work

Shared memory in agent collectives. Generative Agents Park et al. (2023) and multi-agent LLM frameworks share memory streams or context but expose no lock interface and no per-source decomposition of a merged belief, so provenance-conditioned completion is not measurable there. Vector memories (e.g. MemGPT Packer et al. (2023)) merge by embedding arithmetic; the merge is not source-decomposable, which blocks both φ and the masked-replay counterfactual.

Emergent communication. CommNet Sukhbaatar et al. (2016) and TarMAC Das et al. (2019) learn continuous channels; the companion paper shows such channels collapse stage observability. Warp requires an explicit materialization event, which these architectures do not expose.

Stigmergy. Pheromone-style coordination is an *anonymous* warp: evidence transfers without provenance. Our results quantify what anonymity costs (shared-bus warp share is maximal while its warp completion is minimal) and what provenance buys (trust gates, reservations, rollback).

Optimistic concurrency. The rollback primitive is Time Warp Jefferson (1985) transplanted from parallel simulation to memory-driven navigation: locks are optimistic computations, occupancy evidence is a straggler message, anti-$M^\star$ is the anti-message. To our knowledge this transplant is new.

27 Limitations

The provenance analysis is subject to six main limitations. First, the primary experiments use a shared coordinate frame and therefore do not solve cross-agent entity correspondence; Part III addresses this assumption with semantic landmark matching. Second, the evidence is based mainly on controlled grid environments, with a directional VMAS study and a small text-world LLM demonstration rather than a photorealistic benchmark. Third, the sign of counterfactual warp gain is regime-dependent and should not be summarized as a universal benefit of foreign memory. Fourth, the closed-form distance law is specific to the explicit exponential trust–staleness gate and its pruning schedule. Fifth, the co-lock-pressure analysis is exploratory and is reported as mechanism evidence rather than as a confirmed predictive law. Sixth, the LLM experiment uses a limited model and task family. All deviations from the registered design and all failed predictions are retained in Appendix 29.

28 Conclusion

This part introduced semantic warp as a provenance-conditioned annotation of a memory-based commitment. The annotation makes it possible to distinguish direct observation from peer-supported action, to estimate the counterfactual contribution of foreign memory, and to derive an explicit temporal boundary for its authority under a trust–staleness gate.

The experiments show that foreign evidence should not be evaluated only by the direct completion rate of foreign locks. Such locks are fragile under scarce targets, but they can provide substantial downstream value by transporting recipients into self-confirmation range. The same analysis localizes the principal cost of fast sharing to duplicated commitments rather than to evidence quality. A minimal reservation-and-rollback protocol addresses this failure and improves unconditional success across the tested fast cadences.

The resulting conclusion is narrower than a general claim that shared memory is beneficial. Foreign evidence becomes useful when its source, age, and authority are represented explicitly and when its use is coupled to a coordination protocol. Part III extends this result from isolated target commitments to provenance-bearing route structures and removes the shared-frame assumption.

Outlook: the contract as one function among several. The contract perspective suggests a reading we state as a falsifiable program rather than a conclusion: memory is not one mechanism but several functions with distinct failure signatures. The trust$\times$staleness gate studied here implements *memory-for-trust* (ablate it and completion collapses through stale and poisoned commits — the trust-flip and the raw-exchange negative are exactly that signature); it is distinct from *memory-for-return* (own consolidation; its ablation starves candidate generation) and from *memory-for-telling* (what is worth broadcasting at all; its ablation leaves receivers' retrieval unimproved on tags they undersample). Under this reading the right validity criterion for shared memory is not direct verification of content but *ecological viability*: whether a commitment formed on foreign evidence can still be carried to completion under the contract — which is precisely the quantity $P(C^\star \mid W^\star)$ and its law make measurable.

Appendix material of this part

29 Deviations from the pre-registered design

1. Reservation tie-break extended to (lock tick, *distance at lock*, agent id): a pure id tie-break hands simultaneous locks to a distant agent over the target's near-discoverer.
2. Protocol recovery is measured against the scarcity ceiling $\min(m, M)/m$ rather than the $K{=}8$ reference, separating structural from coordination collapse; unconditional metrics are reported alongside (Table 25).
3. Completion breakpoints use the discrete-time refinement of §23 (law constants unchanged; hold-out registered).
4. Collision pressure is operationalized as co-lock count rather than co-recipients (structurally constant $N{-}2$ under broadcast).
5. LLM demo: layered execution (LLM semantics, deterministic locomotion); completion attributed at the affordance radius.

30 Reproducibility

All experiments run on CPU in minutes-to-tens-of-minutes; the LLM demo uses a local Ollama model with deterministic caching.

```
PYTHONPATH=. python experiments/warp/exp_warp_anchor.py
PYTHONPATH=. python experiments/warp/exp_warp_gain.py --seeds 20
PYTHONPATH=. python experiments/warp/analyze_warp_chains.py
PYTHONPATH=. python experiments/warp/exp_warp_gain_individual.py
```

```
PYTHONPATH=. python experiments/warp/exp_warp_age_law.py
PYTHONPATH=. python experiments/warp/exp_warp_age_law_holdout.py
PYTHONPATH=. python experiments/warp/exp_warp_drive.py --seeds 20
PYTHONPATH=. python experiments/warp/exp_warp_cross_session.py --layouts 10
```

31 Full counterfactual sign map

Part III

The Song, Not the Pin: Routes and Meaning-Based Identity

What transfers between agents is the song: edge-provenanced routes, landmark-anchored place identity without shared frames, the vocabulary ablation, and the song anatomy.

Part abstract. Transferring the location of a target does not necessarily transfer an executable navigation strategy. In environments with obstacles, hazards, or private coordinate frames, a recipient may know where a target is believed to be while lacking both the route required to reach it and a reliable correspondence between the sender's places and its own observations. We therefore model the transferable memory unit as a provenance-bearing relational route. Semantic landmarks identify corresponding places, route edges encode transport between them, and edge-level provenance records which parts of the route are known only through another agent. A strict route-warp event is defined as the execution of a connected foreign subpath that the recipient has not previously traversed. Experiments show a clear transition between place-only and route-level transfer: isolated target records fail at the first obstructing structure and can perform worse than unguided exploration, whereas transferred routes remain executable. Because route edges are observed at different times, their trust-weighted evidence decays non-uniformly. The resulting rupture law predicts the first inadmissible edge in all 110 tested regimes and with zero index error on 21 intermediate ruptures. Semantic landmark constellations recover place correspondence across translations and quarter-turn rotations, while ambiguous configurations are rejected rather than materialized as phantom targets. A representation ablation further separates the two informational functions of the route: landmarks provide identity and edges provide transport; only their composition reaches an unseen target. In the tested codec, this composition uses 6.4% of the bits of a full snapshot before protocol metadata. The results support relational, provenance-conditioned route transfer as the appropriate unit for executable foreign memory. The study remains limited to controlled semantic observations and discrete frame transformations; continuous pose estimation and open-ended perceptual landmark discovery are outside its scope.

32 Introduction

Collective navigation memory is commonly represented as a set of places, coordinates, or target descriptors. Such representations can support target retrieval, but they omit two requirements of executable transfer. First, the recipient needs a transport structure that specifies how the relevant places are connected. Second, the sender and recipient must determine whether their local records refer to the same physical places, even when their coordinate frames differ. A shared target label or a nearby coordinate is insufficient when either of these conditions is absent.

This part addresses both requirements by defining a song as a relational route with semantic landmarks, ordered transitions, and edge-level provenance. The term is used as a technical representation: landmarks provide candidate correspondences between local memories, whereas route edges specify the sequence of transitions that makes a target operationally reachable. Provenance indicates which route segments were acquired by the sender and remain unverified by the recipient.

The first question is whether route structure provides measurable value beyond place-only transfer. We evaluate this through controlled mazes in which the target coordinate is correct but the direct path is blocked. The second question concerns temporal validity. Because the observations supporting different route edges have different ages, the transferred path can remain admissible only up to a finite prefix. We derive a rupture index by replaying the trust–staleness gate along the route. The third question is referential: can agents recover the same route when their private frames differ? We use semantic constellations around landmarks to generate mutually consistent correspondences and reject ambiguous matches.

The contributions of this part are as follows:

1. We extend provenance from target commitments to route edges and define strict route warp as the execution of a connected foreign subpath not previously traversed by the recipient.
2. We quantify the performance difference between place-only and route-level transfer and identify the structural conditions under which target coordinates are insufficient.
3. We derive a route-rupture boundary from edge-specific trust–staleness values and validate its predicted failure edge across the registered test regimes.
4. We introduce meaning-based place correspondence through semantic landmark constellations and evaluate frame recovery and fail-closed behavior under ambiguity.
5. We decompose the route representation into identity-bearing landmarks and transport-bearing edges and measure the storage cost of their composition relative to a full snapshot.

The central claim is therefore not that relational or topological navigation is new in itself. The contribution is the composition of route structure, provenance, temporal validity, and semantic place correspondence into a measurable transfer object. The present experiments use controlled semantic vocabularies and discrete translations and quarter-turn rotations. Arbitrary continuous $SE(2)$ estimation, rich visual observations, and learned landmark proposal remain open.

33 Setting

Grid worlds (12×10 to 14×12) with semantic water targets, hazards, and walls; agents with symbolic place memory, peer broadcast on cadence K, and the series' standard planner (target lock = the Q/R/M/C $M^\star$ event; completion $C^\star$). The trust×staleness inclusion rule of the collective-memory companion is used throughout with unchanged constants: evidence mass trust $\cdot\, e^{-\alpha\,\text{age}} \cdot m$, threshold τ=0.30, α=0.05. Warp events follow the companion warp paper: provenance φ frozen at lock time; completion attributed to a continuous lock from a foreign commit (φ=1 at first lock) to success.

34 Route-level transfer

Edge provenance. Broadcast snapshots additionally carry the sender's OWN traversed edges $(a, b, \text{count}, \text{last tick})$; foreign edges are never re-broadcast (the same privacy contract as place snapshots: no agent reads another's graph, transport only). For the path P a planner commits,

$$\varphi_{\text{route}}(P) \;=\; \frac{\sum_{e\in P,\ \pi(e)\neq\text{self}} w(e)}{\sum_{e\in P} w(e)}, \qquad w(e) = \text{trust} \cdot e^{-\alpha\,\text{age}(e)} \cdot \log(1{+}\text{count}(e)),$$

the same weights the route gate uses. A **strict** $RW^\star$ is a committed, traversed, connected sub-path of ≥ 5 edges with $\varphi_{\text{route}} = 1$: the whole verse is someone else's. Place warp is the degenerate case (path length 0: only the target is foreign), giving a two-dimensional provenance plane $(\varphi_{\text{place}}, \varphi_{\text{route}})$.

First strict $RW^\star$. On a comb maze with measured detour D=2.65, a traveler commits and walks a fully foreign 45-edge route to completion in 54 ticks; with the same foreign PLACE evidence but no edges, the standard planner never arrives (300-tick budget), and blind exploration takes 126.

34.1 Performance transition under route transfer

Table 27: Three arms on identical mazes (paired by seed), bucketed by MEASURED detour factor D; 20 seeds per bucket, censored gains (fail = 300). Place-only transfer does not degrade with D — it collapses at the first wall, and is worse than blind exploration there (the known-target manhattan pull pins the planner against walls). Route transfer completes everywhere; every completion is a strict $RW^\star$ with $\varphi_{\text{route}} = 1$.

D	gain route-vs-place [CI]	gain route-vs-blind [CI]	succ r/p/b
1.00	+4.7 [+3.4, +6.0]	+62 [+44, +80]	20/**20**/20
1.38	+281 [+279, +283]	+113 [+77, +155]	20/**0**/16
2.03	+273 [+271, +275]	+123 [+91, +160]	20/**0**/17
3.32	+262 [+259, +264]	+110 [+72, +152]	20/**0**/15

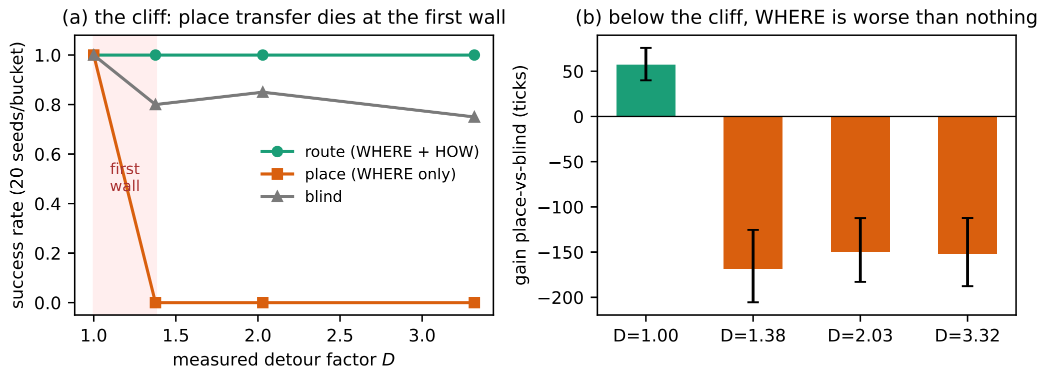

Figure 16: The cliff (paired arms on identical mazes, 20 seeds per measured-detour bucket). **(a)** Place-only transfer does not degrade with detour — it collapses at the FIRST wall, located by the probe bucket between D=1.00 and D=1.38; route transfer completes everywhere. **(b)** Below the cliff, knowing WHERE without HOW is *worse than knowing nothing*: the known-target manhattan pull pins the planner against walls (-150 to -168 ticks vs. blind exploration), while in the open world the same knowledge is worth $+57$.

Figure 16 shows both faces of the result. Two pre-registered predictions failed here, in instructive ways. The negative control at D=1 was registered in ticks ($|\text{gain}| \leq 3$) and came out $+4.7$: the gap is *turn economy* (BFS routes walk straight legs; the dominant-axis planner alternates axes), not route knowledge — in path LENGTH the gap is exactly 0.0 $[0.0, 0.0]$ across all 20 open-grid cells. And "gain grows with D" is unmeasurable: with place success at 0 the gain saturates at the censoring ceiling. The correct form of the law is a step located by the probe bucket at $D \in (1.00, 1.38)$, i.e. the first wall.

35 Rupture law and route-validity boundary

The witness stamps edge i at tick i: its trace ages non-uniformly, the start of the route always older than the end, and a traveler delayed by a_0 *chases a fading song*. Under the standard gate the binding edge is the CURRENT one (the oldest remaining), giving three regimes on the a_0 axis: complete; rupture at a predictable path index; dead on arrival. The predictor is fully deterministic — a no-decay calibration run records the traveler's kinematics (path index per tick, turns included), and the registered prediction replays the closed-form gate over that trajectory, the same device that produced the companion paper's $6/6$ hold-out.

Across 5 mazes $\times$ 11 delays $\times$ 2 trusts (110 cells, registered before execution): **110/110 regime matches; all 21 mid-route ruptures at exactly the predicted edge index (error** 0); and a route trust-flip — at trust 0.7 the single-traversal edge mass gives $\text{age}_{\max} = 9.6 < L$, so *no route ever commits* (0 commits observed; the discrete off-switch of the companion law, now for paths). The rupture of a transferred route is thus not noise but a law: the same constants $(\alpha, \tau, \text{trust})$ that set WHEN a place-warp gate closes set WHERE a route-warp breaks.

36 Hazard preservation along the valid route prefix

The traveler following the witness's path preserves the witness's measured risk profile within the validated route prefix, at a time premium of $+0.7$ ticks (Figure 17a). Composing with §35: in the rupture regime (19/20 worlds) the traveler takes **0 hits before the rupture** and 0.58 $[0.11, 1.32]$ after it, when it falls back on place knowledge and re-enters the field — with all 19 ruptures again at the R2-predicted index (error 0; the edge-exact predictor replicated on a new world class with no adjustment). The song protects while it lasts, and the measured protection ends at the edge predicted by the law ends.

Table 28: Open fields dense with passable-but-punished hazards (density 0.20), the witness's route hazard-avoiding by experience; 20 paired worlds, decay off (the safety axis isolated from the aging axis). Both transfer modes COMPLETE — the wall axis (Table 27) and the hazard axis are independent; they differ in the price.

Arm	hazard hits [CI]	completion	mean t
route (foreign safe path)	**0** (all 20 cells)	20/20	16.6
place (WHERE only)	3.25 $[2.45, 4.10]$	20/20	15.9
blind	28.7	—	—

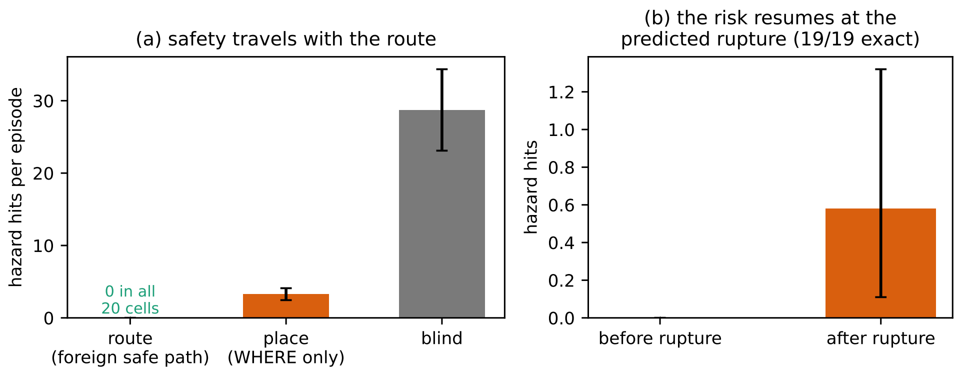

Figure 17: The song protects. **(a)** Hazard hits per episode: the traveler on the witness's route preserves the witness's measured risk profile within the validated route prefix (0 in all 20 worlds); the place-only traveler pays 3.25 hits for the same completion; blind exploration pays 28.7. **(b)** Composition with the rupture law: in the rupture regime the traveler takes zero hits while the song lasts, and the risk resumes after the break — which occurs at the R2-predicted edge index in all $19/19$ cells.

37 Meaning-based place correspondence

Everything so far — and everything in the companion papers — rides on a shared coordinate frame. We now remove it: each agent's private frame is the true grid under a secret translation (later: plus a secret $90°$-multiple rotation), and broadcasts carry private coordinates that are meaningless to the receiver until it decides what the sender's places MEAN.

Two levels of 'the same', and what a landmark is. Cross-agent place identity has two levels that must not be conflated. The *semantic predicate* says two cells belong to the same desired class (`water_source`, GOOD): it travels in every snapshot, and it is what task satisfaction is defined over. But no predicate can decide *entity correspondence* — that two memory entries denote the same world referent — because a predicate is exactly what aliases: all water looks like water. Route-relative description ('n steps past the last GOOD') does not decide it either: it is provenance of a path, not identity of its endpoints (§34–§35 measure what it buys and where it breaks). What decides correspondence in this series is a *landmark*: a place whose local semantic constellation supports a mutually-unique match across agents. Landmark is a *role*, not a cell type — which tag classes can play it is itself measured below (§37.2).

Fingerprints and the handshake. A place fingerprint is the local semantic constellation around a visited cell — which content tags (hazards, walls, water) sit at which relative offsets — translation-invariant and coordinate-free. Matching is *mutually unique* (a pair is accepted only if each side is the other's sole candidate above an exact-equality threshold; the sender knows far more places than the receiver, and one-directional uniqueness aliases distant look-alikes into the receiver's neighborhood), fingerprints need ≥ 2 salient keys, and the frame translation is the modal delta of ≥ 3 agreeing pairs. Landmarks must carry at least one CONTENT tag: border-only ('void') patterns describe the shape of the world, not a place, and are excluded — the lesson of the corner trap below.

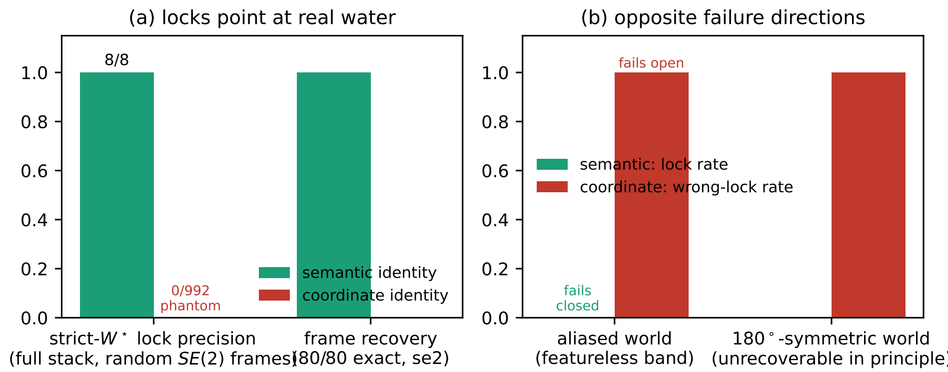

Figure 18: Meaning-based identity under discrete rigid-frame transformations ($\mathbb{Z}^2 \rtimes C_4$: translations and quarter-turn rotations). **(a)** Every strict-$W^\star$ lock the semantic stack makes points at real water $(8/8)$; the coordinate stack makes 992 locks, all phantoms $(0/992)$; frame recovery itself is exact in $80/80$ witness–traveler combinations. **(b)** The failure asymmetry in worlds where identification is impossible (featureless band; a $180°$-symmetric layout where the frame is unrecoverable in principle): semantic identity refuses to lock — fails closed — while coordinate identity keeps confidently locking wrong cells — fails open.

Witness–traveler results. Over 4 offsets $\times$ 2 hazard densities $\times$ 10 seeds: the semantic aligner recovers the secret offset *exactly in 100%* of landmark-world runs (mean 15.1 mutual pairs) and completes with strict $W^\star$ (φ=1) every time; the coordinate baseline under any nonzero offset locks a non-water cell in 100% of runs (its rare 'successes', 6.7%, are the displaced route accidentally crossing the true water). In the aliased world (featureless interior band) the two identity regimes break in OPPOSITE directions: semantic identity makes *zero* locks — it fails closed — while coordinate identity keeps confidently locking wrong cells — it fails open. This asymmetry, not raw accuracy, is the deepest argument for meaning-based identity: when the world does not support identification, the semantic system knows it.

37.1 Integrated correspondence and route-transfer stack

Online alignment in the peer stack. With N=4 agents under scarcity, all in distinct secret frames, alignment develops DURING the episode: agents explore, co-visited landmarks accumulate, and per-sender frames lock in mid-run (onset in 100% of episodes, mean tick ≈ 15). Pooled success: shared-frame oracle 0.456, coordinate under misalignment 0.350, semantic 0.444 — **88% of the gap recovered**, with strict $W^\star$ functioning through recovered frames.

Validity law under recovered frames. Re-running the companion distance law with the trust$\times$staleness gate ON TOP of semantic identity under a frame offset gives breakpoints $20/14/9$ — *the same integers as the shared-frame closed form, exactly*. The gate and the frame recovery are orthogonal layers: the law does not notice that the shared grid is gone.

No shared north: translations and quarter-turn rotations. With secret rotations added, one rotation hypothesis per $90°$ reuses the translation aligner and the winner must *dominate* ($\geq 2\times$ the runner-up's consensus). Results: exact (r^*, δ^*) recovery in **80/80** rotation$\times$offset$\times$seed combinations with 100% strict-$W^\star$ completion; in a $180°$-symmetric world — where the frame is unrecoverable *in principle* — the aligner refuses $(0/10$ locks) while coordinates fail open $(10/10$ wrong locks); and the distance law again does not notice (bp $= 20$ under a $90°$-rotated witness). The translation-only aligner, for contrast, never recovers a rotated frame $(0/60)$ but does not always fail closed: in $2/60$ cells a $180°$-symmetric hazard substructure lets it fail OPEN — the empirical motivation for the dominance rule.

The corner trap. In the full stack, raw success is non-discriminating under rotations (rotated coordinate poison mostly maps off-grid and degrades into pseudo-exploration), so the registered discriminator is *warp-lock precision*: the share of strict $W^\star$ locks pointing at real water. Before the void filter, early border fingerprints (agents start in corners; rectangle borders are $180°$-symmetric) fed wrong-frame hypotheses and semantic precision was 0.159; with borders excluded as non-landmarks it is 1.000 **(**$8/8$**) vs.** 0.000 **(**$0/992$ **phantom locks) for coordinates**, with alignment onset in 100% of episodes and clean regressions on all earlier experiments (Figure 18).

37.2 Causal contribution of the landmark vocabulary

The identity machinery rides on a fixed landmark vocabulary $\Sigma = \{\texttt{wall}, \texttt{hazard}, \texttt{water}, \texttt{void}\}$: the tag classes allowed into a fingerprint. Every claim above is therefore conditional on a hand-chosen alphabet. The last experiment makes the alphabet the treatment: witness–traveler frame recovery (worlds carrying both scattered walls and scattered hazards; 3 offsets $\times$ 12 seeds per arm) under seven vocabularies, from full Σ down to single classes.

Table 29: Landmark-vocabulary ablation (36 cells per arm, 252 total; registered predictions A1–A4, all PASS). Impoverishing the alphabet costs *coverage* — recovery converts into refusal — but never *correctness*: every transported lock, under every vocabulary, goes through the exactly-recovered frame. Attribution = share of consensus-pair key mass under full Σ.

Vocabulary	exact recovery	mean pairs	lock precision	fail-open
full Σ	1.00	33.2	1.00	0
$-$ void	1.00	32.0	1.00	0
$-$ hazards	0.92	11.8	1.00	0
$-$ walls	1.00	15.5	1.00	0
hazards only	1.00	13.7	1.00	0
walls only	0.83	10.8	1.00	0
void only	0.00	0.0	—	0

Four registered predictions, four passes (Table 29). **(A1) The ablation is fail-safe:** in 252 cells there is not one fail-open transport, consistent with the fail-closed asymmetry of §37 holds not just against ambiguous *worlds* but against impoverished *alphabets*. **(A2) Void is scaffolding, not identity:** border patterns carry 0.000 of the consensus key mass, and dropping them changes nothing — consistent with the corner trap, where they actively poisoned rotation hypotheses. **(A3) One content class suffices:** hazards alone recover every frame; walls alone recover 0.83 and refuse the rest. Correspondence needs only ≥ 3 mutually-unique constellations over *some* content class — a bounded-rationality budget, not a rich ontology. **(A4) Attribution is measurable:** hazard constellations carry 0.56 of the consensus mass, walls 0.44. Which features function as landmarks is thus not a modelling assumption that happened to work — it is a measurable property of the world$\times$vocabulary pair, and the machinery degrades along the safe axis whenever the vocabulary is wrong.

37.3 Functional decomposition of the route representation

§37 opened with a claim we have so far only argued: route-relative description (‘n steps past the last GOOD’) is provenance of a path, not identity of its endpoints. The last experiment measures it. A witness compresses its route to the water into a purely *relational* song — a sequence of couplets (σ, e), where σ is the landmark constellation signature and e the exact beat vector from the previous couplet — carrying *no global coordinates whatsoever*. A band-limited traveler who has never seen the water receives it. Four registered arms (bit codec fixed before the runs; 12 paired worlds):

- **Beats only** (the pure ‘n steps from the last GOOD’): usable only by *assuming* the anchor. With a shared anchor it is exact ($11/11$, the negative control); with anchor disagreement it locks every time and is right *never* ($11/11$ fail-open mislocks, displaced by exactly the anchor offset). Beats do not carry identity — they parasitise on the identity of the anchor.
- **Signatures only**: re-identifies co-visited places but has no edges to walk — *zero* transports of the never-seen target. Nodes alone cannot reach the unseen.
- **Signatures $+$ beats**: couplets matched mutually-uniquely, the beat chain checked for metric consistency between matched couplets (a loop-closure test), the target dead-reckoned from the last matched landmark. Coverage 1.0, every transport exact, zero fail-open — at 366 bits per song.

- **Full snapshots** (the machinery of §37): the information upper bound — same correctness at 5,690 bits.

All four registered predictions passed. The song is not a lighter snapshot; it is a different *kind* of object: identity lives in the nodes (landmarks), transport lives in the edges (beats), and their composition delivers full-snapshot transport at 6.4% of the bit cost while inheriting the fail-safe property of §37.2 — everything that goes wrong goes wrong as a refusal. (This 6.4% is the *pure song codec*: nodes and edges only. A deployed collective-memory protocol adds provenance, timestamps, world versions, and a utility certificate; the full-protocol payload ratio, measured in the companion formation paper, is correspondingly larger.)

38 Related work

Map merging and place recognition. Multi-robot SLAM treats inter-map data association as an estimation problem over metric features Cadena et al. (2016); visual place recognition Lowry et al. (2016) matches appearance. Our contribution is not a competing matcher but the measurement layer on top: provenance-conditioned completion through recovered frames, the fail-closed/fail-open asymmetry as a registered prediction, and law invariance under frame removal.

Route sharing. Stigmergy transfers paths anonymously (pheromone gradients); learned communication (e.g. CommNet Sukhbaatar et al. (2016)) transfers policies opaquely. Neither exposes an edge-provenance decomposition, so neither can state — let alone predict — where a transferred route breaks. The rupture predictor is Time Warp Jefferson (1985) thinking applied to memory decay: deterministic replay of a gate over calibrated kinematics.

Songlines. The narrative framing corresponds to a functional distinction in both directions: the song is a sequence (route warp), its places are recognised by meaning (semantic identity), it fades verse by verse (rupture law), and it protects the walker while it lasts (hazard stratification).

39 Limitations

The route-transfer experiments use controlled navigation substrates and a predefined semantic vocabulary. Place correspondence is tested under translations and quarter-turn rotations; this setting should be described as discrete frame recovery rather than arbitrary continuous $SE(2)$ estimation. The semantic fingerprints are sufficiently rich to support exact correspondence in the reported environments, but open-world appearance variation, perceptual uncertainty, and learned symbol alignment remain unresolved. The rupture law is exact for the specified edge-wise trust–staleness mechanism and does not by itself provide a universal safety guarantee. The claim concerning risk is therefore restricted to preservation of the witness's measured hazard profile within the validated route prefix. Finally, the 6.4% figure refers to the pure route representation; a complete deployment cost must additionally include provenance, timestamps, correspondence hypotheses, and coordination metadata.

40 Conclusion

This part identifies a provenance-bearing relational route as the transferable unit of executable navigation memory. A target record by itself can identify a destination without specifying a feasible path, and an edge sequence without a reliable landmark correspondence can transport the agent toward the wrong referent. The experiments show that these functions are distinct: semantic landmarks provide identity, route edges provide transport, and only their composition supports transfer to an unseen target.

The route representation also exposes a temporal boundary. By applying the trust–staleness gate to each edge, the method predicts the route prefix that remains admissible and the edge at which the transfer should be rejected or replanned. In the controlled studies, this rupture index matches the observed failure location exactly. Meaning-based landmark correspondence further removes the shared coordinate assumption for discrete translations and quarter-turn rotations and fails closed when the observations are ambiguous.

The results support a precise interpretation of foreign navigation memory: it must preserve referential identity, transport structure, provenance, and a validity boundary together. The next part embeds these requirements into a broader collective-memory architecture with explicit merge, trust, and staleness rules.

41 Registered predictions and revisions

All predictions were written to disk before execution; the artifact retains both versions wherever an operationalisation was revised. Revisions: R1 negative control re-measured in path length; R1 gain monotonicity re-formulated as cliff detection; W9 full-stack discriminator changed from success to warp-lock precision; W8 strict-$W^\star$ functioning extended to include warp-assisted own completions (taxi chains). The vocabulary ablation (A1–A4, §37.2) and the song-anatomy experiment (S0.1–S0.4, §37.3, bit codec registered with the predictions) passed as registered, with no revisions.

42 Reproducibility

```
PYTHONPATH=. python experiments/warp/exp_route_warp_r0.py
PYTHONPATH=. python experiments/warp/exp_route_warp_r1.py --seeds 20
PYTHONPATH=. python experiments/warp/exp_route_warp_r2.py --seeds 5
PYTHONPATH=. python experiments/warp/exp_route_warp_r3.py --seeds 20
PYTHONPATH=. python experiments/warp/exp_warp_semantic_identity.py
PYTHONPATH=. python experiments/warp/exp_warp_semantic_stack.py --seeds 20
PYTHONPATH=. python experiments/warp/exp_warp_rotation.py --seeds 10
PYTHONPATH=. python experiments/warp/exp_warp_landmark_ablation.py
PYTHONPATH=. python experiments/song_grammar/exp_s0_song_smoke.py
```

Part IV

Collective Semantic Memory: Merge, Trust, and Staleness

The mechanics of peer memory – three explicit rules, the third regime on the M-by-C plane – and the categorical frame in which commonality of interpretation is a parameter, not an assumption.

Part abstract. Collective memory is often treated as a monotone resource: increasing the amount or frequency of shared information is expected to improve performance. The Q/R/M/C experiments show that this assumption fails under semantic target scarcity. Fast sharing improves target materialization but can reduce completion because multiple agents act on the same records. We introduce a minimal collective semantic memory (CSM) that separates three mechanisms: source-aware merge, explicit trust, and staleness-gated consumption. The method retains private memories and peer-indexed views rather than replacing them with a single global archive. In the benchmark, the resulting gate occupies a different region of the materialization–completion plane from unrestricted shared memory and fixed-cadence peer exchange. The benefit arises from selective admissibility rather than from a larger volume of communication. We further remove the shared-frame assumption through semantic place fingerprints and quantify the consistency of the resulting correspondence. The formal analysis interprets local memories as ordered structures, alignment maps as partial semantic correspondences, and collective memory as a coherent gluing of the local views. The trust–cadence phase diagram determines whether the induced correspondence graph contains a connected consensus component spanning the population; disconnected settings produce multiple sub-consensuses rather than the absence of a colimit. A coalgebraic interpretation complements this static account by treating each agent as an ongoing process and the collective as a behavioral system. Finally, controlled LLM experiments compare raw context replay with a consolidated Songlines graph and show that structured memory resolves freshness and cross-frame identity at substantially smaller active-context cost. These results provide a minimal mechanics and mathematical interpretation of collective semantic memory, but they do not establish open-world symbol grounding or a general category-theoretic theory of agent cognition.

43 Introduction

A collection of agents does not obtain a useful collective memory merely by placing all observations in one shared store. The multi-agent measurements in Part I show why: communication improves access to target evidence, but unrestricted or overly frequent sharing can synchronize commitments and reduce task completion. A collective memory must therefore regulate not only what is available, but also which records are admitted into each agent's decision process and how conflicts among local views are resolved.

This part introduces a minimal collective semantic memory (CSM) based on three explicit mechanisms. The merge rule preserves source and support information when records from different agents are combined. The trust rule assigns source-dependent authority to foreign evidence. The staleness rule limits the duration for which an observation can influence retrieval and materialization. Agents retain private memory and construct local merged views; no central cognitive state or single global interpretation is assumed.

The empirical question is whether these mechanisms produce a distinct operating regime rather than merely another communication frequency. We compare independent memory, a shared bus, a central aggregator, fixed-cadence peer exchange, and the proposed CSM under identical navigation and planning components. We then relax the common-coordinate assumption using semantic place fingerprints and evaluate whether the same physical referents can be aligned from local meaning alone.

The formal analysis serves two purposes. First, it clarifies the conditions under which local place memories can be glued coherently. Each realized memory is represented as an ordered category, and partial alignment maps are constructed from observed correspondences. Their colimit represents the merged structure, while connectedness of the accepted correspondence graph determines whether one consensus component spans the population or whether several sub-consensuses remain. Second, a coalgebraic reading represents agents as continuing processes that emit actions and messages over time, complementing the static algebraic description of memory content.

The contributions of this part are as follows:

1. We specify a minimal distributed memory architecture with explicit merge, trust, and staleness rules and evaluate it against four alternative communication topologies.
2. We show that the performance gain is produced by selective consumption of admissible evidence rather than unrestricted sharing.
3. We construct semantic place correspondence across private frames and quantify both task performance and correspondence defect.
4. We provide order-theoretic, categorical, and coalgebraic interpretations that state the invariants and scope of the merge mechanism without treating the formal language as an additional empirical contribution.
5. We compare structured memory with raw context under fixed LLMs and isolate failures involving temporal updates and cross-agent entity identity.

The mathematical account is intended to formalize the implemented merge and alignment operations. It is not used as a decorative claim of novelty, nor does it establish that arbitrary agent societies form a single global mind. The scope remains a controlled family of semantic navigation memories with explicit correspondences and admissibility rules.

44 Memory substrate and task

Each agent observes a grid environment and emits semantic place evidence such as `water_source`, `hazard_edge`, and `open`. A local memory stores place records, tag confidences, visit counts, freshness, and transition statistics. Planner queries retrieve candidate places by matching required, preferred, and penalty tags. A selected candidate is materialized into a concrete cell and executed by a local controller.

The multi-agent task uses a 12×10 grid. Each agent must reach a cell tagged `water_source`. Targets are scarce: $N\in\{3,5,8\}$ agents, $T\leq N$ targets, three layouts, three hazard densities, and 40 seeds for the main cadence sweep. Every run logs Q/R/M/C events per agent:

$$Q^\star \to R^\star \to M^\star \to C^\star,$$

where $M^\star$ means a semantic candidate was locked as an actionable target and $C^\star$ means the target was reached. These logs are not the architectural contribution; they let us see where each topology fails.

45 Four collective-memory topologies

The four topologies share the same local memory and planner. They differ only in where merge happens.

Independent. Each agent owns a private event store and concept graph. There is no communication path.

Shared bus. All observations are appended to a single global event bus. One shared concept graph is rebuilt from the union of all events.

Central aggregator. Each agent owns a private concept graph; a central consensus layer pulls snapshots and computes one trust-weighted report for everyone.

Peer broadcast. Each agent owns a private concept graph and a private merged view. Every K ticks, agents broadcast snapshots; each agent merges its own memory with the latest snapshot from each peer.

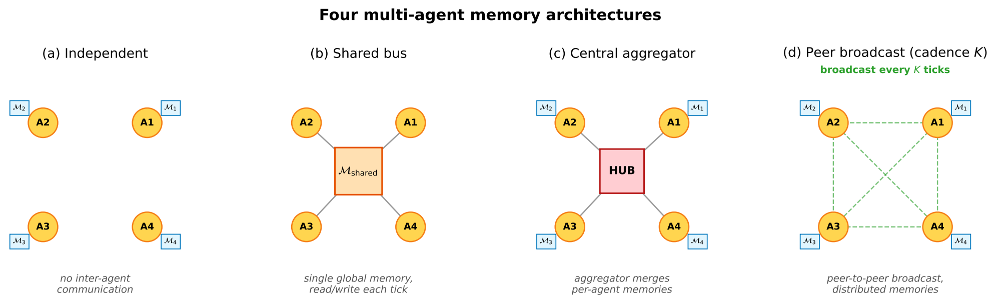

Figure 19: Four memory topologies. **(a)** Independent: per-agent private memory, no exchange. **(b)** Shared bus: one global memory, every agent reads and writes each tick. **(c)** Central aggregator: private memories plus a hub that merges snapshots into a consensus report. **(d)** Peer-to-peer broadcast: private memories with snapshot exchange every K ticks and per-agent merged views.

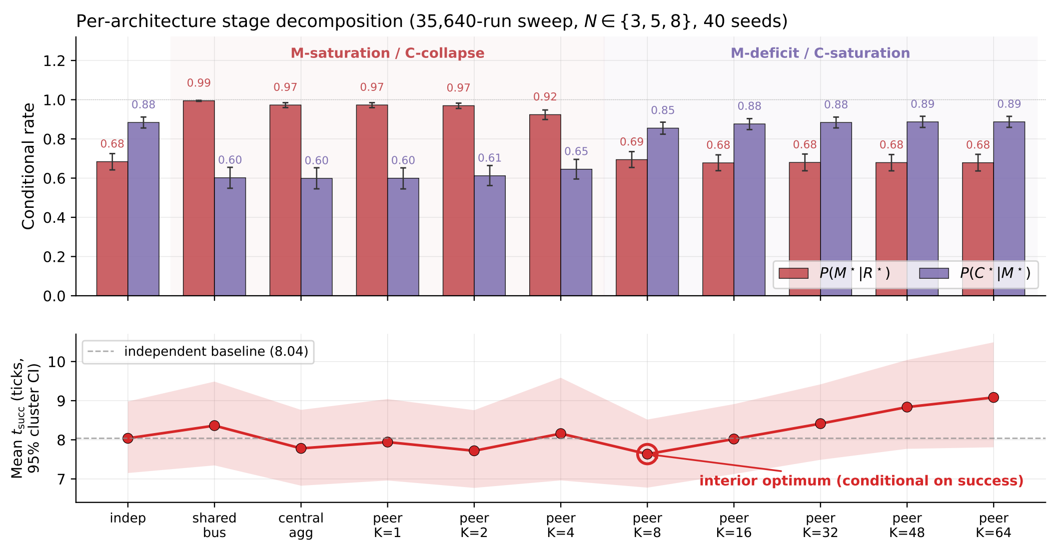

Figure 20: Conditional rates and mean time-to-success from the 35,640-run sweep. Fast sharing improves target materialization but harms completion through contention; slow sharing has the opposite profile. Mean time-to-success has an interior optimum at $K{=}8$.

The key distinction is not the clustering primitive. It is the level at which that primitive is instantiated:

$$\begin{aligned} \text{independent} &: N \text{ graphs, } 0 \text{ mergers} \\ \text{shared bus} &: 1 \text{ graph, } 1 \text{ implicit global merger} \\ \text{central} &: N \text{ graphs } + 1 \text{ central merger} \\ \text{peer} &: N \text{ graphs } + N \text{ private mergers.} \end{aligned}$$

46 Cadence-dependent materialization–completion trade-off

The peer topology has one explicit parameter: broadcast cadence K. The main sweep crosses four architectures and eight peer cadences $K{\in}\{1, 2, 4, 8, 16, 32, 48, 64\}$ for 35,640 runs. Figure 20 summarizes the result.

Fast peer broadcast and centralized sharing live in the M-saturation/C-collapse regime: agents can lock targets reliably, but many locks point to targets that peers also pursue. Independent and slow peer broadcast live in the M-deficit/C-saturation regime: fewer agents lock onto good targets, but those that do are less likely to collide with peer occupancy.

Across all peer runs, Spearman of $P(M^\star|R^\star)$ vs K is -0.608 and of $P(C^\star|M^\star)$ vs K is $+0.420$ (both $p < 10^{-4}$). The stronger signature is visible at fixed cadence: the within-cadence Pearson

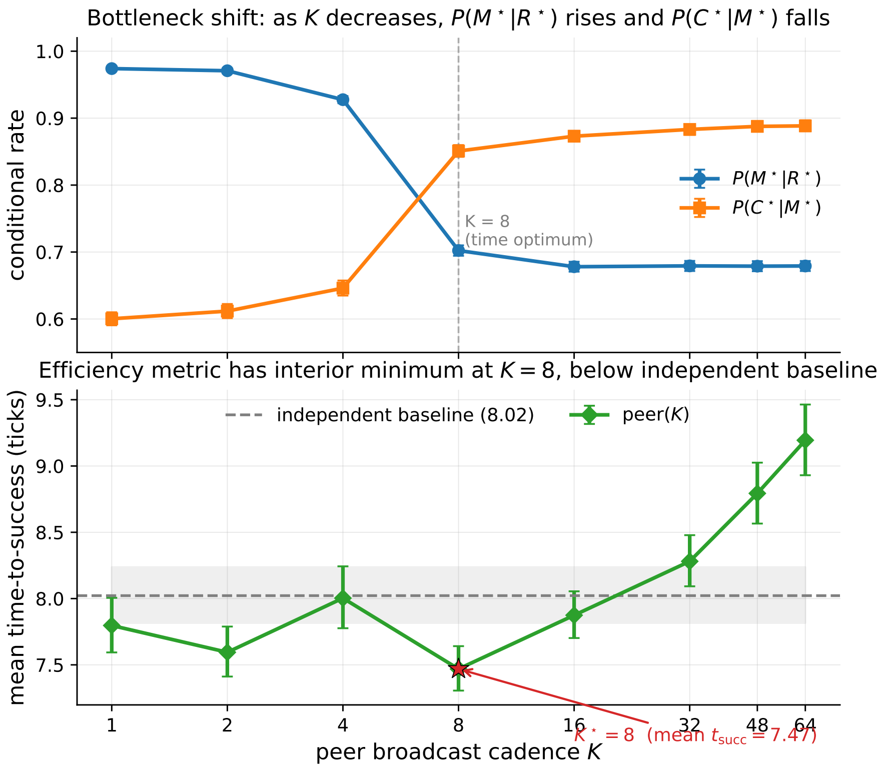

Figure 21: Bottleneck shift across peer cadences. $P(M^{\star}|R^{\star})$ falls and $P(C^{\star}|M^{\star})$ rises as K grows. The practical efficiency metric, mean time-to-success, is minimized at K=8.

correlation between the M and C links peaks at -0.61 for K=4 and decays toward noise as K grows. Cadence therefore does not simply tune "amount of communication." It moves the bottleneck between semantic commitment and shared-resource completion.

47 Minimal collective semantic memory

The cadence sweep shows that fixed-rate snapshot exchange exposes only one control surface. A collective semantic memory should expose at least three:

Merge. How peer evidence about the same semantic place is combined.

Trust. How evidence from different peers is weighted when it conflicts or becomes unreliable.

Staleness. How old evidence decays before it influences retrieval.

The minimal CSM uses peer broadcast at K=8 and overlays three rules:

Merge. Per-place score equals own evidence at weight 1 plus peer evidence at weight $\mathrm{trust}_i \cdot e^{-\alpha \cdot \mathrm{age}}$, with $\alpha = 0.05$. Inclusion threshold $\tau = 0.30$.

Trust. Per-peer EMA on retrieval-success consistency. Peers whose latest snapshot supports the locally locked target gain trust; others decay toward 0.4. Update rate $\beta = 0.10$.

Staleness. Each broadcast snapshot is timestamped; merge weight decays as $e^{-\alpha \cdot \mathrm{age}}$. Snapshots older than $10K$ ticks are pruned.

The implementation is a drop-in replacement for fixed-cadence peer memory: `experiments/collective_semantic_memory/csm_memory.py`.

47.1 Results

We compare CSM against fixed-cadence peer broadcast at $K \in \{1, 4, 8, 16, 64\}$ on scarcity cells $(N, T) \in \{(3, 2), (5, 3), (8, 5)\}$, three layouts, three hazard densities, and 20 seeds: 3,240 runs total.

Table 30: Minimal CSM vs fixed-cadence peer broadcast on the scarcity benchmark. CSM's lower 95% bootstrap CI bound exceeds every peer-K upper CI.

Architecture	$P(M^\star \mid R^\star)$	$P(C^\star \mid M^\star)$	Success rate (95% CI)
peer (K=1)	0.991	0.586	0.577 [0.572, 0.586]
peer (K=4)	0.955	0.623	0.581 [0.572, 0.590]
peer (K=8)	0.730	0.871	0.599 [0.591, 0.607]
peer (K=16)	0.697	0.892	0.597 [0.589, 0.606]
peer (K=64)	0.684	0.900	0.601 [0.594, 0.609]
CSM (minimal)	**0.782**	**0.825**	**0.617 [0.610, 0.624]**

CSM occupies a third regime. Fast broadcasts have excellent materialization but poor completion; slow broadcasts have the reverse. CSM keeps both conditional links above 0.78. The absolute success gain is modest, $+1.7$ to $+4.0$ percentage points, but the stage decomposition shows why it is meaningful: the architecture changes the joint M×C operating point rather than merely moving along the fixed-cadence curve.

Mechanism: gated consumption rather than unrestricted sharing. Provenance annotation of the same runs gives the mechanism. CSM does not share *better* than fixed-cadence peers — it *refuses to act on stale or untrusted foreign evidence*. The inclusion rule of Section 47 implies a closed-form horizon beyond which a peer's evidence can no longer form an intention, $\mathrm{age}_{\max}(\mathrm{trust}_i) = \alpha^{-1} \ln(\mathrm{trust}_i \cdot c/\tau)$, and this horizon is empirically exact: measured intention-formation cutoffs match the formula tick-for-tick across trust levels (23/23.05, 18/18.59, 12/12.84 at $\mathrm{trust} = 1.0/0.8/0.6$), including a registered-prediction hold-out on six never-run cells ($6/6$ exact integer breakpoints) and a discrete trust-flip: $\mathrm{trust} \leq \tau/c$ zeroes foreign-evidence commitments entirely. Fixed-cadence peers have no such gate — their foreign evidence never expires — which is precisely why their completion collapses when sharing is fast. The full provenance-conditioned analysis appears in a companion manuscript; here it serves as the mechanistic reading of Table 30.

48 Place identity from meaning: removing the shared frame

Everything above — and the categorical reading below — assumes a shared spatial basis X: 'the same place' means 'the same (x, y) in a frame the environment hands to every agent'. This section removes that assumption and reports what it was hiding. Each agent now holds a *private* frame (the true grid under a secret translation, and, in the final experiment, a secret 90°-multiple rotation); broadcast snapshots carry the sender's private coordinates, which are meaningless to the receiver until it decides what the sender's places *mean*.

Mechanism. A *place fingerprint* is the local semantic constellation around a visited cell: which content tags (hazards, walls, water) sit at which relative offsets — translation-invariant and coordinate-free. Two agents match places by *mutually unique* fingerprint correspondence (each side is the other's sole candidate; one-directional uniqueness aliases distant look-alikes into the receiver's neighborhood, and border-only patterns are excluded as descriptions of the world's shape rather than of a place). The inter-agent frame is the modal translation of ≥ 3 agreeing matched pairs; under rotations, one hypothesis per 90° reuses the same aligner and the winner must dominate the runner-up's consensus by $2\times$. Foreign evidence — including places the receiver has never seen — is then transported through the recovered frame.

Results. Registered before execution; full protocol in the companion manuscript.

1. **Exact recovery.** Witness–traveler: the secret offset is recovered exactly in 100% of landmark-world runs, and under translations and quarter-turn rotations (the discrete rigid-frame family $\mathbb{Z}^2 \rtimes C_4$) in $80/80$ rotation×offset×seed combinations, with 100% completion on foreign-committed locks.

2. **Failure asymmetry.** Where identification is impossible — a featureless region, or a $180°$-symmetric layout in which the frame is unrecoverable *in principle* — semantic identity makes *zero* commitments (fails closed), while coordinate identity keeps confidently locking wrong cells (100% wrong locks; fails open).
3. **The full stack.** With $N{=}4$ agents in distinct secret frames, alignment develops online (onset in 100% of episodes, mean tick ≈ 15) and recovers 88% of the shared-frame performance gap (success 0.444 vs. 0.456 oracle vs. 0.350 coordinate); under randomly sampled translations and quarter-turn rotations every semantic commitment points at real water ($8/8$) while the coordinate stack produces 992 phantom locks (0 real).
4. **The gate is orthogonal.** Re-running the $\mathrm{age}_{\max}$ analysis above with the CSM inclusion rule ON TOP of semantic identity, under a frame offset and under a $90°$ rotation, reproduces the SAME integer breakpoints ($20/14/9$; $\mathrm{bp} = 20$ rotated) — the trust$\times$staleness gate and frame recovery are independent layers.

The stronger claim promised in earlier drafts' limitations is therefore discharged constructively: place identity can be carried by meaning alone, it composes with merge/trust/staleness unchanged, and — unlike the coordinate convention it replaces — it knows when it does not know.

The categorical indicator: adjunction round-trip defect. Frame recovery is exactly the operation the categorical reading of §49 needs once the shared basis X is dropped: with private frames each agent carries its own place category, and communication is a pair of recovery functors $F = \textsc{align}(A, B)$ (the sender B's frame expressed in receiver A's coordinates) and $G = \textsc{align}(B, A)$. If the two are genuinely adjoint — the frames are glued — the round trip $\eta_X : X \to G(F(X))$ is the identity, so the *adjunction defect*

$$\mathrm{def} = \|F.\mathrm{offset}{+}G.\mathrm{offset}\|_1 \text{ (translation)}, \qquad r_F{+}r_G \equiv 0 \pmod 4 \text{ and } \|R_{r_G}F.\delta{+}G.\delta\|_1 \; (\mathbb{Z}^2 \rtimes C_4),$$

is zero (Mac Lane, 1998). Measured on the recovery method above (`experiments/warp/alignment_defect.py`, 3 offsets $\times$ 10 seeds per cell): the defect is *exactly* 0 in $30/30$ runs for translation frames and $30/30$ for each of the four $90°$ rotations — the recovered transformations are mutual inverses within the implemented discrete transformation family, yielding zero round-trip defect. Where recovery must fail (an agent that has observed only a featureless interior band), the observations are insufficiently discriminative to identify a unique admissible alignment: several transformations are equally consistent, so the runtime leaves the alignment edge undefined in $30/30$ runs and refuses to perform the gluing, with *zero* spurious gluings. The exact discrete aligner therefore either recovers a mutual-inverse pair ($\mathrm{def}{=}0$) or declines to instantiate the edge — and never produces a confident wrong gluing; "failing closed" above is precisely *declining a non-identifiable alignment*. This is the quantity the categorical model computes that a re-description does not: a scalar that is zero iff the collective's local frames are coherently glued, and it is this gluing — not a shared coordinate convention — that §49's merge silently presupposed.

49 Categorical BDI reading of collective semantic memory

This section gives a compact theoretical reading of the water-search example in terms of belief, desire, and intention. The goal is not to introduce a new planner, but to make explicit the mathematical structure already used by the collective semantic memory architecture.

We assume a finite set of agents $A = \{1, \ldots, N\}$, a shared spatial basis X, and a shared semantic alphabet Σ. In the present grid-world implementation, X is the common coordinate frame and Σ contains tags such as `water_source`, `water_candidate`, `hazard_edge`, and `open`. This is an explicit modelling assumption. The agents may have different observations, different memories, and different confidence values, but they interpret places in a common coordinate framework and express semantic evidence in a common tag alphabet. Without the first assumption, peer merge requires a place-alignment problem to be solved first — Section 48 solves it constructively (fingerprint consensus recovers the inter-agent frame), and Section 49.1 states what that construction is, categorically. Symbol alignment (a shared Σ) remains assumed.

What parameterizes commonality of interpretation. It is worth stating explicitly that the categorical setup localizes "how much the agents mean the same thing" in exactly three parameters with three different statuses. *(i) The alphabet* Σ — what counts as the same *predicate* — is a parameter of the category itself (the subscript of $\mathbf{Ctx}_\Sigma$) and is here *assumed*; relaxing it means constructing a translation between $\mathbf{Ctx}_{\Sigma_1}$ and $\mathbf{Ctx}_{\Sigma_2}$, which we leave open. *(ii) The alignment morphisms* of the merge diagram — what counts as the same *place* — were assumed in earlier drafts and are now *constructed* (Sections 48 and 49.1), with their existence an empirically decidable predicate. *(iii) The trust×staleness enrichment* — how much a peer's interpretation *counts* — makes commonality graded rather than binary, with its own closed-form horizon and a discrete trust-flip. The three are ordered by hardness: weights are tunable, place correspondence is solvable from co-observation, and predicate correspondence is genuinely open. The same triple reappears one level up, at the evaluator: scoring retrieval and materialization presupposes a correspondence between memory entities and world referents (see the companion measurement paper's limitations), and grounding *that* correspondence by the mechanisms of Section 48 is future work.

Parameter (i) has, in addition, a measurable *interior*: within a fixed Σ, which tag classes actually carry place correspondence is an empirical property, not a modelling choice. A registered vocabulary ablation in the companion route-transfer paper makes the alphabet itself the treatment and finds the identity machinery *fail-safe in* Σ: impoverishing the vocabulary down to a single content class converts frame recovery into refusal, never into a wrong alignment (zero fail-open transports across 252 cells), and the consensus mass is attributable per class (hazards 0.56, walls 0.44, world borders 0.00). What no mechanism yet does is *select* Σ: here features receive landmark status from design plus the world's cooperation. A rule by which features *earn* that status through participation in successful alignments — the alphabet shaped by coordination rather than assumed — is the sharpest open end of parameter (i), and the natural meeting point with the symbol-alignment program of §49.4.

At time t, agent i maintains a local semantic context $\mathcal{K}_t^i = (G_t^i, \Sigma, I_t^i)$, where G_t^i is the set of place records stored in the agent's private memory, and

$$I_t^i \subseteq G_t^i \times \Sigma$$

is the thresholded incidence relation saying that place $g \in G_t^i$ carries tag $\sigma \in \Sigma$ with sufficient confidence. The graded implementation uses tag confidences and weighted scores; the crisp relation I_t^i is the thresholded skeleton needed for the theoretical description.

This formal context induces a Galois connection between sets of tags and sets of places:

$$\mathrm{ext}_t^i(B) = \{g \in G_t^i : \forall \sigma \in B,\ (g,\sigma) \in I_t^i\}, \qquad \mathrm{int}_t^i(S) = \{\sigma \in \Sigma : \forall g \in S,\ (g,\sigma) \in I_t^i\}.$$

For a water-search task, the desired semantic predicate is $B_{\mathrm{water}} = \{\texttt{water\_source}\}$, or, in the implemented weighted form, a predicate combining positive evidence for `water_source`, `water_candidate`, and `water_nearby` with penalties for `hazard_edge` or nearby risk. Retrieval succeeds for agent i at time t exactly when

$$\mathrm{ext}_t^i(B_{\mathrm{water}}) \neq \varnothing.$$

Thus the retrieval stage is not a coordinate lookup. It is the test that the queried semantic meaning already has a non-empty extent in the agent's realized memory.

We can now define the BDI components.

Belief. The belief state of agent i is the structured semantic memory induced by $\mathcal{K}_t^i$. Equivalently, it is the concept lattice $L_t^i = \mathrm{Concepts}(\mathcal{K}_t^i)$, whose nodes pair extents of places with intensions of tags. A water belief is not the proposition "there is water at coordinate x." It is the region of L_t^i whose intension contains the water predicate. If this region is empty, the agent does not merely choose badly; the relevant semantic object is absent from its current memory.

Desire. Desire is a state-conditioned semantic predicate over the belief lattice. In the water example, the agent's internal state contains a thirst variable. The active desire can be written as

$$D_t^i = \begin{cases} B_{\mathrm{water}}, & \text{if } \mathrm{thirst}_t^i \geq \theta_{\mathrm{on}}, \\ B_{\mathrm{goal}}, & \text{otherwise.} \end{cases}$$

Thus desire is not an additional metaphysical object. It is the rule that selects which semantic predicate should be queried from the current belief structure.

Intention. Intention is the commitment to one materialized candidate. Once desire has selected a predicate and retrieval has produced a non-empty extent, the planner applies a satisficing choice function

$$\chi_t^i : \mathcal{P}(G_t^i) \to G_t^i \cup \{\perp\}, \qquad g_t^{i,\star} = \chi_t^i\big(\operatorname{ext}_t^i(D_t^i)\big),$$

where $\perp$ denotes failure to materialize a target. In Q/R/M/C terms, desire determines the query $Q^\star$, belief supports retrieval $R^\star$, and intention corresponds to materialization $M^\star$. Completion $C^\star$ is downstream of this semantic structure: it depends on whether the physical controller reaches the locked target and whether the target remains available under multi-agent occupancy.

This gives a compact categorical reading. Let $\mathbf{Ctx}_\Sigma$ be the category whose objects are semantic contexts (G, Σ, I) over the fixed alphabet Σ, and whose morphisms preserve semantic incidence. A local memory trajectory for agent i is a time-indexed diagram

$$\mathcal{K}_0^i \to \mathcal{K}_1^i \to \cdots \to \mathcal{K}_t^i \to \mathcal{K}_{t+1}^i \to \cdots$$

in $\mathbf{Ctx}_\Sigma$. Each arrow is induced by observation and memory update. When new evidence is consolidated, the formal context changes; therefore its concept lattice changes. This is the order-theoretic content of memory growth.

In Q/R/M/C terms, an empty extent $\operatorname{ext}_t^i(D_t^i) = \varnothing$ is a belief/retrieval failure, a non-empty extent with no stable lock is a materialization failure, and a successful lock that cannot be reached is a completion failure — the semantic and the physical parts of the agent's state fail in different, separately measurable ways.

The collective case lifts this by one construction. At broadcast times the receiver i holds a diagram of local memory objects $\mathcal{D}_t^i = (\mathcal{K}_t^i, \mathcal{K}_t^{j_1}, \ldots, \mathcal{K}_t^{j_m})$ together with alignment maps, and queries the merged consensus object $\widetilde{\mathcal{K}}_t^i = \mathrm{Merge}_{\text{trust,stale}}(\mathcal{D}_t^i)$, with peer evidence weighted exactly by the rules of Section 47 ($\mathrm{trust}_{j\to i} \cdot e^{-\alpha\,\mathrm{age}_j}$). The same BDI definitions apply to the merged context: intention is now selected from $\operatorname{ext}_{\widetilde{\mathcal{K}}_t^i}(D_t^i)$ rather than from the private extent alone.

This explains why communication cadence produces a bottleneck shift. Faster broadcast increases the freshness of peer evidence. This tends to enlarge or improve the water extent $\operatorname{ext}_{\widetilde{\mathcal{K}}_t^i}(B_{\text{water}})$, so agents can materialize targets more reliably; in Q/R/M/C terms, this raises $P(M^\star \mid R^\star)$. But the same fresh shared belief also synchronizes intentions. Several agents may lock onto the same scarce water source. The memory is semantically accurate, but the physical target is now contested. This lowers $P(C^\star \mid M^\star)$. The bottleneck is therefore physically grounded. It is not a formal artifact of the decomposition: it arises because semantic belief is shared through memory, while completion is constrained by occupancy and scarce resources. Collective semantic memory must therefore regulate not only how much information is shared, but also how it is merged, whose evidence is trusted, and when stale evidence should stop influencing intention formation.

In this sense the "collective" is not a central mind and not a global map but a stabilised family of peer-indexed processes; the next subsection makes that statement precise.

The dual, process-level reading — agents as coalgebras (codata), the collective as a final coalgebra, communication as depth-one corecursion, and the bottleneck shift as a competition between belief- and intention-bisimilarity — is developed in Appendix 52. It is a lens, not a mechanism: nothing downstream depends on it, which is why it lives in the appendix.

49.1 Colimit merge and its invariants

The merge object above can be described more precisely. Working in a timestamp- and provenance-enriched version of $\mathbf{Ctx}_\Sigma$, a snapshot merge from agents $1, \ldots, N$ is a diagram of local memory objects plus alignment maps induced by the place-identity relation. The consensus object $\widetilde{\mathcal{K}}_t^i$ is the colimit of this diagram, implemented concretely by the union-find merge over aligned concepts. This reading gives three useful constraints.

1. **Order independence.** The merged object is unique up to isomorphism, so the result is not an artefact of processing agent snapshots in a particular order.
2. **Chaining is expected.** If concept A aligns with B and B aligns with C, union-find performs the transitive closure. Over-merging is therefore not a bug in the data structure; it is the cost of the chosen equivalence relation.

3. **Decay is not a morphism.** Staleness is kept as a weight field outside the alignment morphisms. Otherwise age-then-merge and merge-then-age need not commute, breaking the clean interpretation of broadcast snapshots as one diagram.

In earlier drafts this section ended by conceding that the model does not answer which perceptual features should identify two places — the alignment maps were assumed, not constructed. That gap is now closed by Section 48, and the closure gives the colimit reading its missing scientific function:

1. **The alignment morphisms are constructed, not assumed.** Fingerprint consensus IS an algorithm that produces the alignment maps of the diagram — mutually-unique landmark matches determine the frame transform through which every local memory object embeds.
2. **Faithful consensus becomes an empirical predicate.** A unique faithful alignment edge is instantiated only when semantic consensus is sufficiently discriminative. The colimit itself still exists for an unaligned or disconnected diagram (§49.2: $\mathbf{Pos}$ is cocomplete), but it then represents separated components — a coproduct of sub-consensuses — rather than a single consensus-spanning memory. In a 180°-symmetric world the available observations do not identify a unique admissible alignment morphism (two incomparable diagrams support equal consensus), and the runtime therefore refuses to instantiate the edge — the fail-closed behavior of Section 48 is exactly this non-instantiation made operational. The predicate is measured, not asserted: the adjunction round-trip defect of §48 is exactly 0 when the pairwise alignment is a strict adjoint (in every recovered translation and 90°-rotation case, $30/30$) and the edge is left undefined when no faithful adjoint exists (fail-closed, $30/30$, with zero spurious gluings); a scalar witness for whether the colimit is consensus-spanning rather than fragmented.
3. **Decay stays outside the morphisms** (invariant 3), and empirically the trust$\times$staleness gate is unchanged by frame recovery — the same integer breakpoints with and without a shared frame — confirming that the two layers commute as the diagram reading requires.

49.2 Collective memory as a coherent gluing of local place categories

The colimit merge of §49.1 *assumed* the alignment maps and §48 *constructs* them; we can now state, as the central categorical result, what object the collective memory is when agents hold *different* frames, and exactly when it exists faithfully.

Setup. Agent i's realized memory is a thin category (a poset): its concept lattice $L_i = \mathrm{Concepts}(\mathcal{K}_i)$ over i's private frame, objects the places, order the extent inclusion of §49. Frame recovery (§48) yields, on the co-observed subposet, a monotone map $F_{ij} : L_i \to L_j$ that transports i's places into j's frame, and its reverse F_{ji}. Because the categories are *thin*, a recovered pair (F_{ij}, F_{ji}) is an *adjoint pair* precisely when it is a Galois connection, with unit $\eta^{ij} : \mathrm{Id}_{L_i} \Rightarrow F_{ji} \circ F_{ij}$ — the adjunction defect measured in §48.

Definition (coherent alignment system). The family $\{F_{ij}\}$ over the communication graph is *coherent* if (i) each pair is an adjoint order-embedding (η^{ij} an isomorphism, i.e. defect 0: no place is aliased onto a distinct one), and (ii) it satisfies the cocycle condition $F_{jk} \circ F_{ij} = F_{ik}$ on triple overlaps.

Proposition (two agents). For an adjoint pair $F \dashv G$ between L_A, L_B, the collage (comma) category $\mathrm{Gl}(F)$ — objects $L_A \sqcup L_B$, cross-hom $\mathrm{Gl}(F)(X, Y) = L_B(FX, Y) \cong L_A(X, GY)$ (Mac Lane, 1998) — contains L_A, L_B as full subposets and is the two-agent collective memory; the defect η_X is the obstruction to X round-tripping to itself, and the gluing collapses distinct places iff η is not an isomorphism.

Theorem (collective memory = colimit of the alignment diagram). Let $\mathbf{D}$ be the diagram in $\mathbf{Pos}$ (equivalently the enriched $\mathbf{Ctx}_\Sigma$) whose vertices are the L_i and whose edges are the F_{ij}. The collective memory is $\mathrm{colim}\,\mathbf{D}$. As $\mathbf{Pos}$ is cocomplete this colimit always exists; it is a *faithful* gluing — every L_i embeds and no two distinct places are identified — *iff* the alignment system is coherent. When all $F_{ij} = \mathrm{id}$ (a shared frame) $\mathbf{D}$ is the constant diagram and $\mathrm{colim}\,\mathbf{D}$ is exactly the union-find colimit merge of §49.1; the general statement removes the shared-frame assumption by replacing identities with the recovered adjoint embeddings.

Argument. Cocompleteness of $\mathbf{Pos}$ realizes the colimit as the disjoint union quotiented by the preorder generated by the F_{ij}-identifications. Each canonical injection $L_i \hookrightarrow \operatorname{colim} \mathbf{D}$ is faithful iff those identifications add no collapse inside an L_i, which holds iff every F_{ij} is an order-embedding (unit iso, defect 0) and the identifications agree across overlaps (cocycle) — i.e. coherence. The shared-frame case is the constant diagram, whose colimit is the componentwise union, i.e. union-find. □

Corollary (two computable witnesses). Coherence is checkable without solving general semantic alignment. (a) The pairwise *adjunction defect* η (§48) is zero iff each edge is a strict adjoint embedding — measured exactly 0 in $30/30$ recovered translation and rotation cases — and guards against *over-merge*. (b) The *connectedness* of the trust×cadence reachability graph (§49.3, Figure 22) decides whether a single $\operatorname{colim} \mathbf{D}$ spans all N or the diagram disconnects into a coproduct of sub-consensuses (*fragmentation*). The two failure modes are categorically distinct and separately measured: defect > 0 is collapse, disconnection is fragmentation — the two axes of Figure 22 and the fail-closed behavior of §48.

Scope. The proofs are the standard cocompleteness of $\mathbf{Pos}$ and the collage construction (Mac Lane, 1998); the theorem's job is to name the object and pin its existence to two measured quantities. The empirical content — that fingerprint recovery actually yields the adjoint embeddings — lives in §48, not in the theorem. One assumption is *not* discharged: a shared tag alphabet Σ; with different alphabets the F_{ij} would have to translate Σ as well, which is beyond the present correspondence mechanism and remains unresolved.

49.3 Protocol parameters as categorical structure: a trust×cadence phase diagram

The reading so far models the *memory* (objects are semantic contexts, merge is their colimit) under a shared frame and full exchange. Its predictive use comes from turning the lens on the *communication structure* and asking *what shape the colimit takes* as a function of the two protocol parameters — so that the category computes a structural fact rather than re-describing an algorithm. (In $\mathbf{Pos}$ the colimit of the diagram always *exists* by cocompleteness, disconnected or not; the structural question is whether it is a single consensus component or a coproduct of sub-consensuses, exactly the dichotomy of §49.2.) Two parameters play two structural roles:

Trust τ gates morphisms. An edge $i \to j$ (agent i incorporates j's evidence) exists only where i's trust in j is at least τ. Raising τ prunes the trust graph.

Cadence K gates composition depth. Over a horizon H there are $R = \lfloor H/K \rfloor$ broadcast rounds, so a composed chain $i \to k \to \cdots \to j$ can be at most R morphisms long. Low K (many rounds) allows deep composition; high K allows only short paths.

A *consensus-spanning* colimit — a single connected consensus component that fuses and serves every agent's evidence — obtains iff the R-round reachability of the τ-thinned trust graph is all-to-all; otherwise the colimit degenerates into a coproduct of sub-consensuses.

Figure 22 sweeps $\tau \times K$ for N=8 agents whose pairwise trust is produced by the system's own EMA rule (`peer_memory.peer_trust.AsymmetricTrust`) under a ring co-observation topology (neighbours co-observe often and earn high trust; distant agents do not), averaged over 12 seeds (`experiments/collective_semantic_memory/phase_diagram_trust_cadence.py`). The result is a genuine phase transition with a diagonal boundary. At low τ the trust graph is dense and the colimit spans all agents at every cadence. Raising τ prunes the long-range morphisms down to a near-neighbour ring; the collective then keeps a spanning colimit *only if* cadence is low enough for multi-hop composition through still-trusted intermediaries to reconnect it — τ=0.55 holds the spanning colimit down to R=2, τ=0.80 needs $R \geq 4$ (the ring diameter), and τ=0.90 fragments throughout. Cadence and trust are therefore not opaque knobs: together they determine a categorical structural invariant, and the phase boundary is exactly where the collective transitions from a single consensus object to fragmented sub-consensuses.

This is the sense in which the categorical model is operational rather than descriptive: each protocol parameter maps to a concrete change in the communication diagram, and the diagram's structural invariant (whether the colimit spans all agents or splits into a coproduct) predicts a qualitative regime of the collective. *Scope.* The trust matrix is produced by the real EMA rule under a modelled

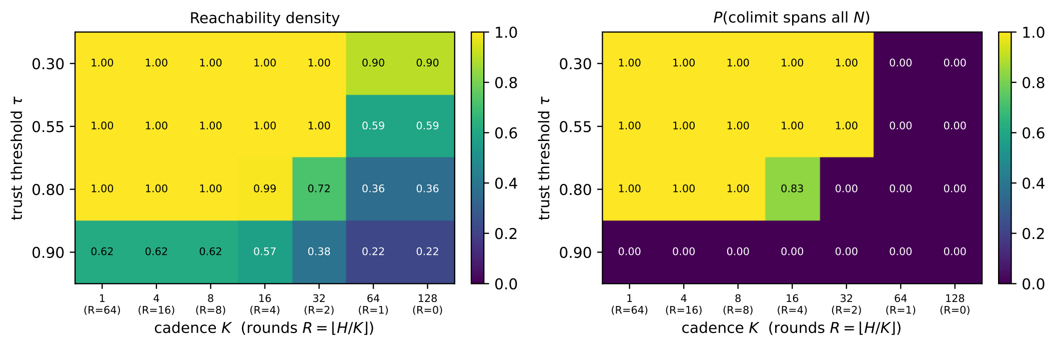

Figure 22: Communication-structure phase diagram over trust threshold τ (rows) and cadence K (columns; $R = \lfloor H/K \rfloor$ broadcast rounds), $N{=}8$ agents on an interaction ring, 12 seeds. **Left:** reachability density (fraction of ordered agent pairs whose evidence can compose within R rounds). **Right:** P(the colimit spans all N), i.e. all-to-all reachability. The diagonal boundary is a structural transition: the collective admits a single consensus object in the low-τ/low-K region and fragments in the high-τ/high-K corner, with higher τ demanding lower K (more rounds) to stay connected.

ring co-observation topology, and "consensus-spanning" is operationalized as all-to-all R-round reachability; establishing quantitatively that crossing this structural boundary predicts the drop in observed Q/R/M/C performance is the natural next experiment.

49.4 Toward symbol alignment: grounding tag alphabets through shared places

The gluing theorem (§49.2) still assumes a shared alphabet Σ. Dropping it, each agent holds a context (G_i, Σ_i, I_i) over its *own* vocabulary, and the right morphism is no longer a functor over one Σ but an *infomorphism* (Barwise and Seligman, 1997): a pair $f : G_A \to G_B$, $\hat{f} : \Sigma_B \to \Sigma_A$ with $I_B(fg,\, y) \Leftrightarrow I_A(g,\, \hat{f}y)$. Infomorphisms compose and have colimits (Barwise–Seligman channels), so the collective memory over heterogeneous alphabets is the *channel core* — the colimit of the infomorphism diagram — which extends §49.2 verbatim (functors over one Σ become infomorphisms over the Σ_i; shared Σ with $\hat{f} = \mathrm{id}$ is the old case).

Grounding mechanism. Symbols align through shared referents: on the places already anchored by frame recovery (§48), the co-occurrence of an A-tag and a B-tag is evidence they name the same feature, and $\hat{f}$ is read off the anchored co-occurrence. (When the alphabets differ so much that fingerprints themselves cannot seed the anchors, one bootstraps from the alphabet-independent structural constellations and alternates recovery of f and $\hat{f}$.) The *symbol-alignment defect* def_Σ — the fraction of anchored (place, y) pairs violating the infomorphism biconditional — is the symbol analogue of the frame adjunction defect of §48.

Preliminary result. A controlled study (`experiments/warp/symbol_alignment.py`; $|\Sigma_A|{=}8$, 120 places, 20 seeds) gives agent B a secret relabelling of A's alphabet with observation noise, and recovers $\hat{f}$ from anchored co-occurrence.

Table 31: Symbol alignment by grounding vs. assuming identical alphabets. Target-ID F1 is identifying A's target feature from B's (translated) evidence. "naive" assumes the alphabets coincide by index; "recovered" learns $\hat{f}$; "oracle" is the true map.

noise	transl. acc.	def_Σ	F1 naive	F1 recovered	F1 oracle
0.00	1.00	0.000	0.56	1.00	1.00
0.10	1.00	0.099	0.55	0.92	0.92
0.30	0.89	0.304	0.56	0.69	0.73

Assuming identical alphabets fails (F1 ≈ 0.55: the relabelling sends B's target tag to the wrong feature); grounding recovers the map (translation accuracy 1.00, def_Σ tracking the noise floor) and reaches oracle-level target identification, robustly down to 15% anchored places (accuracy 0.95). Coarsening two features into one B-tag bounds the achievable accuracy — there is no faithful bijection, so the channel core coarsens rather than errs. Symbol alignment is thus not a wholly open problem but the same grounding-plus-defect recipe one alphabet up; a full treatment (genuine incommensurability, online drift) is the next paper.

50 Limitations and future work

The minimal CSM is evaluated primarily in controlled semantic navigation environments. The merge, trust, and staleness rules are explicitly specified rather than learned, and their transfer to richer perceptual or social settings remains to be established. The semantic place-correspondence mechanism assumes that the candidate vocabulary contains sufficiently discriminative local features; open symbol creation and heterogeneous sensor grounding are not solved in this part. The categorical results formalize coherent gluing in $\mathbf{Pos}$ and should be interpreted accordingly. In particular, a colimit exists for disconnected diagrams; the phase analysis concerns whether one accepted correspondence component spans all agents, not whether a colimit exists at all. The coalgebraic section is an interpretive model of continuing behavior rather than a new control algorithm. Finally, the LLM context comparison uses two model families and controlled tasks. It shows that structured consolidation can resolve specific freshness and identity failures, but it does not establish general superiority over all long-context or production memory architectures.

51 Conclusion

We introduced a minimal collective semantic memory in which merge, trust, and staleness are represented as separate, inspectable mechanisms. The experiments show that collective performance is not a monotone function of shared-information volume or communication frequency. The effective component is the admissibility gate: agents benefit when foreign evidence is merged with source information, weighted by trust, and removed from action authority as it becomes stale.

The same architecture can operate without a shared coordinate frame when local place records are aligned through semantic fingerprints. The formal analysis characterizes this alignment as a coherent gluing of local ordered memories and clarifies the distinction between a single population-spanning consensus component and several sub-consensuses. The coalgebraic interpretation complements this static description by representing the collective as an ongoing behavioral process.

The LLM experiments reinforce the central systems conclusion. Raw context can contain all historical evidence while still failing to resolve which information is current or which records refer to the same world entity. A consolidated graph supplies these relations explicitly and reduces the active context required by the model. The remaining problem is not how to coordinate a memory that already exists, but how agents should learn which observations, distinctions, and route structures deserve to become long-term memory. Part V addresses this formation problem.

52 The coalgebraic layer: individuals as codata, the collective as a final coalgebra

The reading of §49 fixes belief *content*: at a state, the concept lattice $L_t^i = \mathrm{Concepts}(\mathcal{K}_t^i)$ says what is retrievable, and the merge of §49.1 says how peer contents combine into $\widetilde{\mathcal{K}}_t^i$. Both are *algebraic*: they build a belief object. Neither speaks to the agent as a non-terminating *process* that must emit an action and a signal at every tick under a moving environment, nor to the sense in which a *population* comes to share a convention. That process side is *coalgebraic* (Rutten, 2000; Jacobs, 2016). We make it precise on the water-search example and show the two sides are dual, meeting inside a single state transition. (The belief/desire/intention vocabulary follows the BDI agent tradition (Rao and Georgeff, 1995), here given coalgebraic content.)

Signature. Fix an action set Act (move, dig, noop), a per-agent observation set O_i (agent i's local percept — its *unique view*), and an *ostension* set

$$M \;=\; 1 \;+\; (X \times \Sigma),$$

whose values are either *silence* (the summand 1) or a pointer "place x carries tag σ" (whistle-and-point at a water-marked cell). That silence is a first-class value of M matters: the ostension channel *can be informatively empty*, and it is separate from the state update below. This is exactly the observability that a learned real-valued message vector destroys (its channel is never empty and never separates signalling from stepping), and it is why the symbolic stages remain readable.

State. We take the coalgebra carrier to be the H-MDP high-level state of §49,

$$\mathcal{S}_i \;=\; \big\{\, h = (\mathcal{K}^i,\, D^i,\, g^{i,\star},\, s^i) \,\big\},$$

carrying the same information as $(L^i, D^i, g^{i,\star}, s^i)$ since $L^i = \mathrm{Concepts}(\mathcal{K}^i)$. Thus *belief lives in the state*; the lattice is a projection of it.

Definition (codata: the agent's cotype). The observable behavior of agent i is a coalgebra for the endofunctor

$$B_i(S) \;=\; \underbrace{\mathrm{Act}}_{\text{act now}} \;\times\; \underbrace{M}_{\text{signal now}} \;\times\; \underbrace{S^{\,O_i \times M^{A\setminus\{i\}}}}_{\text{continue given (my next percept, peers' signals)}},$$

i.e. a map $c_i \;=\; \big\langle\, \texttt{observe\_act}_i,\; \texttt{observe\_ostension}_i,\; \texttt{step\_intension}_i \,\big\rangle \;:\; \mathcal{S}_i \;\to\; B_i(\mathcal{S}_i)$ with three destructors

$$\texttt{observe\_act}_i : \mathcal{S}_i \to \mathrm{Act}, \quad \texttt{observe\_ostension}_i : \mathcal{S}_i \to M, \quad \texttt{step\_intension}_i : \mathcal{S}_i \to \mathcal{S}_i^{\,O_i \times M^{A\setminus\{i\}}}.$$

Here $M^{A\setminus\{i\}} \;=\; \prod_{j\neq i} M$ is the tuple of peers' ostensions delivered this tick. Operationally, $\texttt{step\_intension}_i(h)(o, (m_j)_{j\neq i})$ (i) folds the percept o and the peer pointers (m_j) into the context by the trust/staleness merge, $\widetilde{\mathcal{K}}^i_t = \mathrm{Merge}_{\mathrm{trust,stale}}(\cdots)$ of §49.1; (ii) recomputes the lattice; (iii) re-evaluates the desire predicate D^i; and (iv) re-locks the intention $g^{i,\star} = \chi^i_t(\mathrm{ext}(D^i))$ of §49. So Merge and χ are *components of one destructor*. The agent *is* the codata generated by c_i: it has no finite terminal plan, only the standing ability to emit an action and an ostension and to continue. This is the "infinite survival generator" of the motivating example.

Definition (indexed family). The system is the A-indexed family $\{(\mathcal{S}_i, c_i)\}_{i\in A}$, a coalgebra in $\mathbf{Set}^A$. The functor B_i depends on the index i through O_i (and through the peer set $A \setminus \{i\}$): the same survival cotype is instantiated at each agent through that agent's local viewpoint. Agent 1 on the dune generates from "I see the horizon"; agent 2 in the hollow generates from "I feel the soil." The index is the viewpoint.

Definition (codependence: the collective coalgebra). Let the environment deliver a joint percept $o = (o_i)_{i\in A} \in \prod_i O_i$. The collective one-tick map on $\prod_i \mathcal{S}_i$ wires each agent's ostension output into the others' step inputs,

$$\Phi\big((h_i)_i\big)(o) \;=\; \Big(\; \texttt{step\_intension}_i(h_i)\big(o_i,\, (\texttt{observe\_ostension}_j(h_j))_{j\neq i}\big) \;\Big)_{i\in A}.$$

Because i's next state is a function of the *destructor outputs* of the others, the family is not N independent processes but one coalgebra for the product functor $\mathbf{B}(S) = \big(\prod_i (\mathrm{Act}\times M)\big) \times S^{\prod_i O_i}$ on $\prod_i \mathcal{S}_i$. This wiring is the *codependence*: agent 1 cannot step without passing agent 2's ostension through its own `step_intension`. Broadcast cadence enters here as a gate: at non-broadcast ticks the delivered tuple (m_j) is the last received (stale) tuple, or silence; K is the delivery period.

Definition (behavior and bisimulation). B_i is a container (polynomial) functor, so it has a final coalgebra $(\nu B_i,\, \mathrm{beh}_i)$ (Aczel, 1988; Rutten, 2000) whose elements are the $\big(O_i \times M^{A\setminus\{i\}}\big)$-branching, $(\mathrm{Act} \times M)$-labelled behavior trees, and there is a unique coalgebra morphism $!_i : \mathcal{S}_i \to \nu B_i$ sending a state to its whole future behavior. A B_i-*bisimulation* is a relation $R \subseteq \mathcal{S}_i \times \mathcal{S}_i$ under which related states emit equal action and equal ostension and, on every input, step to R-related states. Container functors preserve weak pullbacks, so the *greatest bisimulation* $\sim_i$ exists, is an equivalence, and equals $\ker(!_i)$: two states are bisimilar iff they have the same behavior (Sangiorgi, 2011). The same holds for the collective coalgebra $(\prod_i \mathcal{S}_i,\; \langle(\texttt{observe\_act}_i)_i, (\texttt{observe\_ostension}_i)_i, \Phi\rangle)$, with greatest bisimulation $\approx$.

Definition (society / culture). The *collective* is the image of the states in the final coalgebra $\nu\mathbf{B}$, equivalently the quotient $\left(\prod_i \mathcal{S}_i\right)/\approx$. It is neither a central server nor a shared map: it is a bisimilarity class of joint behavior. A *water-search convention* is this invariant read on the water sub-signature. Let $M_{\text{water}} \subseteq M$ be the pointers whose tag lies in the water family, and let $\nu B_i \to \nu B^{\text{w}}$ be the forgetful map to the behavior trees restricted to receiving M_{water} and emitting the corresponding intention lock. The population *holds the convention* when all agents' reachable behaviors agree under this map: hearing "point-and-dig at a water-marked place" drives every agent's intention destructor to lock that place (up to occupancy). Governance is then automatic in both directions. A new agent (the example's fourth agent) is a member of the society exactly when its behavior lands in the class $[\cdot]_\approx$ on M_{water} — otherwise its ostensions are not understood and its own step never converges (top-down); and the class itself was reached by the local, coupled steps of the first agents at one green shrub (bottom-up).

Remark (communication = lazy, depth-one corecursion). The behavior $!_i(h)$ is defined by unfolding c_i indefinitely. On receiving $m_j = \texttt{observe\_ostension}_j(h_j)$, agent i uses only this *depth-one* destructor output inside $\texttt{step\_intension}_i$; it does *not* compute $!_j(h_j) \in \nu B_j$, the peer's full behavior, which is the "I think that you think. . . " regress. Communication is therefore corecursion truncated to the least depth that resolves the receiver's own next step — a finite approximation of the final-coalgebra map — with a relevance-theoretic stopping rule (Sperber and Wilson, 1995) deciding that depth. The regress collapses into one effective action.

Remark (algebra–coalgebra bridge to §49–49.1). Belief is algebraic: the FCA lattice L_t^i built by the colimit merge is the *content at a state*. The agent is coalgebraic: the codata observed by the three destructors is the *process over states*. They are dual and meet in one place — the colimit merge and the satisficing choice χ are the internals of `step_intension`. The memory-growth diagram $\mathcal{K}_0^i \to \mathcal{K}_1^i \to \cdots$ of §49 is precisely the belief-projection of the coalgebra trajectory $h_0^i \mapsto h_1^i \mapsto \cdots$.

Coalgebraic reading of the bottleneck shift. The result of §46 and Table 30 has a one-line reading here. Two kinds of bisimilarity compete. Bisimilarity on the *ostension / belief* sub-signature is the wanted invariant: it is the shared water convention. Bisimilarity on the raw *intention* output over a *scarce* target is the unwanted one: several states mapping to the same $g^\star$ is a collision. Occupancy $\mathcal{O}_{-i}$ is the extra-coalgebraic constraint that penalises the second. Cadence K tunes how hard the coupling Φ drives the collective toward the *full* greatest bisimulation $\approx$: small K delivers fresh ostensions often, enlarging the merged water extent so intention locks reliably ($P(M^\star \mid R^\star)\uparrow$), but over-synchronises the intention destructor so agents collide ($P(C^\star \mid M^\star)\downarrow$). The interior optimum $K{=}8$ is where belief converges without intention collapsing onto one cell; the minimal CSM adds trust and staleness precisely so that `step_intension` reaches belief-bisimilarity (fresh, trusted, aligned evidence) while damping intention-bisimilarity (stale evidence decays before it can herd every agent onto the same lock). We state this as a reading consistent with the data, not as a theorem: the container functors, their final coalgebras, and bisimulation are standard constructions used here as a lens, and the relevance-theoretic stopping rule is an interpretive gloss rather than a fitted mechanism.

53 Structured memory versus raw context under a fixed model

Modern LLMs *accept* long context; the question this appendix measures is whether they turn it into a stable, globally coherent state — and whether the Songlines graph supplies that state instead. In every arm the *same* model answers; only the memory substrate differs: **raw** (the full observation transcript in the prompt — typical usage) vs. **graph** (the same evidence consolidated by the Songlines rule: latest state per place, staleness explicit, O(salient places) prompt). Three axes (`experiments/context_vs_structure/`; HF backend on SLURM GPU nodes and local Ollama as an independent second backend).

Axis 1: volume and integration (one witness log). A witness log over a $30{\times}24$ grid contains a freshness conflict (water seen repeatedly at site A, later dried; one fresher sighting at B). Questions: *fact* (where is water now), *count* (how many waters now), *hop* (nearest hazard to current water).

Two failure layers, both with identified mechanisms. *Cognitive:* the raw arm loses global state at **any** volume — on *count* it answers "2" (17/20) or "3" (3/20) when the truth is 1: it adds up historical mentions instead of resolving freshness, i.e. no consolidation. At large L even single-fact retrieval

Table 32: The motivating example's terms as formal objects of the coalgebraic reading. Each row is the water-search meaning of a construction already used in §52.

Informal term	Formal object	Water-search meaning
Cotype (codata)	B_i-coalgebra $(\mathcal{S}_i, c_i)$	the agent's non-terminating action/signal generator
Indexed type	family over A; dependence of B_i on O_i	the agent's unique local viewpoint (dune vs. hollow)
Codependent type	wiring $\texttt{observe\_ostension}_j \to \texttt{step\_intension}_i$ in Φ	"I step only through your ostension"
Individual (instance)	a state $h \in \mathcal{S}_i$ and its behavior $!_i(h)$	the running agent digging and whistling now
Destructors (cogenerators)	`observe_act`, `observe_ostension`, `step_intension`	step-and-dig; whistle-and-point; fold neighbour's signal into next state
Lazy evaluation	depth-one corecursion + relevance stop	unfold the peer's coalgebra one step, not its whole behavior
Final coinductive core / society	$\nu\mathbf{B}$; greatest bisimulation $\approx$ (on M_{water})	the stable, distributed water-search culture

Table 33: Accuracy by question, arm, and log volume L (lines). Qwen2.5-7B (cluster, 20 seeds/cell) on cells within its compute envelope; Llama-3.1-8B (local, 10 seeds/cell) at large L with *verified full ingestion* (`prompt_eval_count`=25,030 tokens at L=2000; no truncation). Graph prompts stay ≈350 bytes at every L.

Question	Arm	L=50	L=200	L=800	L=2000
fact	raw	1.00 / –	1.00 / –	– / 0.00	– / 0.00
fact	graph	1.00 / –	1.00 / –	1.00 / 1.00	1.00 / 1.00
count	raw	**0.00** / –	**0.00** / –	– / **0.00**	– / **0.00**
count	graph	1.00 / –	1.00 / –	1.00 / 1.00	1.00 / 1.00
hop	raw	0.00 / –	0.05 / –	– / 0.00	– / 0.00
hop	graph	0.00 / –	0.15 / –	0.40 / 0.50	0.30 / 0.60

collapses despite verified full ingestion (a probe with the answer at the very end of a 25k-token prompt is also missed). *Systemic:* on the cluster's 24 GB sm75 GPUs, raw-arm inference at 5–10k tokens exhausts memory through the quadratic attention matrix (marked as compute-envelope cells, "–", never reported as cognition), while the graph arm runs at constant cost at every volume. Raw context loses twice — as cognition and as systems cost; *hop* identifies the empirical boundary: structure repairs retrieval, not arithmetic reasoning.

Axis 2: time (sessions). A persistent world drifts (water relocates, p=0.35/session); after each of 8 sessions the model is asked where the water is now, from **raw replay** (all past transcripts re-injected, the typical chat pattern) vs. the persisted **graph**. Qwen2.5-7B: the graph arm holds 0.62–1.00 across all eight sessions at a flat ≈0.3 kB prompt; the replay arm's valid cells are already at 0.00–0.67 in the earliest sessions and its prompt grows to 25 kB by session 8 (later cells envelope-limited on the cluster; on the local backend, where no envelope exists, replay degrades to 0.00–0.50 by 19 kB — the same pattern).

Axis 3: system size (multi-agent, private frames). Three agents accumulate observations over G sessions, each logging in its *own* secret coordinate frame. Question: "how many DISTINCT water sources exist?" — the same physical water seen by two agents at different private coordinates must be counted once. The raw arm gets all three logs (and is told the frames differ); the graph arm gets the Songlines machinery's merged map (fingerprint frame recovery, §48, plus dedup). Qwen2.5-7B: graph accuracy *rises* with growth — 0.40, 0.90, 1.00, 1.00 at G=1, 2, 4, 8 (≈330 bytes) — because frame alignment locks as fingerprints accumulate (the exact% curve of the growth study); the raw arm (13k–105k chars) does not produce a single correct answer on any envelope-valid cell. Cross-frame identity is precisely what a context window cannot represent and the gluing structure can.

Summary. The same model, given the same evidence, preserves long, drifting, multi-agent context when — and only when — the evidence reaches it as the consolidated Songlines structure: flat-cost prompts, freshness resolved, frames glued. Given the raw stream instead, it fails on integration at 1.3 kB, on freshness at 19 kB with full ingestion, and on cross-frame identity at every tested size. Scope: two model families, one task family per axis, and the compute-envelope cells are reported as such; the graph arm's consolidation here is the deterministic Songlines rule — LLM-driven consolidation is the roadmap's next stage.

54 Reproducibility commands

- Main cadence sweep: `pythonexperiments/big_experiment/exp_cadence_phase.py--modefull--workers8--out_dirtmp/big_experiment_qrmc_40`
- Extended cadences: `pythonexperiments/big_experiment/exp_extra_K.py--workers8--out_dirtmp/big_experiment_extraK`
- Q/R/M/C analysis: `pythonexperiments/big_experiment/analyze_qrmc.py--runs_csvtmp/big_experiment_qrmc_40/runs.csv--out_dirtmp/big_experiment_qrmc_40`
- Minimal CSM benchmark: `python-mexperiments.collective_semantic_memory.run_csm_benchmark--out_dirtmp/csm_benchmark`
- MiniGrid portability sweep: `python-mexperiments.minigrid_multiagent_wrapper.run_portability_sweep--out_dirtmp/minigrid_portability`
- Adjunction round-trip defect of frame recovery (Section 48): `pythonexperiments/warp/alignment_defect.py`
- Context-vs-structure suite (Appendix 53): `experiments/context_vs_structure/run_ctx_vs_struct.py`, `run_persistence_llm.py`, `run_growth_llm.py` (HF backend on GPU or –backend ollama locally)
- Trust×cadence structural phase diagram (Section 49.3, Figure 22): `pythonexperiments/collective_semantic_memory/phase_diagram_trust_cadence.py--figfigures/fig_phase_diagram.pdf`

55 Asset and license note

The experiments use MiniGrid and MiniWorld (Chevalier-Boisvert et al., 2023). Locally implemented benchmark and memory code will be released under the MIT License as part of the artifact package.

Part V

Song Grammar and Utility-Certified Memory: the Ontogenesis of What Is Remembered

Memory formation as a two-axis decision (counterfactual utility times simplicity of analogy): measured deterministically, learned by a contextual bandit that outperforms the hand-designed rule, evolved to a finite budget, and priced against seven policies over heterogeneous embodiments.

Part abstract. The preceding parts assume that semantically structured memory records already exist. This part studies how such records are formed. We model memory formation as a bounded decision among five operations: merge, exception, new schema, repeat, and drop. The decision is based on two independent quantities. Counterfactual marginal utility estimates how a candidate record changes future task cost relative to the current memory, whereas structural analogy estimates whether the candidate can be assimilated into an existing schema without changing its decision consequences. This separation introduces an explicit exception operation for useful but conflicting evidence and retains immutable source episodes for audit and reconstruction. A deterministic prototype confirms the registered behavior of the five-way rule: duplicates have zero marginal utility, exception storage avoids the corruption produced by last-write-wins merging, and similarity alone is insufficient for consolidation. A contextual bandit learns the formation policy and outperforms the hand-designed thresholds (78.1 versus 83.5 task cost), while an evolutionary search selects a finite route-description budget within a fixed grammar family. Long-horizon comparisons show that structural memory substantially reduces storage relative to raw histories and full snapshots, but also demonstrate that consolidation alone does not preserve success under drift. Adding freshness and trust-aware conflict handling reduces this gap. Later experiments close the remaining runtime requirements at prototype scale: a utility estimator matches the exact counterfactual on held-out controlled worlds (Spearman 0.989), receiver-side admission changes long-horizon social exchange from harmful to beneficial (approximately 20% lower cost than independent memory), and the integrated runtime remains superior to raw history under equal memory and communication budgets. Causal landmark selection, graph-based analogy, terminal-reward learning, adversarial provenance, and fail-closed safety are evaluated through separate ablations. A reconstructed active context also outperforms raw conflicting transcripts in controlled LLM tests at 6–12% of the payload. The evidence establishes a complete research prototype for utility-certified, provenance-aware collective memory. It does not establish open-ended perceptual concept formation, broad social LLM generality, or real-robot validity.

56 Introduction: learning what to retain

The earlier parts of the series specify how structured memories are queried, transferred, aligned, merged, and consumed. They do not specify how an observation becomes a durable memory item. This is a separate problem. An effective long-term memory must distinguish between repeated evidence, useful exceptions, novel schemas, short-lived observations, and records whose apparent similarity would produce an incorrect action if merged.

The need for an explicit formation mechanism is already visible in the Q/R/M/C diagnosis. In the single-agent benchmark, candidate ranking is accurate when candidates exist, but the memory frequently contains no eligible record. The limitation is therefore not only retrieval; it is the preceding process that determines which observations are consolidated into queryable structure. In a collective, the same formation decision additionally affects communication cost, cross-agent transfer, and the possibility of propagating stale or misleading evidence.

We formulate memory formation through two independent axes. The first is counterfactual marginal utility: the expected change in task or team cost produced by adding a candidate record to the current memory. The second is structural assimilability: the cost of mapping the candidate onto an existing schema while preserving decision-relevant relations. Utility answers whether the record is worth retaining. Analogy answers whether it should update an existing schema or create a distinct one. A single importance, novelty, or similarity score cannot represent both questions.

The resulting controller selects among five operations:

Utility	Structural relation	Operation
high	compatible with an existing schema	**MERGE**
high	similar but decision-conflicting	**EXCEPTION**
high	not assimilable	**NEW_SCHEMA**
low	redundant with stored evidence	**REPEAT**
low	unsupported and non-redundant	**DROP**

The exception operation is essential. A useful observation that contradicts a stored schema should not overwrite the original evidence, but it should not be discarded either. It is retained as a context-dependent counterexample and resolved during later retrieval or schema splitting. This design is supported by an immutable episodic layer and by provenance links from schemas to their source episodes.

The empirical program proceeds in four stages. First, deterministic replay tests the five-way formation rule under controlled conflicts. Second, a contextual bandit and a terminal-reward learner evaluate whether the formation policy can be recovered from coordination feedback rather than fixed by the designer. Third, evolutionary search and structural matching test whether the route grammar and analogy mechanism can be compressed without losing decision quality. Fourth, an integrated multi-agent runtime evaluates receiver-side admission, world-time staleness, equal budgets, perceptual noise, adversarial provenance, continuous transfer, and LLM active-context reconstruction.

The contributions of this part are as follows:

1. We define a two-axis memory-formation model that separates counterfactual usefulness from structural assimilability and includes an explicit exception operation.
2. We specify a conditional memory record with provenance, per-role utility, applicability conditions, analogical mappings, and failure cases, together with a receiver-side admission process.
3. We show that the formation policy is learnable and that its parameters can be optimized within a fixed grammar family.
4. We evaluate the method over horizons of up to 1000 episodes, identify staleness as the principal long-horizon failure of consolidation-only policies, and measure the contribution of freshness and trust-aware conflict handling.
5. We integrate formation, communication, quarantine, admission, alignment, route consumption, reservation, rupture, and rollback into a single runtime and compare it with independent, raw-history, heuristic, and structurally ablated memories under equal budgets.
6. We evaluate utility estimation without oracle replay, causal landmark promotion from a fixed candidate set, graph-based analogy, LLM context reconstruction, adversarial provenance, and fail-closed safety on discrete, continuous, and VMAS substrates.

The scope is that of an end-to-end research prototype. The utility estimator is calibrated against exact counterfactual replay in controlled environments; its validity in external stochastic systems remains open. Landmark promotion operates over a predefined candidate superset rather than generating new perceptual concepts. The grammar parameters evolve within a fixed representation family. These restrictions are retained explicitly in the claims and limitations.

57 The model

Counterfactual marginal utility. For memory item m, agent i, intent ι and current memory M:

$$U_{i,\iota}(m \mid M) \;=\; \mathrm{cost}_i(M,\iota) \;-\; \mathrm{cost}_i(M \cup \{m\},\iota),$$

measured by *deterministic replay* with the item switched on and off — the source-masked replay device of the warp analysis, generalised from "foreign evidence on/off" to "this memory on/off." Utility is contextual (per agent and intent) and marginal (relative to what is already stored): a duplicate of a stored schema has $U = 0$ *by construction*, so repeats are detected without any similarity heuristic. The team-level version subtracts collision, communication and false-transfer costs.

Analogy cost. A new episode e maps onto a stored schema s through ϕ with cost

$$C_{\text{analog}}(e,s) = L(\phi) + \lambda_d\, D_{\text{decision}}(e,s) + \lambda_a\, \text{Defect}(\phi), \qquad A(e,s) = e^{-C_{\text{analog}}},$$

where L counts structure outside the correspondence (here: couplets outside the signature LCS), D_{decision} measures how differently the two songs *decide* — operationalized as the distance between their end-to-end beat displacements, i.e. where they ultimately send the walker — and Defect penalises broken relational structure. Descriptive similarity and decision agreement are different quantities, and D_{decision} captures the second; this is the rate-distortion intuition of Zou et al. (2026) imported into a multi-agent, structural setting.

The memory unit is a conditional program, not a fact. We fix a single canonical record for the whole part,

$$m = (G, C, E, U, P, T, F, A, H),$$

with G the relational graph/song, C applicability conditions, E expected effects, U the role-conditioned utility profile, P provenance (references to immutable source episodes), T observation and world time, F failures and exceptions, A admission and authority state, and H alignment requirements (and any temporary alignment hypotheses). Two things are deliberately *not* stored fields: the known analogical maps $\mathcal{A}_m$ and the concrete inter-frame correspondence $\Gamma_{ij,t}$ are both re-derived at consumption (from G and the receiver's own observations) rather than carried in the record — H records only what a valid correspondence would have to satisfy, not a fixed transform. On exchange, a schema travels with a **utility certificate** (conditions, ΔV, uncertainty, evidence ids, $\mathcal{A}_m$, failures); the receiver transports the song through its own mapping and recomputes its own utility. For heterogeneous embodiments this is essential: a route through a hazard field can have $U_{\text{robust}} > 0$ and $U_{\text{fragile}} < 0$ simultaneously, so a memory item carries a utility function over roles, not one weight.

Four storage layers. (1) an *immutable episodic store* — source episodes are never rewritten (continuous rewriting first helps and then hurts (Zhang et al., 2026b); primary evidence anchors reconsolidation); (2) the *analogical schema graph* — songs, landmarks, exceptions, governed by the two-axis matrix; (3) a *procedural layer* for schemas promoted to policy fragments; (4) a *coordination layer* for social rules (reservations, roles, trust, transfer conditions). An agent's active context is a reconstruction — relevant schemas, minimal supporting episodes, current exceptions and the coordination contract — not a replay of history.

57.1 The runtime as one algorithm

Every experiment below is a configuration of one per-agent loop; we state it once so the method is reconstructable from the paper alone. A record is the canonical $m = (G, C, E, U, P, T, F, A, H)$ of §57 — song graph, applicability, expected effect, per-role utility, provenance, observation/world time, failures and exceptions, admission/authority state, and alignment requirements H. The concrete inter-frame correspondence $\Gamma_{ij,t}$ is re-derived at consumption and never stored.

Algorithm 1 (Songline Memory Runtime).
In: observation o, intent ι, local memory M, quarantine Q.
1. extract candidate song from o; 2. estimate marginal utility $\widehat{U}$ (§65, UE1); 3. find structurally closest schema (analogy §63); 4. compute analogy cost + decision distortion; 5. choose DROP / REPEAT / MERGE / EXCEPTION / NEW_SCHEMA; 6. append to immutable episodic store; 7. update schema graph; 8. build provenance + utility certificate; 9. broadcast on cadence, within wire budget; 10. place received records in Q; 11. on visit, validate world-time, role, local utility; 12. promote Q-record to retrieval/planning/action authority; 13. retrieve an admissible song (support$\times$staleness order); 14. recover landmark correspondence (consensus $\geq k$ + loop closure); 15. reserve target/route; 16. execute safe prefix ($\geq$ 5 couplets verified); 17. detect rupture; 18. commit / replan / rollback.

Quantities in one place. Marginal utility $U_{i,\iota}(m|M) = \text{cost}_i(M,\iota) - \text{cost}_i(M \cup \{m\},\iota)$; analogy cost $C = L(\phi) + \lambda_d D_{\text{decision}} + \lambda_a \text{Defect}(\phi)$; retrieval score $\text{support} \cdot e^{-\alpha\,\text{age}}$ (trust-flip $\times 0.25$ on an excepted parent); world-clock admissibility $\text{ver}(m) \geq \text{known\_ver}(\text{fam})$; rupture index $= \arg\min_e \text{trust} \cdot e^{-\alpha\,\text{age}(e)} < \tau$; objective $w_s\,\text{succ} + w_t(-t) - w_b\,\text{bits} - w_c\,\text{collisions}$ (weights w_s, w_t, w_b, w_c distinct from the staleness rate α above and from Part I's conditional product Φ).

Hyperparameters (frozen; test seeds never used to tune). utility threshold U_{thr}=5 (validation, U1); analogy share ≥ 0.4 and distortion ≥ 3 (validation, U1); staleness α=0.05, τ=0.30 (fixed, Part II); song budget $v_{\max}$=10 (*evolved*, U3); anchor consensus k=3 and safe-prefix depth 5 (safety design, N1/C1). Selection method and per-parameter sensitivity are tabulated in Appendix 69.

58 U1: the matrix, measured deterministically

Eight valid grid-world families, five candidate memories per family — one per matrix cell, in sequence: a useful novel song (NEW), its exact duplicate (REPEAT), a song from a structurally different world (NOVEL), a song to a goal irrelevant to the water intent (IRRELEVANT), and a song from a variant world whose water has been secretly moved (CONFLICT). Two single-axis baselines run on the same stream: similarity-only (store if novel, last-write-wins merge if similar) and utility-only (store if useful, consume newest-first). Four predictions were registered; the first registration failed on two construction defects (an "irrelevant" goal that accidentally lay en route to the water; consumers spawning inside walls), both documented, and the corrected registration passed $4/4$:

- **Repeats are free by construction:** the duplicate's marginal utility is exactly 0 in every world; final memory is 3 schemas vs. 5 for append-everything.
- **Similarity is not a consolidation criterion:** the similarity-only baseline incorporates the irrelevant song and silently corrupts the original world by overwriting its schema with the conflict variant (62.6 steps and phantom-first, vs. 12.3 **steps first-try** for the two-axis memory).
- **Exceptions beat both overwrite and recency:** with the exception stored, the variant world costs 20.5 vs. 70.4 without it, while the original stays first-try correct; utility-only's recency order phantoms first on the original (24.0 steps).
- **Certificates are recomputable:** a receiver at a different position recovers positive utility from the certificate's evidence in every world.

59 U2: the matrix is learnable — and improves itself

A LinUCB contextual bandit replaces the hand thresholds. Features: $(U, \text{share}, D, \text{novelty}, \text{memory size})$; actions: the five operations; reward: the *measured* cost change of the memory on a two-world probe (the candidate's world and the nearest schema's origin world — the second term is what teaches the controller not to overwrite), minus a small storage tax.

The first registration failed on two evaluation defects that are themselves findings. The final battery was circular (each policy was scored on the worlds it chose to store, making random indistinguishable from designed), and two pairs of actions were *reward-equivalent*: NEW and EXCEPTION had the identical structural consequence, as did DROP and REPEAT — and the bandit recovered exactly that quotient, confusing actions strictly *within* equivalence classes while learning the drop cell at $40/43$. Bookkeeping without consequences is unlearnable. Version 2 made the battery independent and gave support a consequence (consumption order). Result:

	learned	hand matrix	random
held-out battery cost	**78.1**	83.5	105.9

The learned policy reduces task cost by 6.5% relative to the hand-designed policy. Its majority actions agree with the hand matrix in $3/5$ cells (a registered miss); the disagreements are two context-dependent equivalences (merging a duplicate $\equiv$ repeat; repeating junk $\equiv$ drop) and one *principled, profitable deviation*: where the hand matrix says MERGE, the controller appends a new version and lets support ordering arbitrate — it independently discovered the immutable-store principle (§57), the same lesson the consolidation literature reaches from failure analyses (Yang et al., 2026; Zhang et al., 2026b).

60 U3: optimization of grammar parameters within a fixed family

An evolution strategy (30 generations $\times$ 16 genomes) selects the grammar θ = $(\text{couplet gap}, \text{share threshold}, \text{distortion threshold}, \text{utility threshold}, \text{couplet budget } v_{\max})$, with fitness measured on maps the genome never sang on (cost + λ·bits). Both registered predictions passed: the evolved genome matches the hand grammar on hold-out within registration tolerance, and it selects a *finite* budget ($v_{\max} = 10$) with no cost regression — bounded rationality obtained by selection, not imposition. Beyond registration: selection converged to the neighborhood of the hand-designed thresholds (share $0.74 \to 0.46$ vs. hand 0.4; distortion 3.9 vs. hand 3) while choosing denser couplets and a lower utility bar — "store more, but shorter." The designed constants of the matrix sit near a selected parameter region rather than being arbitrary.

61 U7: seven policies, two bodies, three horizons

The decisive comparison runs one mixed episode stream (new families, exact repeats, appearance variants with resampled hazard texture over persistent walls-and-water structure, and water-moved conflicts) through seven team-memory policies, consumed by two embodiments — *fragile* (hazards cost 12 steps) and *robust* (hazards cost 1, steps cost 2) — and evaluates on the families' latest true worlds plus fresh appearance variants never streamed. Policies: raw long context (append all, newest-first), embedding-style signature similarity, recency-importance-relevance (Park et al., 2023), full-snapshot exchange, utility-only, analogy-only, and the full two-axis memory with role-profiled certificates.

Table 34: Mean fragile-role battery cost by stream length (12 seeds) and stored bits at 1000 episodes. All four registered predictions failed; the corresponding failure mechanisms are analyzed in the text.

Policy	10 eps	100 eps	1000 eps	bits @1000
raw long context	128.9	**53.3**	**139.8**	339,453
utility-only	130.5	53.6	144.6	162,107
two-axis (UCSM)	132.8	60.8	145.2	**153,774**
signature similarity	161.1	214.7	232.1	186,843
recency-importance-relevance	190.7	225.4	250.1	339,453
full snapshots	209.4	246.6	264.5	5,198,886

Failure analysis. The registered dominance, sublinear-bits, role-preservation and false-merge predictions all failed. Three mechanisms carry the content. *First*, the fail-closed identity layer makes bloat cheap in steps: stale or foreign songs simply produce no anchors and hence no targets, so the raw archive is filtered without an explicit storage penalty at consumption time and pays only in bits and matching compute. *Second*, sublinear memory growth is impossible without cross-family abstraction: in an open stream where 30% of episodes found new structures, consolidation caps within-family growth ($2.2\times$ fewer bits than raw, $34\times$ fewer than snapshots) but cannot compress novelty it has no schema for. *Third*, the long-horizon failure mode — phantom-first rates near 0.95 for every song policy at 1000 episodes — is a *staleness* cascade from accumulated conflicts, which no storage policy fixes by ordering alone; it is the native territory of the series' trust$\times$staleness law, which none of the seven policies implemented.

Results stable across horizons. At every scale, structural memory (identity + beats, whatever the consolidation rule) beats the mainstream heuristics — embedding similarity, recency-importance-relevance scoring (Park et al., 2023; Xu et al., 2025), and snapshots — by 1.7–$4.5\times$; and the two-axis memory is the cheapest in bits among song policies everywhere.

U7b: testing the staleness diagnosis. An eighth policy — the same two-axis consolidation consumed newest-first — was registered and run. At 100 episodes it *passes*: it matches or beats raw on both roles ($52.0/67.9$ vs. $53.6/68.9$) at 54% of raw's bits, closing the entire consolidation-overhead gap by ordering alone. At 1000 episodes it fails by $\sim 3\%$: under strong appearance drift, raw's redundancy retains residual value — older variants of a song still anchor where the newest one fails to. The program formula that survives: **consolidation buys bits, freshness buys success, and they compose at moderate horizons; the next lever is a real age gate and cross-family schemas, not better ordering.**

62 Addressing staleness, landmark selection, and abstraction

A second registered wave attacked the three most concrete gaps the first wave localized.

Staleness in consumption (U7c). The series' trust$\times$staleness contract was wired into consumption: schemas ordered by support $\cdot\, e^{-\alpha\,\mathrm{age}}$, plus the discrete *trust-flip* — a schema that has accumulated an exception is demoted, so the counterexample is tried before its discredited parent. At 1000 episodes this closes the robust role entirely (214.2 vs. raw's 214.3) and cuts the fragile-role gap threefold (-0.9% vs. -3% for freshness ordering alone); the strict both-roles registration still fails — the failure is retained and analyzed: residual redundancy value under appearance drift survives even the gate. The composition ladder is now measured: consolidation alone -14%, +freshness -3%, +trust-flip -0.9%.

Causally promoted landmarks (E1). The causal loop *propose* $\rightarrow$ *ablate* $\rightarrow$ *measure* $\rightarrow$ *promote* was implemented and evaluated: starting from a feature superset including a derived candidate and deliberately harmful frame-variant junk (cell parity), the loop caught the junk causally ($\Delta = -0.67$; its harm manifested as refusals — zero fail-open across all ablation rounds) and reached a minimal vocabulary at perfect score ($0.33 \rightarrow 1.0$). The registered expectation that content classes would survive failed in the most instructive way: the surviving pair was $\{\texttt{wall}, \texttt{void}\}$ — the border feature that carries *zero* consensus mass in the attribution measurement turns out to be causally necessary as an *anti-aliasing guard* once the vocabulary is impoverished. Causal ablation and attribution mass are different quantities; the value of a feature is non-additive and contingent on the rest of the vocabulary.

First-order cross-family abstraction (X1). A shared signature codebook — constellations recurring across families stored once and referenced by index, lossless by construction — was added to the storage codec. Registered and passed: the codebook's bits growth ($7.91\times$ per decade) is smaller than the plain codec's ($8.42\times$), and the absolute saving grows with scale (-10% at 100 episodes, -16% at 1000): cross-family motifs exist and reduce storage through lossless reuse of recurring constellations. Sublinearity is approached, not achieved — full schema-level abstraction (slots and roles, not just shared constellations) remains the open half.

The LLM end-to-end cycle (L1). The full promised loop — observation $\rightarrow$ candidate memory $\rightarrow$ controller $\rightarrow$ schema graph $\rightarrow$ reconstructed active context $\rightarrow$ LLM action — was implemented on live models (the W4 division of labour: the LLM commits the semantic target, a deterministic waypoint primitive walks). Formation is deterministic UCSM over 4- or 12-episode histories in which the water is moved once or twice, so the stale targets hold up to a 10:2 majority of the raw transcript; a fresh LLM session then commits under three context arms (no memory / full transcript / schema graph with certificates and exceptions). Registered and measured: memory is needed (the no-memory arm is 0.0 everywhere); reconstruction beats replay — `llama3.1-8B` degrades on the raw transcript from 0.42 to 0.08 as history grows (conflict chains $0.17 \rightarrow 0.00$) while the active context holds 1.0 throughout at 6–12% of the transcript payload; `Qwen2.5-3B` reads the short transcript perfectly (1.0) yet drops to 0.375 on 12-episode chains, with the active context again at 1.0. A 4B thinking model fails the supersede chain in every arm — a capability floor below the task, reported for completeness. The transcript's failure mode is specifically the stale majority, not length (benign 12-episode logs stay readable to the stronger model): raw context reproduces the U7 phantom cascade in prompt space, and the reconstructed schema graph is its prompt-space staleness gate.

63 Extensions: structural matching, social exchange, and delayed credit

The remaining three gaps were evaluated at scale. One was resolved within the registered criterion, one produced a near-miss, and one returned the strongest negative result of the program.

Graph-matching analogy (G1). The LCS proxy was replaced by structural alignment: Needleman–Wunsch over couplets with partial node similarity (Jaccard over signature keys) and a *relational defect* — beat-chain consistency of the aligned pairs (do the two songs walk the same skeleton?). On appearance-heavy streams (300 episodes, 12 seeds) the graph controller recognises "the same structure in a new texture" that LCS blindly filed as new: 24% fewer schemas and bits (122 vs. 162 items; 42k vs. 55k bits) at cost within 5% on both roles and no safety price (phantom +0.012). The registered schema threshold ($\leq 0.75\times$) was missed by 0.01 — reported as a failed registration whose every substantive clause held.

Terminal credit (M1). The immediate-reward bandit was retired: an evolution-strategies meta-learner optimizes the same policy class against a *terminal-only* reward — one number per full stream, after dozens of interleaved decisions. On held-out streams the meta-learned policy beats both the densely-shaped bandit (-97.95 vs. -99.69) and the hand matrix (-99.95), and on hand-EXCEPTION candidates it takes store-class actions in 91% of cases with no replace-majority — the third independent recovery of the immutable-store principle (after the hand-designed controller and the bandit), this time from a single delayed scalar.

Long-horizon social exchange without admission (S1). Six agents of mixed embodiments, each with its OWN memory, acting simultaneously in a world whose families evolve globally, exchanging certificates on a cadence. At 20 episodes communication helps; at 300 episodes **independence wins outright** (fragile 148 vs. 170; robust 171 vs. 238 for every communicating arm). Mechanism: receivers accumulate hundreds of foreign schemas (741 vs. 221 items) whose world-referents drift stale faster than role-gating or recency can filter — broadcast time is not world time. Three of four registered predictions failed on this; the one that passed is reservations (S1.3: duplicate commitments to zero at no cost). This is the series' central law — *the gate, not the exchange, carries the value of collective memory* — re-derived in the formation substrate: certificates without a world-clock staleness contract are not merely insufficient, they are net harmful at horizon, exactly as raw exchange was for the LLM collective in Part II (§25).

The world-clock contract (S2) — necessary, not sufficient. S1's diagnosis was implemented: every family carries a state version bumped when its world actually changes; certificates carry (family, version); version news travels in certificate metadata even when the schema is rejected; admissibility is gated on world clocks at both consumption and consolidation. On paired assignment streams (12 seeds, 300 episodes) the gate improves every communicating arm (robust $238 \to 220$, fragile $170 \to 163$) and reservations still eliminate duplicate commitments without additional task cost — but the S1 reversal does not reverse: independent agents beat the gated arm on *every* seed, and memories barely shrink (688 vs. 708 items). The gate recovers only $\sim 27\%$ of the communication penalty. Two registered negatives now bracket the problem from both sides: message-time filtering (S1) and world-time filtering (S2) both fail to make social exchange net-positive at horizon. The residual mechanism is not temporal at all: it is *cross-family aliasing from sheer foreign volume* — hundreds of schemas about families the receiver rarely visits still partially match its local constellations and emit phantom targets. What the data now names is ADMISSION control, not staleness control: accept foreign knowledge only where it has demonstrated utility on one's own visits — the certificate is testimony, and testimony, the two negatives jointly say, is not admissible evidence.

Admission control (S3) — the reversal reverses. The principle the two negatives pointed at was implemented: every foreign certificate is QUARANTINED on arrival (version metadata still travels); when the agent visits family f, quarantined certificates about f are validated ON THE SPOT — the foreign song's marginal utility is replayed on the actual current world under the agent's own role; passers are admitted with their utility *measured, not told*, and are usable in that same visit; failers are discarded; knowledge about never-visited families never enters consumption. All four registered predictions pass (12 seeds, paired streams). **Social exchange is net-positive at horizon for the first time**: the full contract beats independence on every seed and both roles (fragile 118 vs. 148, -20%; robust 134 vs. 171, -21%). The mechanism is monotone — testimony $162/220 \to$ relevance-only admission $136/169 \to$ demonstrated-utility admission $118/134$ — hygiene finally materialises (admitted memory 233 vs. 688 items, $0.34\times$, with 592 junk certificates held harmlessly in quarantine), and reservations compose (zero duplicate commitments at $+0.6\%$ cost). The social ladder of the program is thereby measured end to end: exchange without contracts is harmful (S1), world-clock staleness is necessary but insufficient (S2), and admission by demonstrated utility on one's own visits — testimony is not evidence — resolves the evaluated social-memory setting (S3).

64 One method: the integrated runtime and its benchmark

Every mechanism above was validated in its own controlled experiment; the central integration question is whether these mechanisms remain effective when composed. We therefore built **Songline Memory Runtime v1**: the one canonical record type $m = (G, C, E, U, P, T, F, A, H)$ of §57 (song graph, applicability, expected effect, per-role utility, provenance, time and world version, failures/exceptions, admission state, and alignment requirements; frame mappings $\Gamma_{ij,t}$ re-derived at consumption, never stored), one cycle from observation to execution, and every mechanism as a configuration flag — so the benchmark's arms and ablations are configurations of one artifact, not adjacent codebases. Wire and storage accounting uses the full codec (certificate 64, provenance 48, time/version 32, reservation 24 bits).

Single-run integration (R1). The ten-point completion test — route discovered in a private frame; song received; certificate held in quarantine; admission by the receiver's own measured utility; landmark correspondence; route execution; rupture fallback after the world changes; reservation deferral; superseded versions never used — passes within a single execution of the integrated runtime, $10/10$ scenarios.

The integration benchmark (I1). Six agents of mixed embodiments, two intents (resource and rest), private frames, globally drifting families, 300 episodes, 12 paired test seeds (numbered $100+$, never used to tune anything; all thresholds frozen from the earlier waves). Group cost (fragile/robust): full Songline $\mathbf{120.0/125.1}$ dominates independent (137.3/146.5), raw shared history (160.9/204.4), graph-without-provenance (146.3/174.5) and the song ladder without admission (146–157/174–196); it is simultaneously the only arm with zero duplicate commitments, the lowest phantom-first rate and a wire cost below raw's (55 vs. 69 KB). The advantage grows with scale — -11%, -13%, -14% at $N = 4, 6, 8$ (registered H3: **pass**) — and survives 10% and even 20% perception noise (-13–15%; the advantage clause of H2: **pass**).

Ablation matrix and failure modes. Removing landmarks or beats is catastrophic (success 0, cost worst: fail-closed by construction); removing admission costs $+22\%/+38\%$ — the component with the largest measured contribution, confirming S3; removing the world clock costs $+9\%$ on the robust role; removing reservations restores duplicate commitments; removing the utility gate bloats wire and memory by $\sim 40\%$ at equal cost. But three ablations — provenance, exceptions, the immutable layer — are cost-free *in this substrate*: registered H1 therefore **fails** as stated, and the mechanism is worth the failure. Their value was demonstrated in the settings that exercise them (corruption on return in U1, gossip explosion in S1, audit and reconsolidation); the live-visit drifting benchmark never punishes their removal. In one artifact they are free insurance, not dead weight — but the composition claim must be scoped to the mechanisms the substrate actually stresses.

Utility without the oracle (UE1). A ridge estimator over runtime-available features only (anchoring, local plan distance, song shape, memory counts) reaches Spearman 0.989 and sign accuracy 0.97 against the true counterfactual on held-out worlds, and the full controller driven by $\widehat{U}$ alone matches the oracle version (cost ratio $0.965/0.977$) — the critical experiment: the system survives estimation.

The margin lesson (N1). Under independent false-negative $\times$ false-positive perception noise, coverage degrades gracefully ($0.96 \to 0.92$ at $10\%/10\%$: registered **pass**), but the fail-closed contract does NOT survive naive margin softening: with a plain cosine margin the fail-open rate reaches 0.21 inside the 20% envelope (registered N1.1: **fail**). Exact-equality matching was necessary in this configuration; the named fix is consensus-of-anchors verification (the identity layer's $\geq$ 3-pair consensus rule, which single-anchor song consumption bypasses) — margin machinery, not a lower threshold.

65 Integrated evaluation under equal budgets, adversarial provenance, and continuous transfer

We evaluate whether each mechanism produces its predicted failure mode in at least one targeted or integrated setting: at equal budgets the full method should beat the best baseline on unseen dynamic worlds without hurting safety, the advantage should survive noisy perception, private frames, heterogeneity and long-horizon drift, and removing a mechanism should reproduce its characteristic failure somewhere in the suite. We do *not* require every component to affect aggregate cost in every substrate — a mechanism whose value is shown in a specialized experiment (corruption for provenance, gossip for the immutable layer) need not also move the integration benchmark. We report each clause below.

Statistical footing. On the integration benchmark the full method beats independent on $12/12$ **paired test seeds** (mean team gain 13.6%, sign-test $p = 0.0002$), and its hazard contacts are *lower* (5.06 vs. 5.66 per visit) — the advantage does not buy speed with safety.

Equal budgets (B1). Capping wire and memory at the full method's own footprint (55/45 KB) and giving the raw-history baseline the same, the full method obtains $119.8/124.5$ versus $158.7/199.9$; and raw does not catch up at $2\times$ or even $4\times$ the budget ($159.3/201.1$ throughout). Raw's deficit is not a storage shortage that a bigger cap fixes — it is the absence of consolidation and admission. (On the continuous substrate raw is not merely worse but computationally intractable at the full horizon — $\mathrm{O}(E^2)$ matching over an unconsolidated store — a cost the wire/byte accounting alone does not capture.)

Adversarial provenance (P1). With one poisoning peer that corrupts its own songs and launders mutated foreign records under their original origin, the full method degrades $+1.2\%$ while the defense ordering holds (full $\leq$ no-provenance $\leq$ no-admission). The instructive negative: the unprotected arm barely degrades under poison *because it is already saturated with phantoms from ordinary drift* (159.7 clean). The dominant adversary of collective memory is world entropy, not a malicious agent — and admission, which gates on demonstrated utility at the receiver's own visit, absorbs the liar and the stale alike; origin-bound receipt blocks the laundering at no measurable cost.

Safety under noise (N1, revised). The fail-open bar $< 1\%$ is not met by a softened cosine margin (up to 0.21). The named fix — require a consensus of ≥ 2 mutually-matched anchors whose pairwise displacements agree with the beat chain (the identity layer's loop-closure rule, which single-anchor song consumption bypassed) — drives **fail-open to** 0.000 **across the entire** $20\%\times20\%$ **envelope**, trading coverage ($0.88 \rightarrow 0.58$ at $10\%/10\%$). Safety is bought with refusals, as the fail-closed contract demands.

A second substrate (C1). Formation and consumption were moved to a continuous box with real-valued feature positions, per-observation jitter, soft constellation matching (a positional tolerance replacing exact key equality), unimodal anchor clustering with centroid refinement, and tolerance-based loop closure — the runtime otherwise unchanged. The advantage transfers on both roles and against both baselines — at a matched feasible horizon full $245.5/71.4 <$ raw $256.9/101.2 <$ (fragile) and $<$ independent $255.3/75.9$ (C1.3), and at the full horizon full $244.7/74.7 <$ independent $256.8/79.6$ (C1.1) — confirming the effect is not a grid artifact. Continuous social-scale safety needs a *three-layer* defense, each layer catching what the previous misses: anchor consensus (≥ 3 mutually-matched, loop-closed anchors) reduces per-transport aliasing ($0.79 \rightarrow 0.15$); single-commitment gating (walk only the most-anchored target, never a phantom candidate list) reaches 0.04; and a *safe-prefix verification* — the traveler physically walks the song's first five couplets and requires each observed constellation to match, so a foreign-family phantom that diverges from the actual terrain is refused before commitment — drives fail-open to $\mathbf{0.0014}$ on held-out test seeds ($554\times$ below the raw continuous rate, $10/12$ seeds exactly zero), *under* the 1% safety bar, while first-lock success *rises* (phantoms that used to be committed are now filtered). The prefix check is the proposed "successful use on a short safe prefix" promoted to a hard gate; it also lets the consensus threshold relax (recovering coverage) by catching the lone wrong anchors a consensus count cannot. Calibration was frozen on dev families (5001+); test seeds (100+) were never touched.

External substrate (VMAS physics). To test whether the effect depends on the authors' own grid and synthetic-continuous substrates, we transplanted not the diagnostic logger but the *whole runtime* (formation, certificate exchange, quarantine, receiver-side admission, frame-free continuous alignment, safe-prefix consumption) onto VMAS, a third-party continuous-physics multi-agent simulator. Holonomic agents move under a proportional controller in a landmark-populated box; a receiver with no shared frame recovers a witnessed water target purely by matching landmark constellations. Against a memory-free baseline given a dense box-tiling search and a longer horizon — so it *can* find the target by exhaustion, making the contrast a genuine speed-up rather than a floor — the full runtime reaches the target in 254.5 versus 543.5 team steps (-53.2%, $12/12$ paired seeds, sign test $p = 0.0005$), with transport fail-open $\mathbf{0.0000}$: agents that cannot anchor *refuse* and fall back to the same blind search rather than committing a phantom. Only the substrate density (landmark count, witness-sampling rate, consensus threshold matched to the constellation geometry) was recalibrated; every mechanism flag is unchanged from the grid configuration. The effect and its fail-closed safety therefore transfer to a third, externally implemented substrate.

66 Related work

Zou et al. (2026) argue memory should preserve decision-relevant differences rather than descriptive similarity and gate merging by decision distortion; we adopt the intuition and add the coordination side: heterogeneous receivers, per-role utility profiles, and certificates recomputed by the receiver. Generative Agents (Park et al., 2023) score recency, importance and relevance — our importance is a measured counterfactual, and the U7 comparison quantifies the gap. MemRL (Zhang et al., 2026a) learns a utility of *using* stored episodes; Mem-α (Wang et al., 2025) learns construction operations with RL on question-answering reward — our controller learns *formation* under coordination reward with corruption probes. MemGAS (Xu et al., 2025) selects granularity; TrustMem (Yang et al., 2026) verifies consolidation transitions (coverage, preservation, faithfulness) — our exception operation and immutable episodic layer attack the same failure from the mechanism side, and our learned controller rediscovers append-not-overwrite from reward alone. Zhang et al. (2026b) show continuous rewriting eventually corrupts; our U1 measures the same corruption in the similarity baseline and quantifies the benefit of the exception mechanism. To our knowledge no prior system combines structural analogy, causal utility, heterogeneous embodiments, coordination consequences, emergent landmark status and immutable provenance in one measured loop.

67 Limitations and open problems

The present results establish an integrated prototype, but several limitations remain. First, the utility estimator is trained and validated in environments where exact paired replay is available for calibration. External stochastic environments require additional causal or off-policy validation. Second, the graph-matching analogy represents route structure but does not yet include a complete model of roles, affordances, preconditions, and causal effects. Third, landmark promotion is causal only within a predefined feature superset; open-ended proposal and naming of new concepts from learned representations remain unresolved. Fourth, the cross-family codebook captures repeated motifs but does not yet induce higher-order schemas with variables and role slots. Fifth, the social LLM experiments address active-context reconstruction rather than the complete multi-agent admission process. Sixth, the continuous and VMAS studies remain simulated and do not establish robustness under real sensors, actuation uncertainty, or robot hardware. Finally, the evaluation criterion is task and coordination utility under the implemented contract; it should not be interpreted as direct verification of the truth of every stored proposition.

68 Conclusion

This part introduced a two-axis model of long-term memory formation in which counterfactual utility determines whether an observation should be retained and structural analogy determines how it should be integrated. The resulting controller distinguishes merging, exception storage, new-schema creation, repetition, and candidate rejection (dropping). Deterministic experiments show that the exception operation prevents the corruption caused by similarity-based overwriting, while learned and evolutionary controllers recover interpretable design principles, including append rather than overwrite and a finite route-description budget.

The long-horizon evaluation separates the functions of the main components. Structural consolidation reduces memory cost, but freshness and trust-aware consumption are required to preserve task performance under drift. Unregulated social exchange is harmful at long horizons; a world-time contract is necessary but insufficient; and receiver-side admission based on locally demonstrated utility changes the outcome in favor of collective exchange. The integrated runtime combines these mechanisms with provenance, private-frame alignment, route execution, reservation, rupture detection, and rollback. Under the tested equal-budget conditions, it outperforms raw history and independent memory, and the safety gates reduce fail-open behavior below the predefined threshold on the controlled substrates.

The current result is therefore a complete research prototype of utility-certified, provenance-aware collective memory. It should not be interpreted as a universal solution to long-context memory for all LLM or robotic systems. Open-ended concept formation, richer causal analogy, broad social LLM evaluation, and real-world robotic validation remain the principal extensions.

Appendix material of this part

69 Reproducibility

Every central result has a command (all in `experiments/song_grammar/`, run with `PYTHONPATH=.`):

```
# formation core
exp_u1_ucsm_smoke.py                       # U1 two-axis matrix
exp_u2_bandit.py    --train-seeds 0 24 --eval-seeds 100 110 --episodes 40
exp_u3_evolution.py --generations 30 --pop 16 --episodes 25
exp_e1_landmark_emergence.py               # E1 causal promotion
# scale, learning, LLM
exp_u7_seven_arms.py --seeds 0 12 --episodes 1000   # U7/U7b/U7c/X1
exp_m1_metarl.py    --generations 40 --pop 32 --episodes 30
exp_g1_graph_analogy.py --seeds 0 12 --episodes 300
exp_l1_llm_endtoend.py --backend hf --model Qwen/Qwen2.5-7B-Instruct \
    --mode long --layouts 12          # L1/L1b (also llama3.1 via ollama)
# social ladder
exp_s1_social.py  --seeds 0 12 --episodes 300 --agents 6
exp_s2_worldclock.py --seeds 0 12 --episodes 300 --agents 6
exp_s3_admission.py  --seeds 0 12 --episodes 300 --agents 6
# integration + acceptance test
exp_r1_runtime_checklist.py                # R1 single-run integration
exp_i1_integration.py --config songline_full --seeds 100 112 \
    --episodes 300 --agents 6         # I1 (17 configs; +--noise +--agents)
exp_ue1_utility_estimator.py --train-seeds 0 20 --test-seeds 100 110
exp_n1_noise.py   --seeds 30 --consensus 2                # N1v2
# B1 equal-budget: run the cap at 1x, 2x, 4x the full method's footprint
exp_i1_integration.py --config raw_history --seeds 100 112 --episodes 300 \
    --agents 6 --mem-cap-kb 45 --wire-cap-kb 55    # 1x
exp_i1_integration.py --config raw_history --seeds 100 112 --episodes 300 \
    --agents 6 --mem-cap-kb 90 --wire-cap-kb 110   # 2x
exp_i1_integration.py --config raw_history --seeds 100 112 --episodes 300 \
    --agents 6 --mem-cap-kb 180 --wire-cap-kb 220  # 4x
# P1 adversarial provenance (one poisoning peer)
exp_i1_integration.py --config songline_full --seeds 100 112 --episodes 300 \
    --agents 6 --poison 1 --launder 1              # P1
exp_c1_continuous.py --config songline_safe --seeds 100 112 \
    --episodes 300 --agents 6         # C1 continuous + three-layer safety
# external substrate: the whole runtime on VMAS physics (V2.1, -53.2%)
exp_v2_vmas_runtime.py --arm songline_safe --seeds 0 12 --agents 4 \
    --max-steps 600 --out tmp/v2_full
exp_v2_vmas_runtime.py --arm independent   --seeds 0 12 --agents 4 \
    --max-steps 600 --out tmp/v2_full
# main equal-budget benchmark, 30-seed rerun (BU.1/BU.2), run ON SPHINX:
#   bash cluster/submit_bench30.sh          # 6 policies x 30 seeds (100-129)
#   python scripts/aggregate_bench30.py tmp/cluster/song_grammar/bench30
```

Registered predictions, per-run rows and verdicts sit beside each output; failed first registrations (U1 v1, U2 v1, N1 v1, C1-safe depth-3) are retained with revision notes. Cluster launcher: `cluster/submit_song_grammar.sh`. Full method freeze, claim–evidence matrix and per-verdict source map: `docs/SONGLINES_V1_FREEZE.md`, `docs/CLAIM_EVIDENCE_MATRIX.md`, `docs/SERIES_VERDICTS.md`. Design documents: `docs/FRONTIER_{SONG_GRAMMAR,UCSM}_*.md`, `docs/MEMORY_REFRAMES_2026-07-26.md`.

Tables, figures, environment, and expected cost. All tables and figures in this part regenerate from the raw run outputs with `python scripts/aggregate_song_grammar.py` (long CSV) and `scripts/make_figures.py`; `scripts/reproduce_core.sh` chains the central deterministic blocks end to end. The pinned environment is `environment.yml` (local: Python 3.9, numpy 2.0.2; cluster base: Python 3.13, numpy 2.5.1, `vmas` 1.5.2); the GPU LLM arms use the cluster `gpu-env` (torch cu124). Expected cost of the heavy blocks: U7 at 1000 episodes $\approx$ 24–48 h per seed-shard on a CPU node; the L1/ALFWorld HuggingFace arms run on a 24 GB SLURM GPU node (7B fp16); the VMAS external runtime is CPU, $\approx$4 s for all 12 seeds; the 30-seed benchmark is 18 CPU shards, $\approx$1.5 h wall-clock; the local grid sweeps and Ollama studies are minutes-to-hours on the workstation. Test seeds (100+) were frozen before any tuning.

References

M. Andrychowicz, F. Wolski, A. Ray, J. Schneider, R. Fong, P. Welinder, B. McGrew, J. Tobin, P. Abbeel, and W. Zaremba. Hindsight experience replay. In *NeurIPS*, 2017.

R. Bělohlávek. Fuzzy Galois connections. *Mathematical Logic Quarterly*, 45(4):497–504, 1999.

J. Carmona, B. van Dongen, A. Solti, and M. Weidlich. *Conformance Checking: Relating Processes and Models*. Springer, 2018.

S. Es, J. James, L. Espinosa-Anke, and S. Schockaert. RAGAs: Automated evaluation of retrieval augmented generation. In *EACL System Demonstrations*, 2024.

M. Leucker and C. Schallhart. A brief account of runtime verification. *Journal of Logic and Algebraic Programming*, 78(5):293–303, 2009.

H. Lightman, V. Kosaraju, Y. Burda, H. Edwards, B. Baker, T. Lee, J. Leike, J. Schulman, I. Sutskever, and K. Cobbe. Let's verify step by step. In *ICLR*, 2024.

X. Lin, Y. Wang, T. S. T. Kwok, D. Guo, S. A. Nale, C. Fleming, and G. Cheng. REFLECT: Intervention-supported error attribution for silent failures in LLM agent traces. *arXiv preprint arXiv:2606.09071*, 2026.

C. Ma, J. Zhang, Z. Zhu, C. Yang, Y. Yang, Y. Jin, Z. Lan, L. Kong, and J. He. AgentBoard: An analytical evaluation board of multi-turn LLM agents. In *NeurIPS*, 2024.

D. Ru, L. Qiu, X. Hu, T. Zhang, P. Shi, S. Chang, et al. RAGChecker: A fine-grained framework for diagnosing retrieval-augmented generation. In *NeurIPS Datasets and Benchmarks*, 2024.

W. M. P. van der Aalst. *Process Mining: Data Science in Action*. Springer, 2016.

M. Bettini, R. Kortvelesy, J. Blumenkamp, and A. Prorok. VMAS: A vectorized multi-agent simulator for collective robot learning. In *16th International Symposium on Distributed Autonomous Robotic Systems (DARS)*, 2022. arXiv:2207.03530.

S. Boyd, A. Ghosh, B. Prabhakar, and D. Shah. Randomized gossip algorithms. *IEEE Trans. Information Theory*, 52(6):2508–2530, 2006.

M. Chevalier-Boisvert, D. Bhoopchand, H. Abdolmaleki, and A. Mordatch. BabyAI: A platform to study the sample efficiency of grounded language learning. In *ICLR*, 2019.

M. Chevalier-Boisvert, B. Lahlou, S. Willems, and P. S. Castro. MiniGrid and MiniWorld: Modular reinforcement learning environments. *arXiv:2306.13831*, 2023.

P. Chhikara, D. Khant, S. Aryan, T. Singh, and D. Yadav. Mem0: Building production-ready AI agents with scalable long-term memory. *arXiv:2504.19413*, 2025.

P. Dayan. Improving generalization for temporal difference learning: The successor representation. *Neural Computation*, 5(4):613–624, 1993.

M. Deitke, E. V. Wijmans, D. Schwenk, O. Salvador, K. M. Ehsani, et al. ProcTHOR: Large-scale embodied AI using procedural generation. In *NeurIPS Datasets and Benchmarks*, 2022.

B. Efron. Bootstrap methods: Another look at the jackknife. *Annals of Statistics*, 7(1):1–26, 1979.

J. N. Foerster, Y. M. Assael, N. de Freitas, and S. Whiteson. Learning to communicate with deep multi-agent reinforcement learning. In *NeurIPS*, 2016.

J. N. Foerster, G. Farquhar, T. Afouras, N. Nardelli, and S. Whiteson. Counterfactual multi-agent policy gradients. In *AAAI*, 2018.

B. Ganter and R. Wille. *Formal Concept Analysis: Mathematical Foundations*. Springer, 1999.

D. Ghosh, A. Gupta, A. Reddy, J. Fu, S. Levine, and C. Finn. Learning to reach goals via iterated supervised learning. In *ICLR*, 2021.

G. Gigerenzer and D. G. Goldstein. Reasoning the fast and frugal way: Models of bounded rationality. *Psychological Review*, 103(4):650–669, 1996.

S. Gupta, J. Davidson, S. Levine, R. Sukthankar, and J. Malik. Cognitive mapping and planning for visual navigation. In *CVPR*, 2017.

S. Holm. A simple sequentially rejective multiple test procedure. *Scandinavian Journal of Statistics*, 6(2):65–70, 1979.

E. Kolve, R. Mottaghi, D. Gordon, Y. Zhu, A. Gupta, A. Farhadi, and D. Fox. AI2-THOR: An interactive 3D environment for visual AI. *arXiv:1712.05474*, 2017.

B. Kuipers. The spatial semantic hierarchy. *Artificial Intelligence*, 119(1–2):191–233, 2000.

F. Lieder and T. L. Griffiths. Resource-rational analysis: Understanding human cognition as the optimal use of limited computational resources. *Behavioral and Brain Sciences*, 43:e1, 2020.

R. Lowe, Y. Wu, A. Tamar, J. Harb, P. Abbeel, and I. Mordatch. Multi-agent actor-critic for mixed cooperative-competitive environments. In *NeurIPS*, 2017.

S. Mac Lane. *Categories for the Working Mathematician*. Springer, 2nd edition, 1998.

I. Momennejad, E. M. Russek, J. H. Cheong, M. M. Botvinick, N. Daw, and S. J. Gershman. The successor representation in human reinforcement learning. *Nature Human Behaviour*, 1:680–692, 2017.

J. O'Keefe and L. Nadel. *The Hippocampus as a Cognitive Map*. Oxford University Press, 1978.

C. Packer, V. Fang, S. Patil, K. Lin, S. Wooders, and J. Gonzalez. MemGPT: Towards LLMs as operating systems. *arXiv:2310.08560*, 2023.

J. S. Park, J. C. O'Brien, C. J. Cai, M. R. Morris, P. Liang, and M. S. Bernstein. Generative agents: Interactive simulacra of human behavior. *arXiv:2304.03442*, 2023.

T. Rashid, M. Samvelyan, C. Schroeder de Witt, G. Farquhar, J. Foerster, and S. Whiteson. QMIX: Monotonic value function factorisation for deep multi-agent reinforcement learning. In *ICML*, 2018.

N. Savinov, A. Dosovitskiy, and V. Koltun. Semi-parametric topological memory for navigation. In *ICLR*, 2018.

M. Savva, A. Kadian, O. Maksymets, Y. Zhao, E. Wijmans, B. Jain, J. Straub, J. Liu, V. Koltun, J. Malik, et al. Habitat: A platform for embodied AI research. In *ICCV*, 2019.

J. Schulman, P. Moritz, S. Levine, M. Jordan, and P. Abbeel. High-dimensional continuous control using generalized advantage estimation. In *ICLR*, 2016. arXiv:1506.02438.

J. Schulman, F. Wolski, P. Dhariwal, A. Radford, and O. Klimov. Proximal policy optimization algorithms. *arXiv:1707.06347*, 2017.

M. Shridhar, X. Yuan, M.-A. Côté, Y. Bisk, A. Trischler, and M. Hausknecht. ALFWorld: Aligning text and embodied environments for interactive learning. In *ICLR*, 2021.

N. Shinn, F. Cassano, A. Berman, and A. G. Labash. Reflexion: Language agents with verbal reinforcement learning. In *NeurIPS*, 2023.

H. A. Simon. A behavioral model of rational choice. *Quarterly Journal of Economics*, 69(1):99–118, 1955.

H. A. Simon. Rational choice and the structure of the environment. *Psychological Review*, 63(2):129–138, 1956.

K. L. Stachenfeld, M. M. Botvinick, and S. J. Gershman. The hippocampus as a predictive map. *Nature Neuroscience*, 20(11):1643–1653, 2017.

S. Sukhbaatar, A. Szlam, and R. Fergus. Learning multiagent communication with backpropagation. In *NeurIPS*, 2016.

R. S. Sutton, D. Precup, and S. Singh. Between MDPs and semi-MDPs: A framework for temporal abstraction in reinforcement learning. *Artificial Intelligence*, 112(1–2):181–211, 1999.

E. C. Tolman. Cognitive maps in rats and men. *Psychological Review*, 55(4):189–208, 1948.

G. Wang, Y. Xie, Z. Jiang, A. Mandlekar, C. Xie, A. Yu, and P. Liang. Voyager: An open-ended embodied agent with large language models. *arXiv:2305.16291*, 2023.

Q. Wu, G. Bansal, J. Zhang, Y. Wu, B. Li, E. Zhu, L. Jiang, X. Zhang, S. Zhang, J. Liu, A. H. Awadallah, R. W. White, D. Burger, and C. Wang. AutoGen: Enabling next-gen LLM applications via multi-agent conversation. *arXiv:2308.08155*, 2023.

F. Wilcoxon. Individual comparisons by ranking methods. *Biometrics Bulletin*, 1(6):80–83, 1945.

R. J. Williams. Simple statistical gradient-following algorithms for connectionist reinforcement learning. *Machine Learning*, 8(3–4):229–256, 1992.

S. Yao, J. Zhao, D. Yu, N. Du, I. Shafran, K. Narasimhan, and Y. Cao. ReAct: Synergizing reasoning and acting in language models. In *ICLR*, 2023.

L. Xiao, S. Boyd, and S. Lall. A scheme for robust distributed sensor fusion based on average consensus. *IPSN*, 2007.

C. Yu, A. Velu, E. Vinitsky, J. Gao, Y. Wang, A. Bayen, and Y. Wu. The surprising effectiveness of PPO in cooperative multi-agent games. In *NeurIPS Datasets and Benchmarks Track*, 2022.

A. Das, T. Gervet, J. Romoff, D. Batra, D. Parikh, M. Rabbat, and J. Pineau. TarMAC: Targeted multi-agent communication. In *ICML*, 2019.

D. R. Jefferson. Virtual time. *ACM Transactions on Programming Languages and Systems*, 7(3):404–425, 1985.

C. Cadena, L. Carlone, H. Carrillo, Y. Latif, D. Scaramuzza, J. Neira, I. Reid, and J. J. Leonard. Past, present, and future of simultaneous localization and mapping: Toward the robust-perception age. *IEEE Transactions on Robotics*, 32(6):1309–1332, 2016.

S. Lowry, N. Sünderhauf, P. Newman, J. J. Leonard, D. Cox, P. Corke, and M. J. Milford. Visual place recognition: A survey. *IEEE Transactions on Robotics*, 32(1):1–19, 2016.

P. Aczel. *Non-Well-Founded Sets*. CSLI Lecture Notes 14, 1988.

J. Barwise and J. Seligman. *Information Flow: The Logic of Distributed Systems*. Cambridge University Press, 1997.

B. Jacobs. *Introduction to Coalgebra: Towards Mathematics of States and Observation*. Cambridge University Press, 2016.

A. S. Rao and M. P. Georgeff. BDI agents: From theory to practice. In *ICMAS*, 1995.

J. J. M. M. Rutten. Universal coalgebra: a theory of systems. *Theoretical Computer Science*, 249(1):3–80, 2000.

D. Sangiorgi. *Introduction to Bisimulation and Coinduction*. Cambridge University Press, 2011.

D. Sperber and D. Wilson. *Relevance: Communication and Cognition*. Blackwell, 2nd edition, 1995.

M. Zou, Z. Guo, L. Liang, Z. Wang, Q. Wang, Q. Wen, I. King, L. Qu, and Z. Xu. Remember the decision, not the description: A rate-distortion framework for agent memory. *arXiv preprint arXiv:2605.10870*, 2026.

J. S. Park, J. O'Brien, C. J. Cai, M. R. Morris, P. Liang, and M. S. Bernstein. Generative agents: Interactive simulacra of human behavior. In *UIST*, 2023.

S. Zhang, J. Wang, R. Zhou, J. Liao, Y. Feng, Z. Li, Y. Zheng, W. Zhang, Y. Wen, Z. Li, F. Xiong, Y. Qi, B. Tang, and M. Wen. MemRL: Self-evolving agents via runtime reinforcement learning on episodic memory. *arXiv preprint arXiv:2601.03192*, 2026.

Y. Wang, R. Takanobu, Z. Liang, Y. Mao, Y. Hu, J. McAuley, and X. Wu. Mem-α: Learning memory construction via reinforcement learning. *arXiv preprint arXiv:2509.25911*, 2025.

D. Xu, Y. Wen, P. Jia, Y. Zhang, W. Zhang, Y. Wang, H. Guo, R. Tang, X. Zhao, E. Chen, and T. Xu. From single to multi-granularity: Toward long-term memory association and selection of conversational agents. *arXiv preprint arXiv:2505.19549*, 2025.

T. Yang, S. Paul, V. Srinivasan, V. Kulkarni, and S. Chappidi. TrustMem: Learning trustworthy memory consolidation for LLM agents with long-term memory. *arXiv preprint arXiv:2606.25161*, 2026.

D. Zhang, Y. Lin, Z. Wu, Y. Sun, B. Li, D. Li, and H. Peng. Useful memories become faulty when continuously updated by LLMs. *arXiv preprint arXiv:2605.12978*, 2026.